01. INTRODUCTION

DURATION : 04:33

STUDY SHEET

COURSE SUMMARY

In this introduction to biodynamic embryology, we delve deeply into the process of human development, highlighting the intrinsic power of the foetus. Osteopaths are encouraged to relinquish conscious will to actively participate in this ballet of dynamic growth. By exploring fundamental interactions, molecular and tissue synchronicity, and key scenes of embryonic development, this session offers an enriched understanding of human construction. Practitioners will learn how these principles can revolutionise their therapeutic approach.

BIODYNAMIC EMBRYOLOGY: A QUEST FOR SENSORY HONESTY

This seminar is dedicated to **biodynamic embryology**, an approach that invites us to rethink our perception of human development and our role as osteopaths. This journey to the heart of genesis leads us to a profound search for **honesty** in our felt sense, pushing us to explore a context where **conscious will** gives way to **participation**.

BEYOND WILL: THE INTRINSIC POWER OF THE FOETUS

In the embryonic process, we are confronted with an **unheard-of power**. The developing foetus is endowed with an oriented force, a capacity to build itself with **remarkable precision**, making very few errors. It reveals a complex ballet of **interactions**, **inductions**, and **synchronicities** that guide its growth.

Our role, as practitioners, is not to perform a purely voluntary act, but to engage in an act of **participation**. We must surrender to this process, working with the **algorithms** inherent in development, these repetitive and fundamental patterns that underpin the entire construction of the being.

GROWTH: AN UNPRECEDENTED PERSPECTIVE

When we study anatomy and embryology through books, we often miss the **dynamic** aspect of growth. This deficiency can give us erroneous perceptions. Let's observe the construction of man carefully:

- **Man is absolutely not built around a bone structure.** Bone appears much later in development.
- **We do not build ourselves from the dense, but from the fluid.**

The origin of this fluid, its own construction, seems to come from an almost **electrical**, **electromagnetic** level. Pushing this reflection to its extreme, we could even speak of a **photonic** level, a luminous level. It is as if light itself could induce matter – this is precisely what we are going to explore.

THE FUNDAMENTAL SCENES OF DEVELOPMENT

We will journey through embryonic development in several **scenes**, from the very first moments:

1. **The first scene:** The beginnings of life.

2. **The second scene:** The emergence of the first structures.

3. **The third scene:** The complexification of forms.

4. **The fourth scene:** The deepening of systems.

We will observe how events unfold simultaneously in different places, revealing astonishing **synchronisms**.

SYNCHRONICITIES: FROM MOLECULAR TO TISSUE

These synchronicities are not limited to the macroscopic scale. They manifest at all levels:

- **Molecular:** Brain processes are the theatre of precise molecular synchronicities.
- **Cellular:** The coordination of cells is orchestrated by common rhythms.
- **Tissue:** Events occurring at the level of the crista galli, for example, can unfold simultaneously at the level of the palatine.

The essential thing is to grasp this notion of **dynamic growth**.

FROM FERTILISATION TO THE NOTOCHORDAL AXIS: THE EMERGENCE OF STRUCTURE

We will follow development from the first cell, from **fertilisation**, and even what happens before fertilisation on an electrical plane. This journey will lead us to:

- **The appearance of a first midline:** The notochordal axis.
- **The establishment of primitive fluidic cavities:**

* The first primitive peritoneal cavity.

* The amniotic cavity.

* The vitelline cavity (third chamber).

- **The emergence of movement within the embryonic tissue:** A wave that will form the notochordal process.

This notochordal process is crucial because it will induce the **neurological process**. We will study in detail the dynamic development of the brain and its practical implications in osteopathy.

PRACTICAL IMPLICATION IN OSTEOPATHY: THE INVITATION TO PARTICIPATION

Understand this absence of conscious will in development. When you are in the maternal womb, the growth process is ineluctable. You cannot say "stop, cease". The body grows **autonomously** and **powerfully**.

It is precisely at this level that we are invited to participate as osteopaths. Our intervention is not an imposition, but a **deep listening** and **accompaniment** of the inherent movement of life. Biodynamic embryology offers us the keys to understand and honour this **intrinsic wisdom** of the developing body.

02. GENERALITIES

DURATION : 08:38

STUDY SHEET

COURSE SUMMARY

This session addresses the importance of the environment in the development and well-being of individuals, highlighting the impact of emotional and physical contexts. Embryology is presented as anatomy in motion, integrating the notions of time and memory, essential for understanding bodily blockages. The concepts of body, mind, and spirituality are explored, establishing a link between osteopathy and epigenetics, while emphasising the importance of an integrative approach in therapeutic practice.

BIODYNAMIC EMBRYOLOGY: A HOLISTIC APPROACH TO LIFE

THE CRUCIAL INFLUENCE OF ENVIRONMENT AND SPACE

The **environment** in which we evolve is the driving force of our development. Take a seed: if you put it in the fridge, nothing will happen. But if you plant it in the earth, with light, sun, and warmth, it can develop.

Similarly, a complicated **emotional environment**, or any other, will have a major influence on us. We are talking more and more about **burnout**, and the strength of the environment accounts for almost **90%** of this power.

It is essential to consider our patients' **living environment**. I very often ask: "Where do you live? What are you experiencing? What have you experienced?" This is an anamnesis aimed at understanding the environmental "layer" surrounding the person.

To develop, we need two things:

- A **favourable environment**
- **Space**

Restoring space to the mind, tissues, and emotions is fundamental.

EMBRYOLOGY: FUNCTIONAL ANATOMY AND THE TEMPORAL DIMENSION

Embryology is functional anatomy. What does that mean?

- It is anatomy in **motion**.
- It is anatomy that uses the **4 to 5 dimensions** we know: up, down, left, and the fourth dimension, that of **chronos**, of time.

You will need to be able to observe this small developing embryo, its movements, and even through sections.

THE NOTION OF TIME AND MEMORY

This notion of **time** is also a notion of **memory**. You carry time within you; you are a concentration of time. Sometimes, in the body, there are specific areas where this concentration of time is blocked.

A word, a gesture, a look can block a person their entire life, and this can be found in the tissues. You may have noticed that after therapeutic work, an **emotional release** occurs. You have, in fact, released a concentration of time, allowing it to flow again.

In embryology, this notion is crucial because it allows us to rediscover this time. If you remember the film we watched this morning, there were movements where we moved forward, then suddenly, movements where we moved backward, like a wave.

If you observe such a movement, it means there are **points of support**. We will use these as therapeutic tools, because the body knows its points of support. If you give it a good point of support, it knows where it can develop, it knows how to rebuild itself.

THE UNITY OF BODY, MIND, AND SPIRITUALITY

There is the notion of time, tissues, fluids. Everything expresses itself in a **constellation**. It is said that "mind, body, spirit" are united.

Osteopathy is a science based on the art of touch, which offers a true understanding of the living body. It tells the story of anatomy linked to the movement of the spirit and mind.

One of the goals of osteopathy is to improve **exchanges** and to have an impact down to the **cellular** level. I even think we go further, with impacts on the human plane. We now see this through phenomena of thought and **epigenetics**. We have a very profound impact.

THE PATH OF LEARNING: FROM EMBRYOLOGY TO SPIRITUALITY

My professor proposed a path of study to me:

1. **Embryology**: Understanding embryology is understanding anatomy.

2. **Anatomy**: It gives us physiology.

3. **Physiology**: We must also study its pathology, its pathophysiology.

4. **Psychology**

5. **Philosophy**

6. **Spirituality**

Conception is how life develops. Anatomy is also how I die. Death, on a developmental level, is useful. If there is no cellular death, I prefer to speak of **transformation**. It is a continuous transformation.

THE LIFE FORCE AND TRANSFORMATION

This power, already established in the first cell, is present and continues until the end of your life, until your last breath. You have this ability to take the breath of life, to return it, and still have the energy to take it again. We participate in this at every moment. This is where transformation lies.

Sometimes, people are stuck in time and come looking for information about it. Very often, after attending these courses, something happens in their **predictions**. This is because we go into a space where the mind does not intervene directly, but where a modification occurs within.

We will observe the development of the "perfect" embryo, the **blueprint**, which maintains its midline, without deviation. This is the space of my conception. Sometimes, the midline is well aligned, but other times, one can be deviated from conception or at birth. This is **trauma**.

These points must be recognised. Non-recognition develops all compensatory activities. I must, at some point, through the quality of my presence, through the quality of touch, through what the tissue informs me, be able to recognise the impacts. And there, a **correction** can occur. One just needs to be present for the touch, because there is no will. This is what is truly important.

SELF-REGULATION AND THE FORCE OF NATURE

The body has a capacity for **self-regulation** and maintaining its health. This notion of the force of nature is about reconnecting with that. We see very clearly that if human impact disappears, nature quickly reclaims its rights. We have been somewhat cut off from this.

In this development, nature is there. The less we intervene, the better.

Another principle is the notion of **reciprocity** of structure and function. All these principles remind us that the body must be considered in its **totality**. We must realise that I can use this in my practice.

What I ask is simply that the body be **adjusted**. There are specialised drawings for little fingers, which is great. At the same time, they don't see that the lower or upper limb is an expansion of a **peritoneal sac**.

Atoms form **molecules**, molecules form **cells**, cells form **tissues**, tissues form **organs**. Ultimately, I am a bundle of atoms that move. But before the atom, there is also this whole notion of **energy**. What we perceive is a set of energy, a bundle of energy that sometimes needs to be structured a little, and which also thinks.

ONTOGENESIS AND PHYLOGENESIS

ONTOGENESIS

"Onto" means "being", and "genesis" means "formation". **Ontogenesis** is the formation of a being. We will study the ontogenesis of man, and especially his **morphogenesis**, that is, the formation of his shape.

PHYLOGENESIS

Phylogenesis traces the history of the development of all species through time.

We will study human ontogenesis, focusing primarily on the form and power of human development, and observing the impacts this can have at the practical level.

03. CHRONOLOGY OF THE DIFFERENT SYSTEMS

DURATION : 07:33 STUDY SHEET

COURSE SUMMARY

This video explores the timeline of communication systems within cells, highlighting the different stages from autocrine communication to the creation of complex systems. It reveals how the cell, through autocrine communication, becomes aware of its state, and then interacts with its neighbours via paracrine communication. As the systems evolve, structures such as the primitive digestive tube, the connective, circulatory, and endocrine systems emerge, leading to the development of the first organs, notably the primitive heart, and a complexification of the nervous systems. This improved understanding helps osteopathic practitioners grasp the interconnectedness between the different layers of the organism.

THE EVOLUTION OF CELLULAR COMMUNICATION: AN EMBRYOLOGICAL PERSPECTIVE

In **biodynamic osteopathy**, understanding the **chronology of appearance** of different communication systems is fundamental. This approach illuminates how organisms develop and adapt, from the **primitive cell** to the **complex human being**.

AUTOCRINE COMMUNICATION: KNOWING ONESELF

The very first form of communication to appear is **autocrine communication**.

Imagine a small cell. Around it, there is an environment, a membrane, and another environment. What does this cell do? It ejects a fluid outside itself.

Rapidly, this cell develops a small receptor on its membrane, called a **glycocalyx**, which is actually sugar. This receptor allows it to retrieve the fluid it has ejected and reincorporate it.

This means the cell is informed about itself. It's almost philosophical: before you can interact with the outside world, you need to know what you are.

It's a bit like the exercise we did this morning: taking three minutes to settle, connect with your body, your breath, your mind, and observe your state. Recognising this state is a form of autocrine communication, but at a cellular level. This is the first step.

PARACRINE COMMUNICATION: EXCHANGE WITH THE NEIGHBOUR

Next, the system evolves to develop **paracrine communication**.

It is based on the same principle as autocrine communication. I have two cells: one cell emits a small fluid, and the neighbouring cell, right next to it, receives this fluid.

This is paracrine communication, "next to". It is the fastest communication that exists in the body and is continuously used in your **homeostasis** system.

THE EMERGENCE OF SYSTEMS: FROM NOURISHMENT TO COMPLEXITY

The system continues to evolve. The fundamental idea is the **constant search for nourishment**.

I have one cell, then a group of cells. They function well thanks to autocrine and paracrine communications. In a group, we are stronger, more efficient.

THE PRIMITIVE DIGESTIVE TUBE

Primarily, to be efficient in the search and assimilation of nourishment, a **tube** develops. This is an adaptation pattern. I have created a system to distribute information when it arrives.

This is called the **primitive digestive tube**.

THE CONNECTIVE SYSTEM

It is only after the appearance of a primitive digestive system that a network, the **connective system**, develops between my cells.

THE CIRCULATORY SYSTEM

And it is only when I have a connective network that I will develop a **circulatory system**, in order to further improve exchanges.

THE ENDOCRINE SYSTEM

It is only when I have a circulatory system that I will be able to develop an **endocrine system**. Why? Because the cell can release a substance that will travel into the connective system, then into the circulatory system, and be distributed at a distance.

THE FIRST ORGANS AND THE CARDIAC SYSTEM

After functioning well in my network, I start looking for more specific things. I create **first organs**.

One of the very first organs to develop is the **primitive heart**. I develop within my network a system to distribute better, more quickly, a contractile system.

THE PRIMITIVE NEURO-ENTERIC SYSTEM

Then, a system called the **primitive neuro-enteric system** appears. This means I will develop a system where information enters, I need to close, then afterwards I will need to open: this is the **parasympathetic** system.

Then I will start to become more complex, by developing an **orthosympathetic** system.

MOVEMENT AND SENSES

Next, I will need to improve my **movement** in space. In this movement, I will need to develop a **sensory system**.

Then only develop a much more centred, **skeletal** system, and finally my limbs.

CHRONOLOGY AND CLINICAL IMPLICATIONS

This chronology of appearance is almost superimposed between **phylogenesis** (the evolution of species) and **ontogenesis** (the development of the individual).

For example, a child presenting with a motor development problem: it is essential to examine what happens before. Many practitioners, faced with this, work on the cranial. However, by observing the chronology, it is often necessary to go back to the digestive, connective, or circulatory system.

This implies that your **first brain** is enteric, it is not at all here (in the head). The cephalic brain developed logically, initially for itself. We have developed it so much that we have lost sight of this connection, we are cut off.

This chronology is important and must be learned in this sense. We will see that in ontogenetic development, this will follow, even if we do not always see it outwardly.

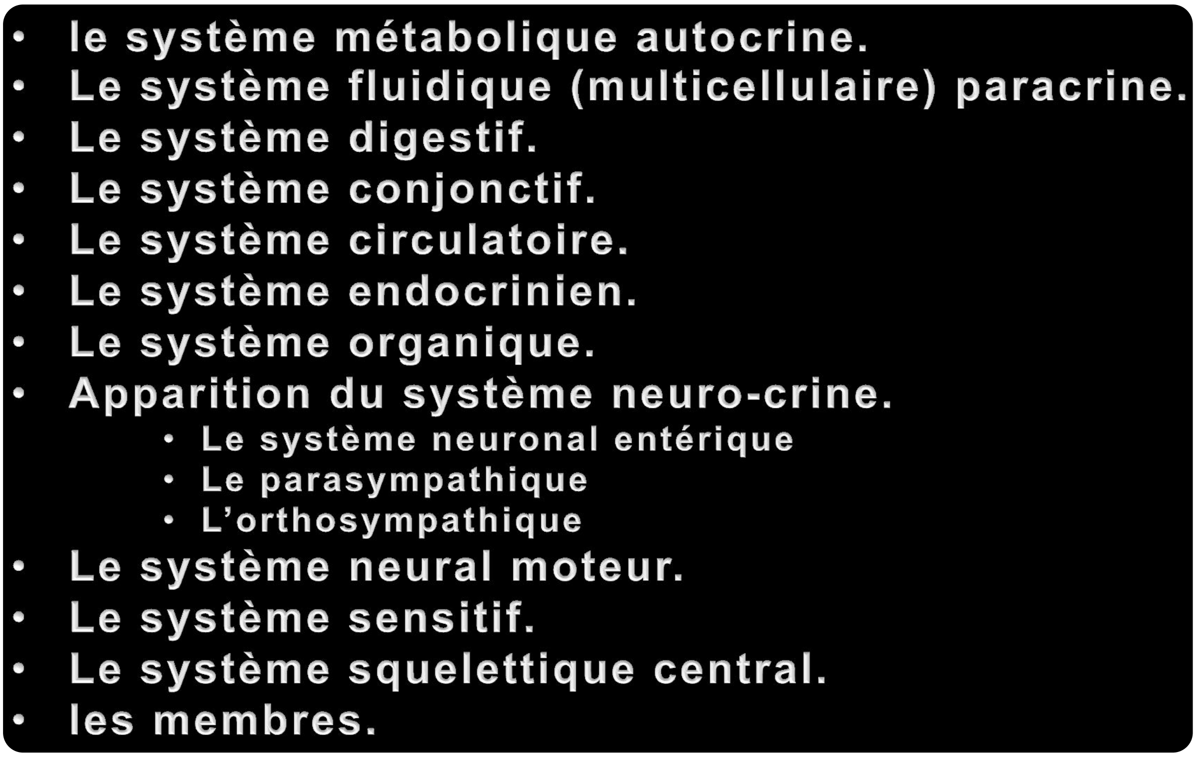


FIG. — Biological and anatomical systems

04. NOTIONS OF EPIGENETICS, TENSEGRITY, ELECTROMAGNETISM

DURATION : 06:25 STUDY SHEET

COURSE SUMMARY

This video explores the concepts of epigenetics, tensegrity, and electromagnetism, revealing how the environment modulates our genetic code. As osteopaths, it is crucial to understand that our actions and interactions can alter cellular behaviour and transmit transgenerational information. The concept of tensegrity reminds us that every touch is an integral vibration, influenced by various fields and stimuli. Together, these ideas provide an essential framework for a more conscious and holistic therapeutic approach.

NOTIONS OF EPIGENETICS, TENSEGRITY AND ELECTROMAGNETISM

THE IMPACT OF ENVIRONMENT ON GENETICS

Initially, our understanding of **genetics** led us to believe that everything was predetermined. It was thought that **DNA** contained all the necessary information to generate an organism. However, research has revealed a surprising truth: only **5 to 10%** of our genes are actually used, the rest having long been considered "useless".

EPIGENETICS: WHEN ENVIRONMENT REDEFINES THE CODE

With advancing knowledge, we have discovered the colossal impact of the **environment** on genetic behaviour. All these environmental systems can modify this behaviour at an ultra-rapid pace.

Imagine a DNA strand, composed of molecules and amino acids forming **genetic codons** that enable protein production. This system is not static; it continuously modifies and adapts. A change in environment, an emotional shock, or a powerful event can alter this structure.

OUR THERAPEUTIC RESPONSIBILITY

As **osteopaths**, our knowledge of anatomy and its function allows us to realise the profound impact of embryology. As Anthony Schiller states, this impact extends to the cellular level and even beyond. We have a responsibility: to be mindful of our actions and how we touch. Every interaction has profound repercussions.

THE CELL: A UNIVERSE OF TENSEGRITY

Let's take the example of a **cell**. It is connected to other cells by **intercellular adhesion molecules**, which maintain the structure. What we have discovered is that the cell possesses a **cytoskeleton**. This internal skeleton is connected to the membrane and extends to the nucleus, forming lines of force and precise structures within the cell.

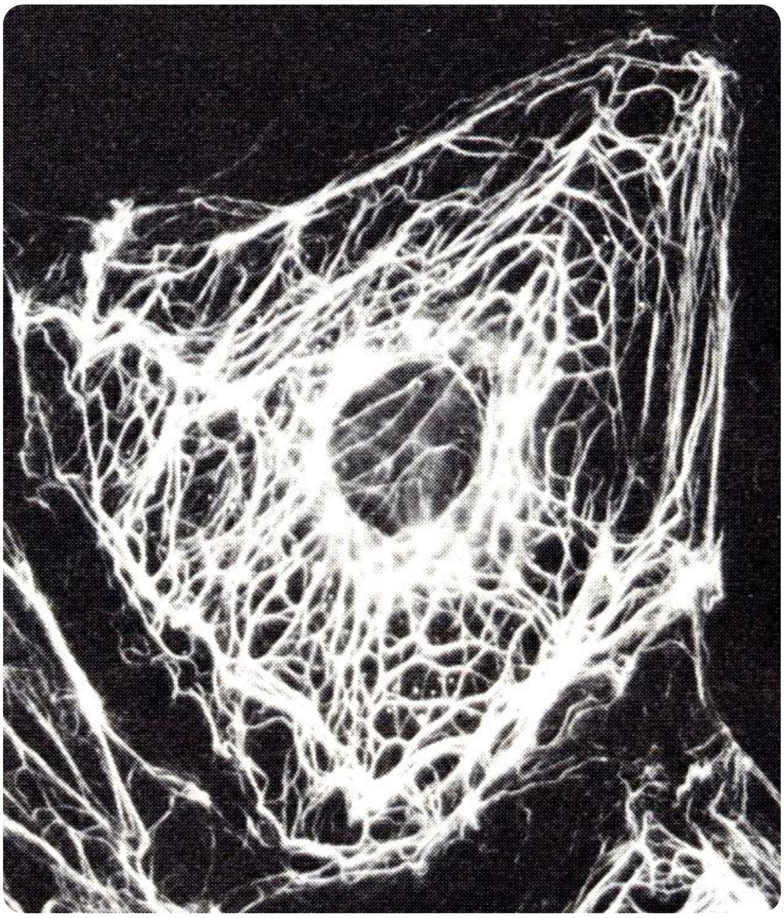


FIG. — Cell cytoskeleton

THE CONCEPT OF TENSEGRITY

This is where the fundamental concept of **tensegrity** in osteopathy comes in. If I touch someone, I touch the entirety of their body. It is a frequency, a vibration that propagates throughout the entire organism.

Let's observe a cell with its **extracellular matrix** and the junction between the cell and its external environment. What happens at this junction contains profound information.

ENVIRONMENTAL INFLUENCES ON THE CELL

An **electric field**, an **electromagnetic field**, light, temperature, pH, mechanical stress, shocks, deformations, and hydrostatic pressures – all of these influence cellular behaviour. This is not a small impact, but a global environment that acts on the system.

TRANSGENERATIONAL INHERITANCE AND FUNDAMENTAL FORCES

We have also discovered that our genetic programme carries **transgenerational** impact in our behaviours.

THE IMPRINT OF PAST GENERATIONS

Traumas experienced by our ancestors can be carried and manifest. It has been observed that we can carry information in our genetic code for up to **three generations**, sometimes even more. We are sometimes here to release this information.

Genetic studies have been able to trace back up to three generations to identify behaviours or appearances that resurface. This suggests that our technique should not be a simple manipulation, but a frequency, a vibration that interacts with this profound information.

THE FOUR FUNDAMENTAL FORCES OF LIFE

The presence of elementary particles implies forces in life. We currently recognise four major forces:

- **The weak force**
- **The strong force**
- **The electromagnetic force**
- **The force of gravity**

The weak and strong forces are nuclear forces that are beyond us. To give you an image, if one of these forces were removed from the universe, everything would collapse like a house of cards.

THE IMPORTANCE OF THE ELECTROMAGNETIC FORCE

The **electromagnetic force** is particularly interesting because it allows for the interactions of systems. It is responsible for bonding and allows matter to exist in its form. The organism must therefore involve and possess systems to call upon these forces. Where does the electromagnetic force come from? It is there before us. When matter appears, it must have this capacity.

THE POLARISED HUMAN AND MAXWELL'S LAW

This brings us to the concept of the **polarised human**. **Maxwell's Law** is very simple: if you have an electric field, you always have an electromagnetic field around it.

In other words, an electric field inevitably generates a surrounding electromagnetic field. This interaction is fundamental to understanding how our body functions and interacts with its environment.

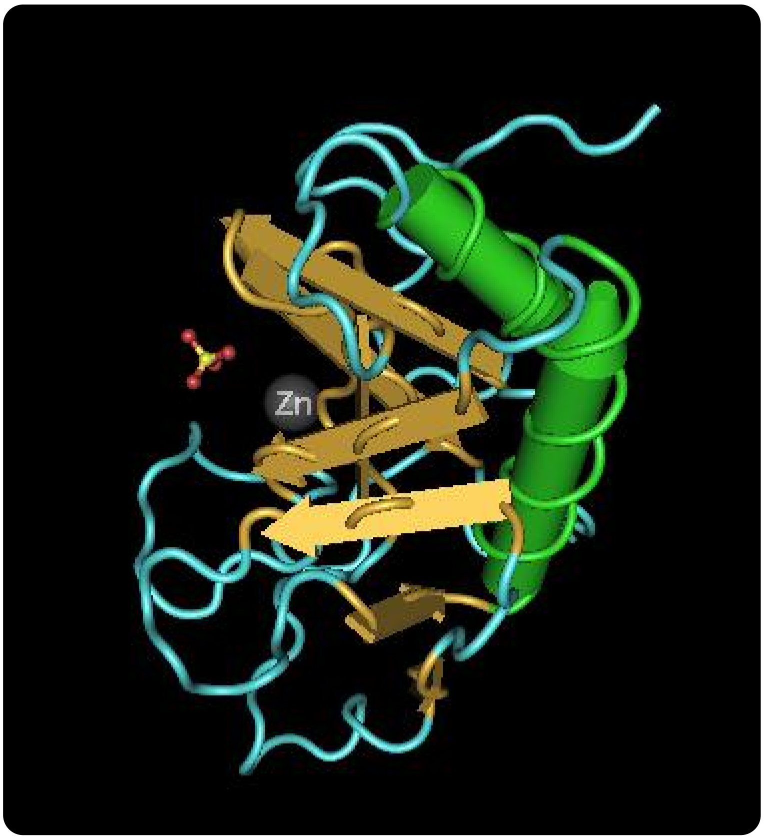


FIG. — Protein, Zinc, Sulfate

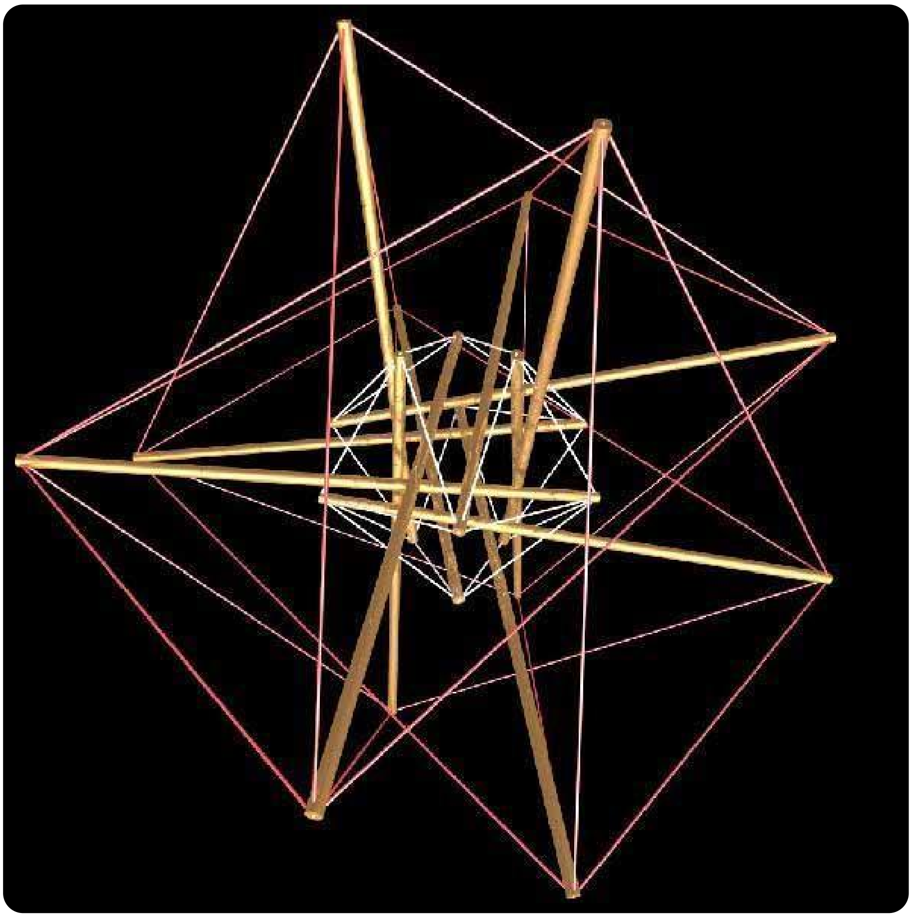


FIG. — Tensegrity structure

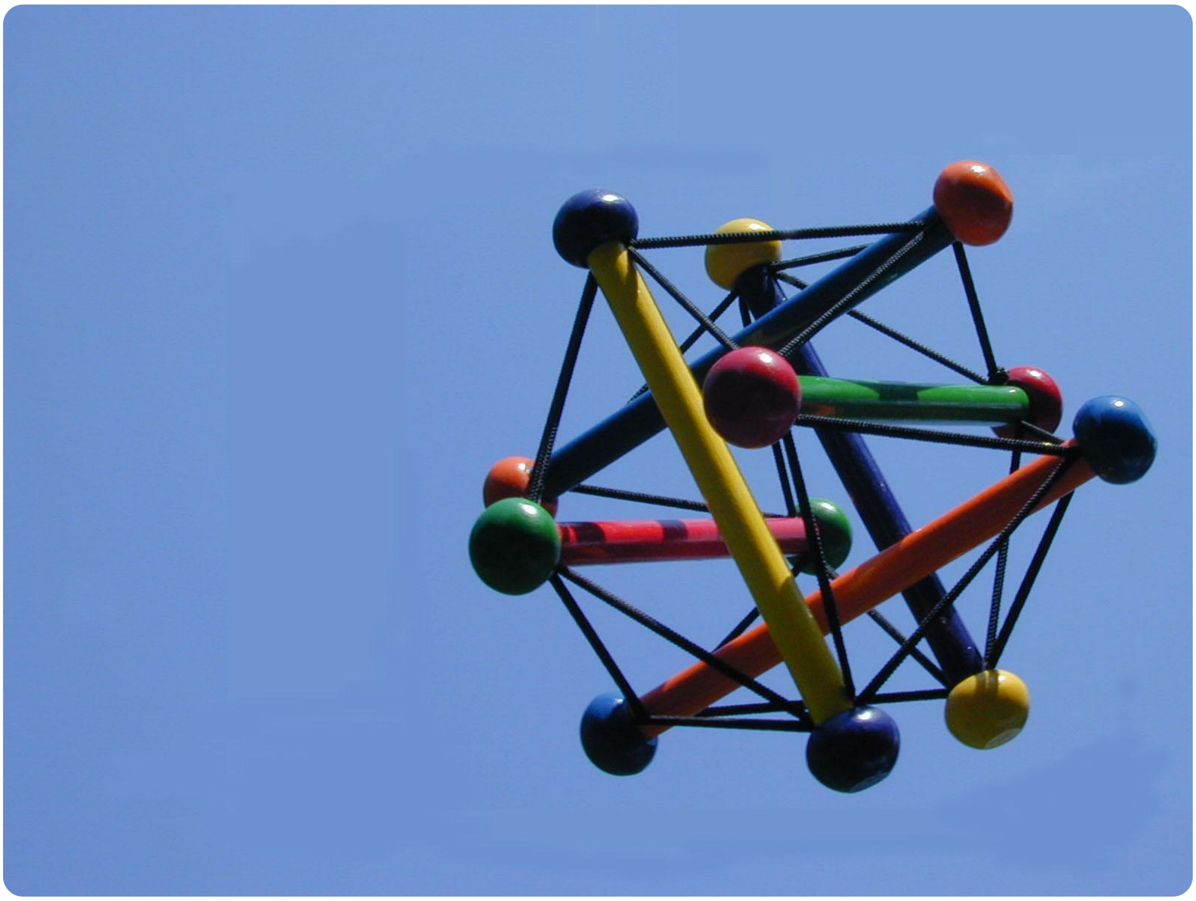
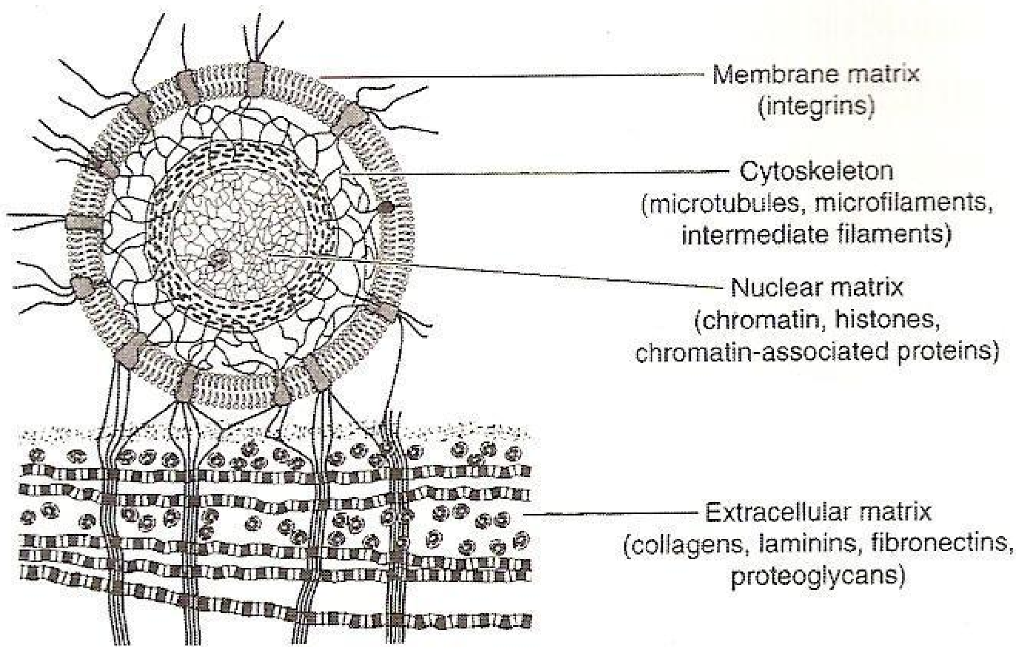


FIG. — *Tensegrity*



4.14 Tissue matrix system as described by Pienta and Coffey, 1991. (Reproduced from Oschman JL (2000). *Energy Medicine.* New York: Churchill Livingstone. Fig. 4.6, p. 66. Reprinted with permission from Elsevier Inc. and Medical Hypotheses.)

FIG. — Tissue matrix system

THE ECTODERM

5. The Cell: Metabolic Fields and Differentiation

DURATION : 11:37

PEDAGOGICAL SHEET

COURSE SUMMARY

This session on biodynamic embryology teaches the fundamentals of cellular development, explaining the trajectory from stem cell to differentiation. With essential concepts such as cleavage, polarizing substances, and metabolic fields, students will understand how cells communicate and interact. Emphasis is placed on the importance of polarization in oocyte formation, as well as on the types of cellular communication, illustrating concrete examples to enrich the understanding of embryogenesis.

PLANCHES ANATOMIQUES & SCHÉMAS (2)

FIG. 1

FIG. 1 FIG. 1

FIG. 2

FIG. 2 FIG. 2

THE ECTODERM

6. The 3 Embryological Tissues

DURATION : 47:51

PEDAGOGICAL SHEET

COURSE SUMMARY

This session explores the three embryological tissues: the ectoderm, endoderm, and mesoderm, linking them to biodynamic concepts. By introducing Blechschmidt's classification, the teaching highlights the relationship between boundary and interior tissues, illustrated by concrete examples such as the eardrum. The importance of connective tissues, their nutritive role for epithelial tissues, and the dynamics of congestion and desiccation are discussed. Understanding these elements will enable you to integrate a holistic approach into your therapeutic practice.

PLANCHES ANATOMIQUES & SCHÉMAS (2)

FIG. 1

FIG. 1 FIG. 1

FIG. 2

FIG. 2 FIG. 2

07. CHRONOLOGY, GROWTH, SYNCHRONICITIES

DURATION : 03:59

STUDY SHEET

COURSE SUMMARY

In this video, we explore the essential concepts of Biodynamic Embryology, focusing on the timeline of embryonic development, from the earliest stages of conception to the major transformations of the second month. We distinguish between motility, intrinsic to the tissue, and mobility, linked to respiration, highlighting the importance of treating the environment rather than just the organs. Notions of potency within the tissue and the importance of the diaphragm are also discussed, offering therapists tools to work effectively with embryonic movement and growth.

BIODYNAMIC EMBRYOLOGY: CHRONOLOGY, GROWTH, SYNCHRONICITY

DEFINING THE EMBRYO: DAYS, CARNEGIE, AND GROWTH

To understand the **embryo**, we can define it in terms of **days** or according to the **Carnegie stages**. In this course, we will focus on days.

When **synchronised** phenomena occur, such as the closure of the **neuropores** or the integration of the **coeloms**, we will give you precise points in time, for example, at the moment of the **heart's** awareness. Otherwise, we will speak in days or weeks, and we will also address notions of **size**.

What is truly important for you is the notion of **growth**. You must understand that the embryo is a **developmental movement**. It is a movement that develops from the inside out, and not the other way around.

MOTILITY AND MOBILITY: TWO DISTINCT MOVEMENTS

This is the fundamental difference between what we call **motility** and **mobility**:

- **Motility** is an embryonic response. It follows the movement of the intrinsic development of the tissue.
- **Mobility** is a movement purely related to respiration and motor function.

It is crucial to make this distinction clear. When I move my arm, it is very different from the intrinsic movement at the level of my **liver**.

If I want to rework a tissue, I bring it back into its motility. This is where its **functional force** lies, where it will reorganise itself. This is the work we do: to return to this tissue so that it can reorganise and regain its mobility.

It is always about following the movement, but for that, you need **points of support** and to have the movement in your hand.

TREATING THE ENVIRONMENT, NOT JUST THE ORGAN

I do not treat an **organ**, I treat an **environment**. However, my action can sometimes influence mobility and motor function.

In the context of mobility, the master is the **diaphragm**. That is why we will learn how this diaphragm is built.

THE POWER OF TISSUE: A REVELATION

One day, my teacher told me: "Marc, one day you will feel the **power** in the tissue." For me, this power was very intellectual. I could understand that a developing embryo represents incredible power, a movement that never stops.

He also told me that sometimes, in **"stillness"** (moments of immobility), in certain points of support, this power is also present. It is incredible, and at that moment, something happens. However, you cannot create this all the time. You have to try to be present, to have a good connection, and at some point, the system adjusts.

He gave me an image: "It's as if you were in a small boat in the sea. You are just in the water, and you don't see it, but beneath you, a **whale** is coming. And you feel, you feel all of a sudden, it's a bit like that sensation."

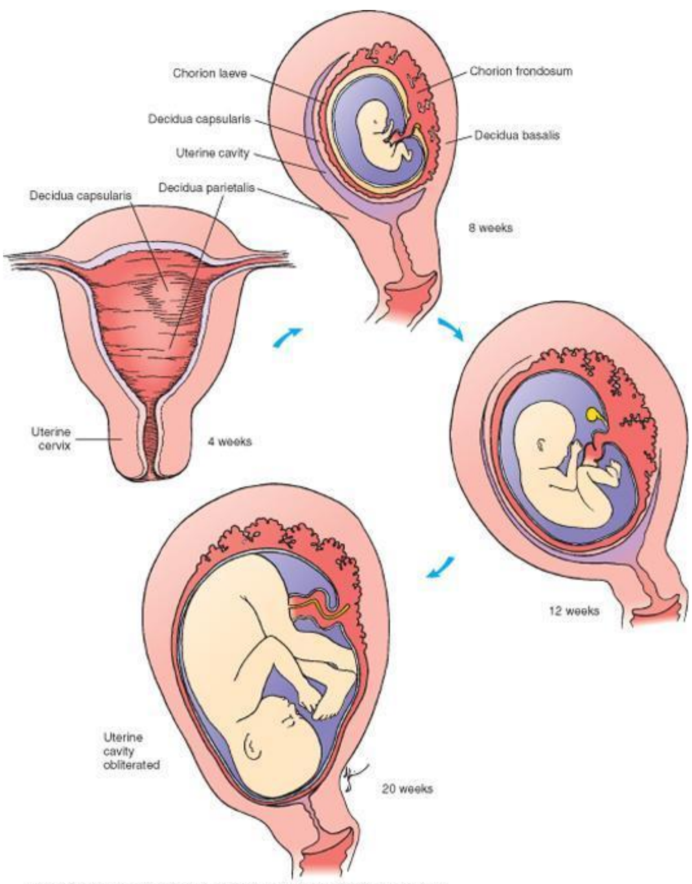
FROM OVUM TO SECOND MONTH: AN INCREDIBLE TRANSFORMATION

Look at what we are on the first day: a small **mess**. And look at what will happen at the end of the **second month**. In just two months, an incredible transformation takes place.

From the beginning of the **twentieth day**, a form is already beginning to take shape. The potential for life is there.

Initially, it is simply an **ovum** that has received the genetic heritage from the father, the **sperm**. The two heritages are meeting. This is roughly the first image of your life.

We then observe what is called the **zona pellucida** and **nurse cells**. There are several important layers of cells that are being put in place.



Schoenwolf et al: Larsen's Human Embryology, 4th Edition.

FIG. — Fetal uterine development

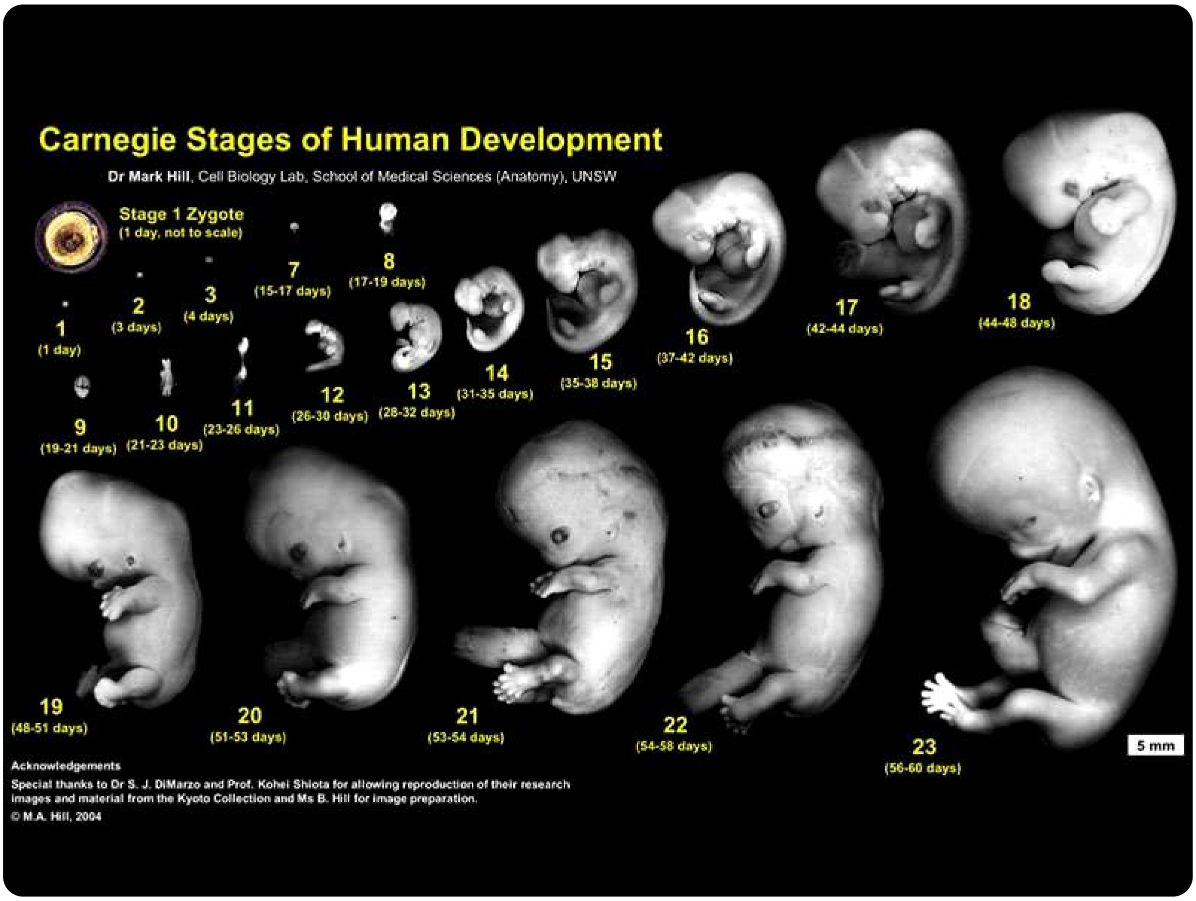


FIG. — Human embryonic development

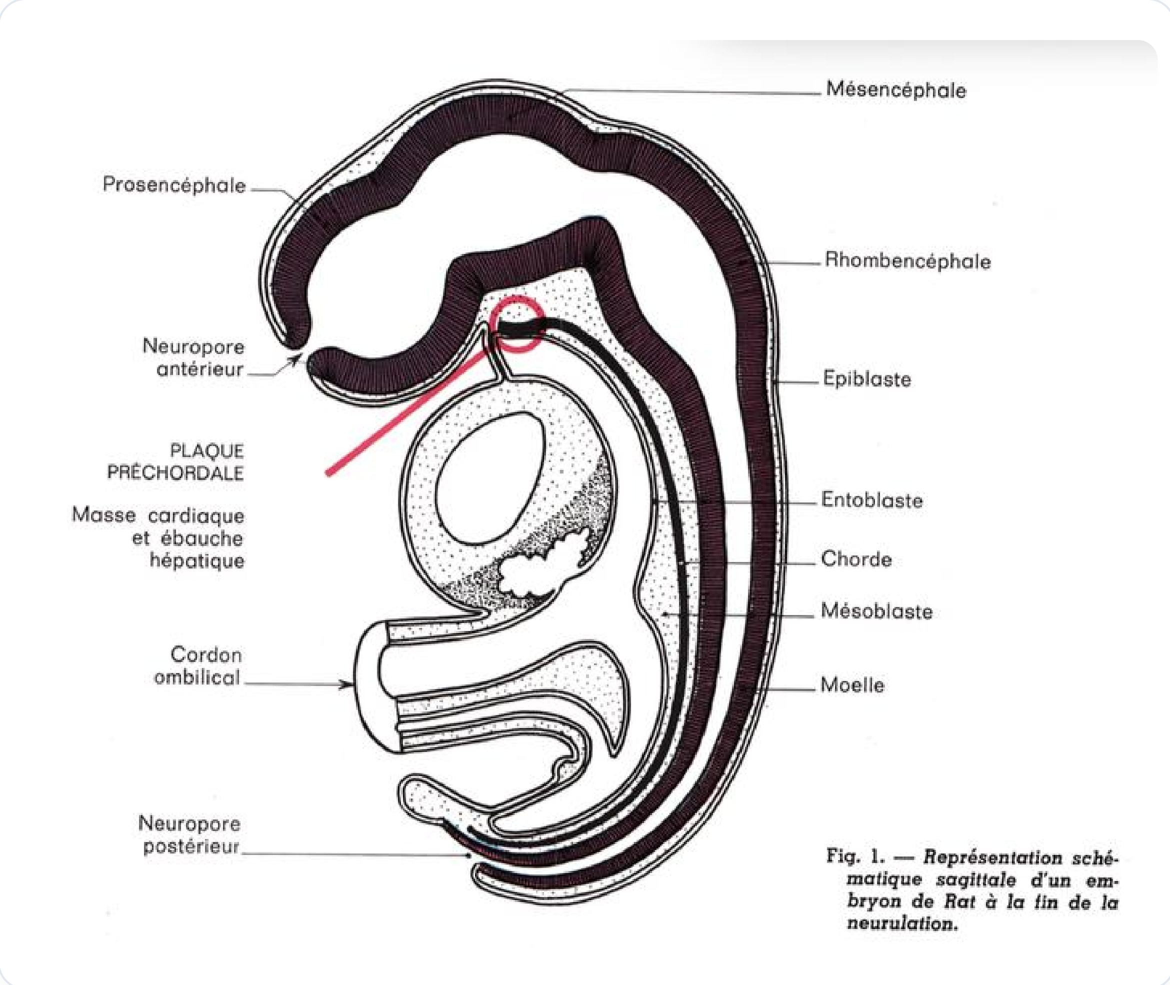


FIG. — Embryo sagittal neurulation

FIG. — Embryo, 6-7 weeks

08. PRACTICE: POTENCY IN MOTION

DURATION : 05:51

STUDY SHEET

COURSE SUMMARY

In this video, we explore the fascinating relationship between cellular polarity and embryology, starting with the notion of polarity at the cellular level, namely the necessity of an eccentric nucleus to promote metabolic exchange. We then delve into the process of oocyte polarisation and the crucial interactions with the spermatozoon, illustrating how this encounter initiates a systemic reorganisation. By addressing the body's electrical fields, particularly that of the spine, we learn to anchor our practice by connecting to our own bodily experience. The session concludes with advice on supporting the health process, reiterating the importance of remaining focused on health rather than pathology.

PRACTICE: POTENCY IN MOTION

CELLULAR POLARITY

When a cell has its nucleus centrally located, it is not **polarised**. For it to exchange with its environment, an **asymmetry** is necessary.

This possibility of exchange applies to the entire cellular environment. It is crucial to remember that in all the cells of our body, the nucleus is always **eccentric**.

This eccentricity is an image of polarity. Indeed, a cell with an eccentric nucleus exhibits increased **metabolic activity** at that location. This increases metabolic force, which is the very expression of polarity.

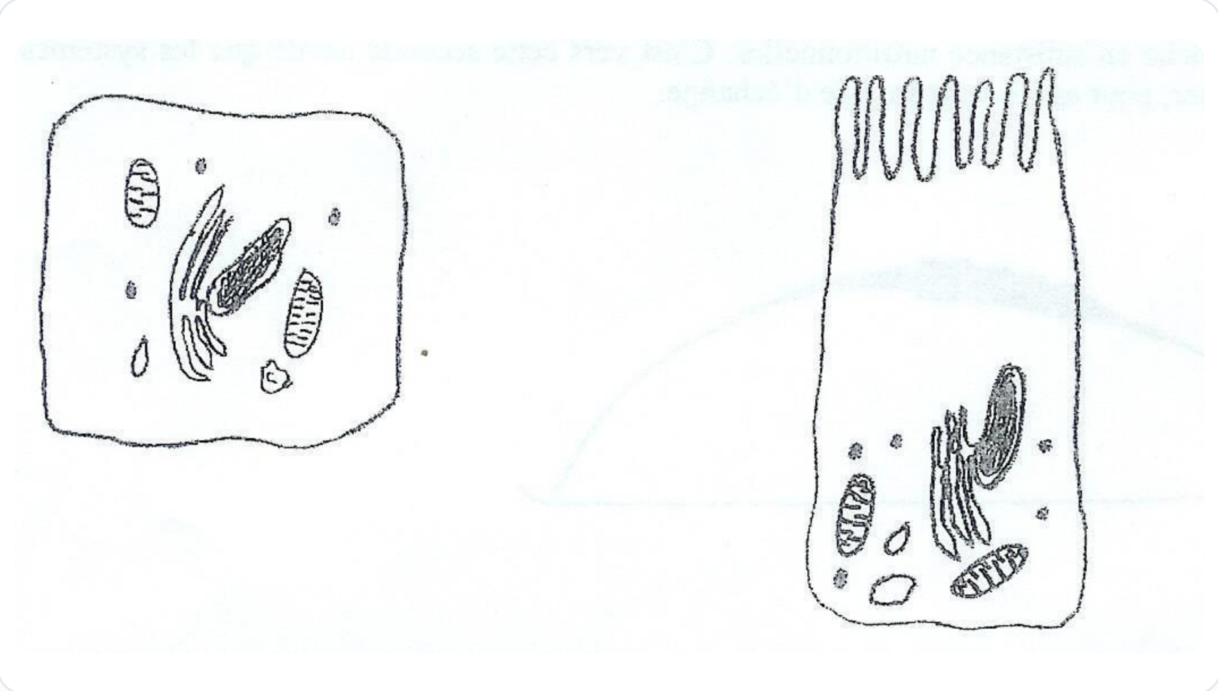


FIG. — Enterocyte microvilli

OOCYTE POLARISATION

At the **oocyte** stage, it is polarised by small surrounding cells that release substances. These substances create what is known as the **Wolpert's flag**, leading to molecular and atomic concentrations within the system, which polarises it. There is therefore a force, a **plus** and a **minus**.

THE ENCOUNTER WITH THE SPERM

When the sperm arrives, it touches the oocyte membrane. It releases a hormone called **ZP3**, a specific substance to be aware of.

The polarity inside the embryo functions like a **compass** with a north and a south, but this orientation is relative to space and not to the external environment.

Upon the arrival of the sperm, a global **reorganisation** of the system occurs. The electrical field reorganises, and a membrane information is received, leading to **cortical degraulation** to allow the passage of a single sperm.

This encounter between the two heritages is a true **miracle**. The oocyte then becomes polarised, and the fusion of the two heritages forms a nucleus, a field. The nucleus positions itself on the electrical axis of the system, its eccentricity being its response to the whole.

ELECTRICAL FIELDS OF THE BODY

The largest electrical field in our body is located along the **spinal column**. It is a true potency.

From the brain, a powerful electrical field unfolds. There is also an electrical field at the terrestrial level and another at the heart level. However, the most important axis remains that of the spinal column, a crucial place of neurological exchange right down to the fingertips, with a particular concentration around this area.

PRACTICE: BECOMING A CELL

We will now practice by imagining that we are a cell.

Take your place on the table, as a cell. Do not worry about how you touch others, the important thing is to touch gently.

Visualise that we start from a point, in the void, to be attached by a **net** in the universe. Everything develops and returns around us. We start from a **pedicle** towards a cord, everything integrates here. The navel stays in its place, sometimes it is us who are not in the right place.

We do not decide our position, we simply return to this connection with what is called "the anterior heaven" and "the posterior heaven".

By placing ourselves like this, we will observe how the body organises itself.

ACCOMPANYING THE PROCESS

It is essential to learn to recognise this process in practice. At some point, a process may set in, and it must be accompanied.

It is **health** that begins to work, and we must accompany it. We tend to be hypnotised by the lesion, drawn to it.

In the work we are going to do with embryology and biodynamics, it is important to say: "I saw you, but I am not going with you". We recognise the lesion, but we remain focused on health.

We remain in the dynamic of health, on the health **fulcrum**.

09. GENERALITIES CONTINUED

DURATION : 05:52 STUDY SHEET

COURSE SUMMARY

This video covers the fundamental aspects of embryonic development, including the concept of the 'formative field' and the importance of energy without matter. Mechanisms such as mechanotransduction, polarity, and various environmental factors like heat and light are explored. You will learn how cell location influences their differentiation, as well as the differences between embryonic tissues, including ectodermal, endodermal, and mesodermal derivatives. The video concludes with an overview of the phases of embryogenesis, which are profoundly essential for any therapeutic practice.

GENERALITIES ON EMBRYONIC DEVELOPMENT

The **potential for embryonic development** is comparable to a **"formative field"**, an expression that describes a hypothetical field containing **energy without matter**. This concept is complex and includes elements such as nuclear impact, **tensegrity**, and the importance of the **matrix**.

We have examined mechanisms that go beyond a simple electromagnetic field. Factors such as **temperature**, **heat**, **mechanical stresses**, **hydrostatic pressures**, **electric fields**, and the **oxidation-reduction system** all play a crucial role in the system's **homeostasis**.

The presence of **elementary particles**, **electromagnetic energy**, the polarised human, the electromagnetic field, **Maxwell's law**, as well as the notion of **space** and **time** are also essential. **Light** is another important factor to consider.

CELLULAR DIFFERENTIATION

How do a multitude of cells differentiate? We have observed that even minimal **changes** are linked to **mechanotransduction**.

A significant input of light modifies the parameters. When an electric field is present, there is always an electromagnetic field, which ensures the system's **cohesion** and provides a **reference**. Thus, a cell at a given location does not have the same spatial reference as another cell.

A cell's **genetic heritage** expresses itself differently at different times, even if all cells initially possess the same heritage. Cell layers will express themselves in different spaces and times, according to the **chronology** and **location** relative to the midline and the electric field.

The electromagnetic field provides the **information**. It is the position relative to this field that releases the information. Other parameters, such as **acidity**, are also involved, but **polarity** is the first major and very important component.

ORIGIN AND LOCALISATION

Where do **polarity**, **age**, and these forces come from? They existed long before us and will continue to exist after us.

The **energy of vibratory matter**, concentrated like light, induces matter, fluid, electricity, and **densification**, before dissipating. What makes one cell **hepatic** and another **cerebral** is its **localisation**. It is this localisation in space and time that determines differentiation.

Sometimes, cell clusters can be displaced and redifferentiate, but this depends on the timing of development.

TISSUES AND EMBRYONIC DERIVATIVES

You now have a good understanding of the difference between an **epithelial tissue** and a **connective tissue**. You can also distinguish between **ectodermal**, **endodermal**, and **mesodermal** tissue.

Let's take the **tongue**. It contains **muscle** (mesoderm) and is lined with epithelial tissue. It is a mixture of **endodermal** (epithelium) and **mesodermal** (muscle) tissues.

The **pancreas** is an endodermal derivative. All hormonal structures derive from epithelial tissue, also called **eminence tissue**. In other words, all **glands**, whether **exocrine** or **endocrine**, always originate from epithelial tissue, deriving from the ectoderm or endoderm, never from the mesoderm.

This is an important point to remember, now that we have clarified the origins. The **neurocranium**, **viscerocranium**, and **cranial base** represent a balance between the neurocranium and viscerocranium.

PHASES OF EMBRYONIC DEVELOPMENT

Tissue derivatives form from the first three weeks. By the beginning of the third week, the process is well underway. We can divide development into three main periods:

- The free period of **embryogenesis**.
- The free period of **organogenesis**.
- The free period of **periocephaly** (in two months).

Within two to three months, all **noble structures** are already formed. After that, it is mainly about **maturation** and **growth**.

It is essential to emphasise the notion of **growth** and the fact that **motility** is a movement always present in the body, defined by the developmental system. This movement becomes more complex in different situations.

Growth and the **autogenetic range** are fundamental processes.

10. GAMETOGENESIS

DURATION : 02:38 STUDY SHEET

COURSE SUMMARY

This session explores gametogenesis, detailing the crucial differences between the oocyte and the spermatozoon, both cellularly and functionally. Students discover oocyte production, their origin during foetal life, as well as the importance of nurse and follicular cells in embryonic development. The impact of temperature on fertility is also discussed, highlighting the need for warmth for the ovaries and coolness for the testes, as well as the implications for fertility problems in women. Finally, the anatomical links between organs and the arterial system are presented, highlighting the axes of trophic communication and the role of osteopathy in fertility management.

GAMETOGENESIS

COMPARISON OF OOCYTE AND SPERMATOZOON

The **cell volume** of the oocyte is very large, in contrast to that of the spermatozoon which is very small.

We can say that the ovum is in **dilation**, in **opening**, completely in **extension** and open.

Whereas the spermatozoon is in **internal rotation**, **desiccated** and **mummified**.

The **DNA** concentration is very high for spermatozoa.

The **motility** of the oocyte is absent, while the spermatozoon is **active**.

CELL PRODUCTION

The number of oocytes produced is approximately **400**. Primary oocytes are formed during the **foetal period**.

This means that, when you were in your mother's womb, you were already making your own oocytes.

NURSE AND FOLLICULAR CELLS

We distinguish between **nurse cells** and **follicular cells**.

One cell is of the **germline**, the other is of the **somatic line**. One is transmitted through development and the other is given by the mother.

These cells will create **axes**, representing a path for the transmission of **genetic** information, of an **epigenetic** type and of **transgenerational formation**.

TEMPERATURE AND FERTILITY

The **ovaries** need to be inside the body because they require **heat**.

In contrast, the **testicles** have descended outside because they need a cooler **temperature**.

IMPACT ON FEMALE FERTILITY

This is an important point to consider when women consult for **fertility** problems.

Sometimes, when working at the level of the **sacrum**, one can feel a **coldness**, which may indicate a problem.

ANATOMICAL AND PHYSIOLOGICAL LINKS

We will discuss this in relation to the **duodenum**, the **arteries** and the "plicas" or **folds** at the level of the duodenum.

These structures allow the passage of **trophic** information via the arterial system, particularly the **ovarian** or **testicular arteries**.

Fertility often needs to be treated higher up, on the framework of the **facial pentagon** of the **pancreas**.

11. FERTILISATION

DURATION : 14:26 STUDY SHEET

COURSE SUMMARY

This session deeply explores the fertilisation process, from ovulation to the reception of the sperm by the oocyte. The concepts of ovarian mobility, natural barriers to fertilisation, and membrane recognition through ZP3 are discussed to understand the complex dynamics of reproduction. The session also highlights the implications of in vitro fertilisation techniques, emphasising essential emotional and energetic links between mother and child, as well as the importance of knowing one's personal history for holistic health.

FERTILISATION

OVULATION AND OVARIAN MOBILITY

Ovulation is a crucial process, and it is essential to understand **ovarian mobility**. The fimbriae of the ovary play a key role in this mechanism.

The anatomy of the lesser pelvis, uterus, and ovaries is complex. Structures such as the **broad ligaments**, **round ligaments**, and **lumbo-ovarian ligaments** are found there, all composed of the same tissue.

The embryology and anatomy of the uterus reveal that it is a fold of the **posterior parietal peritoneum**. This structure must be mobile to capture the **oocyte**. A hemispherical movement, linked to motility, is necessary for this capture.

If we consider the digestive system, **Tol's fascia** inserts into the posterior parietal peritoneum. In cases of **appendicitis**, for example, a loss of energy or fascial disturbance can affect ovarian mobility and prevent oocyte capture.

The peritoneum must be free to allow fertilisation. If the oocyte is not captured, it can remain in the pelvic area, leading to dangerous **ectopic pregnancies**. There is a direct link between the vagina and the peritoneal cavity, which highlights the importance of this mobility.

NATURAL BARRIERS TO FERTILISATION

In males, fertilisation is complicated by the absence of connection between the reproductive structures, due to the **Müllerian tubercle** and the **Wolffian duct**.

In females, several natural barriers exist. The first is **vaginal acidity**, which helps destroy unwanted bacteria. Then, a **mucus plug** at the cervix plays a crucial role. This plug opens once a month, allowing sperm to pass. If it does not open, passage is blocked.

Once in the uterus, sperm must navigate through **cilia** that help them move forward. They are attracted by heat, sugar, and electrical signals. However, it is not a single sperm that succeeds, but a group that manages to reach the oocyte.

MEMBRANE RECOGNITION: ZP3

When the sperm reaches the oocyte, it must go to the most optimal area, where the **oocyte** is ready to receive the sperm. On the oocyte membrane are **glycoproteins** called **ZP3**, which act as a recognition key.

This membrane recognition is essential: if it fails, fertilisation cannot occur.

IN VITRO FERTILISATION AND ITS IMPLICATIONS

In vitro fertilisation (IVF) is a common method, particularly **ICSI** (Intracytoplasmic Sperm Injection), where sperm is directly injected into the oocyte. This technique bypasses natural recognition, which can lead to long-term fertility problems.

Children born from IVF may present complications. During consultations, it is important to ask questions about conception, especially about any miscarriages. Uteri can be charged with residual energy related to these events, requiring energetic cleansing.

It is crucial to recognise the child, establishing a bond between mother and child. Mothers must establish physical and emotional contact with their child to foster this connection.

THE IMPORTANCE OF KNOWING ONE'S HISTORY

Knowing one's history is fundamental. A child born from sperm donation, for example, may carry a lie within them, unaware of their origins. Tissues do not lie, and an osteopath can feel this truth through their hands.

THE ROLE OF OOCYTE POLARISATION

The **oocyte** is polarised. When the sperm arrives, it causes a reorganisation of the oocyte's axes, creating dorsal-ventral and apico-caudal axes. This symbolic moment is a connection with the primitive heart.

ATTACK ON THE OOCYTE MEMBRANE

Sperm must pass through several layers to reach the **zona pellucida**. Once on the membrane, membrane recognition must occur. The sperm that successfully binds releases an acrosomal acid, allowing its entry into the oocyte.

THE FIRST CELL OF LIFE

The meeting of the genetic heritage of the two cells forms the first cell of life. This dynamic process gives rise to the first two **blastomeres**, which are the first germ-mother cells.

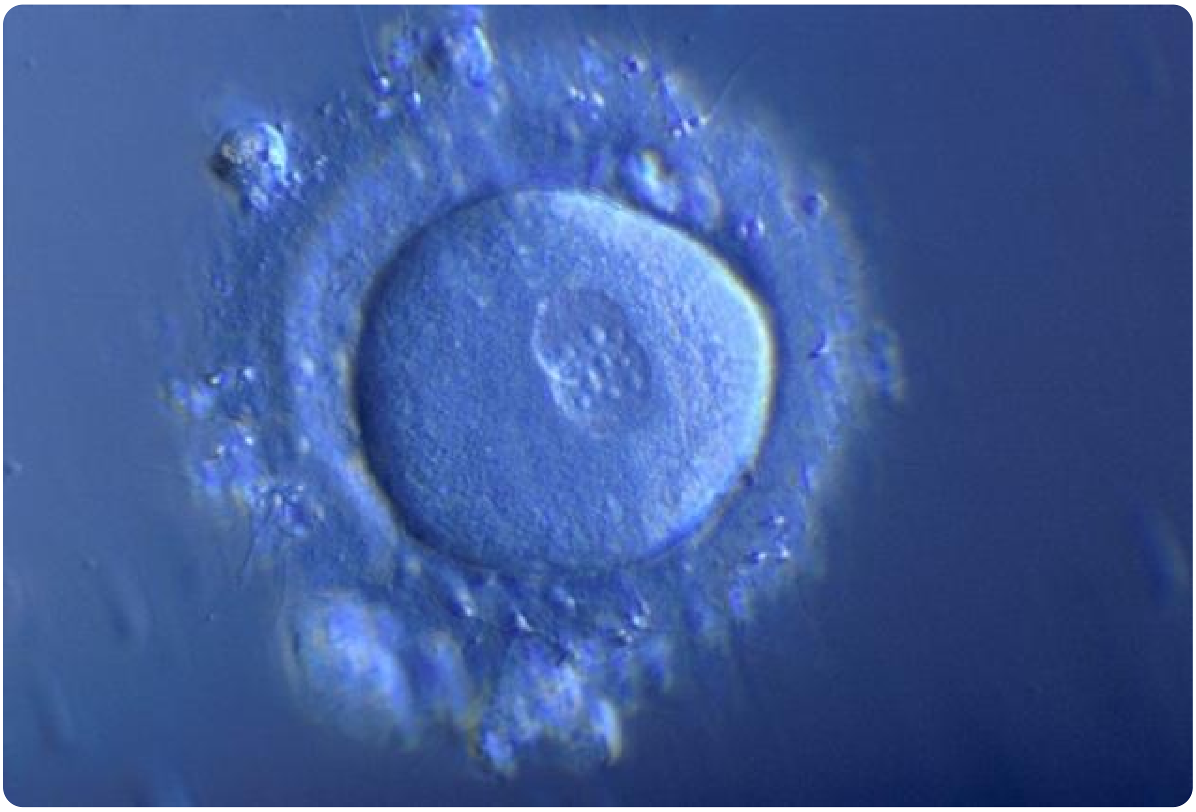


FIG. — Fertilized egg

This research on polarity and the first phases of embryonic development is essential for understanding the foundations of life.

12. ALGORITHM OF DEVELOPMENT

DURATION : 10:38 STUDY SHEET

COURSE SUMMARY

This session explores the fundamental algorithm of embryonic development, detailing the transition from oocyte to zygote and the crucial early stages of cell division. The concepts of polarity, asymmetric cell divisions, and increased metabolic potential are discussed, illustrating how a single cell begins to multiply and organise itself. The session also highlights the hatching phase, where accumulated energy reaches a critical threshold, resulting in embryo migration. A fascinating overview of the basic mechanisms underlying embryonic growth, essential for therapeutic practitioners.

12. DEVELOPMENTAL ALGORITHM: FROM OOCYTE TO BLASTOCOEL

For the first three to four days of development, the **oocyte** we observe is no longer an oocyte, but is now called a **zygote**.

It is crucial to understand a **fundamental algorithm** that will constantly repeat throughout embryonic development. This process relies on the concepts of **growth** and **developmental movement** within this growth.

POLARITY AND ASYMMETRICAL CELL DIVISIONS

Here we have a clear image of **polarity**. The first two cells of the zygote will divide extremely rapidly and increasingly to form a cluster of small cells.

These first two living cells respond to a polarity, which means they will not initially divide at the same rate. This difference in division speed is a manifestation of polarity.

With each cell division, the **membrane** increases in surface area, and there is a slight loss of fluid outwards, a small **exudate**. This contributes to the formation of a mass of cells, with the appearance of a first small **fluid-filled cavity** within it.

Initially, during the first few days, the whole multiplies but does not grow in size. Due to cell multiplication and the small loss of water, there is an overall increase in the membrane and cells within a confined space.

This implies an increase in overall **metabolism** and, consequently, an increase in the developing **potential energy**.

Let's recap for a better understanding: initially, we have the first two cells in a confined space, enveloped by a membrane called the **zona pellucida**.

These two cells each have a **nucleus**, never centred, always off-centre. Given cell A and cell B in a plane, depending on the polarity, one cell, closer to a metabolic field, will divide slightly faster than the other.

- **Difference in division speed:** The cell closest to a metabolic field divides faster.
- **Increase in exchange surface:** Each cell division or cleavage leads to an increase in membrane length, thus increasing the exchange surface.
- **Exudate:** A little fluid escapes to the side, forming an exudate.

It is important to note that this division is never symmetrical in the body, which demonstrates that there is no **perfect symmetry**, but rather **harmony**.

THE INCREASE IN METABOLIC POTENTIAL

Initially, we have two cells and two nuclei, offering a certain exchange capacity. Then, there are three cells and three nuclei, which means a greater metabolic exchange force.

The **energetic** and metabolic potential is more or less strong depending on the position in the cell, with more powerful areas. Each division represents a significant transformation and opening.

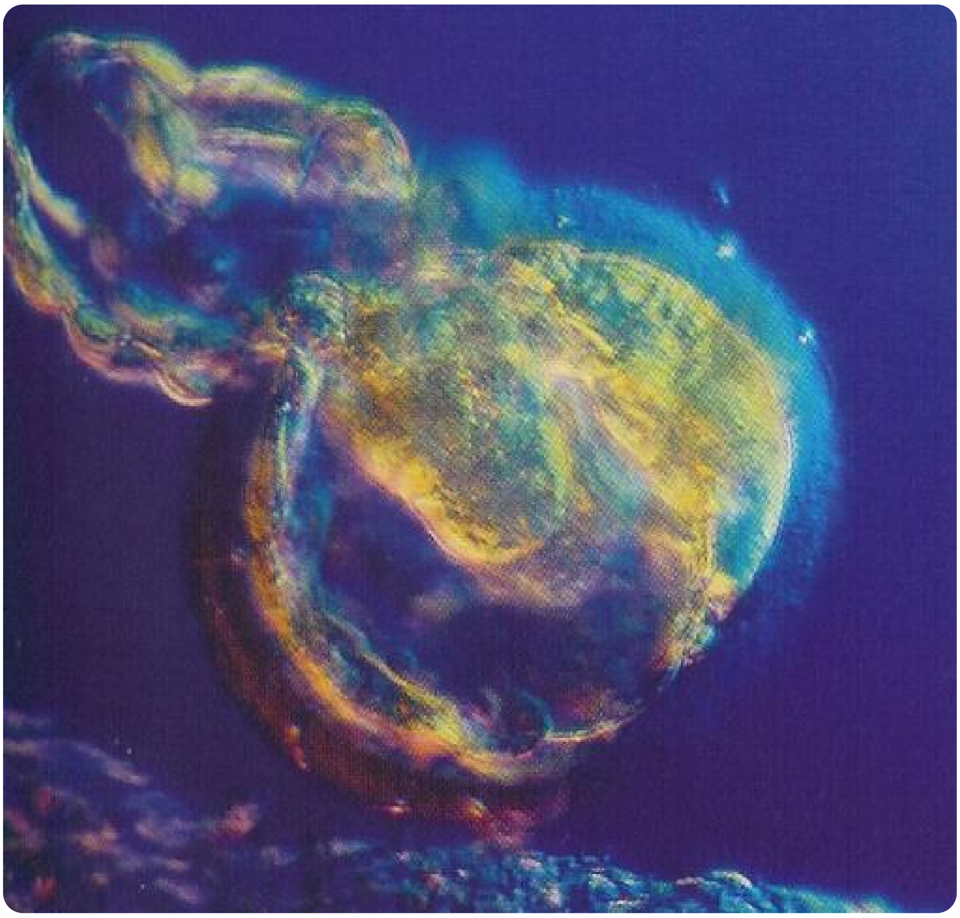


FIG. — Segmenting blastocyst

THE CONCENTRATION PHASE AND HATCHING

During the first three to four days, the whole divides but does not grow, remaining in the same space. It is as if an internal power is concentrating, creating a mass of **metabolic energy** ready to "explode" and seek new trophic forces.

So we have a mass energy, a potential energy that builds up, a very intense metabolic power that concentrates for three days.

This concentration reaches a sufficient level around the 3rd-4th day to cause a phenomenon of **hatching**. The cell, globally enclosed in a tissue, undergoes a rupture due to the accumulated force, allowing the embryo to migrate.

Just before hatching, within this layer, each cell division causes:

- An overall increase in membrane size.
- An increase in the number of cells, and therefore in metabolism, force, and power.
- An exudate.

The very first cavity that appears, filled with water, joins and forms a small cavity.

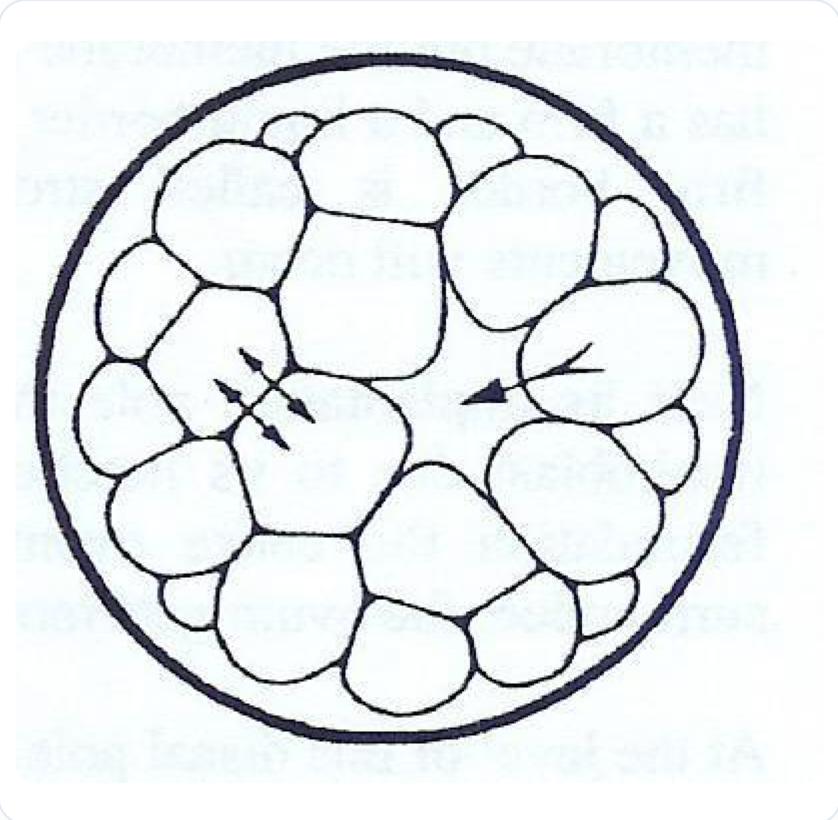


FIG. — Blastocyst cavitation

THE BLASTOCOEL: FIRST FLUID CAVITY

This small cavity is called the **blastocoel**. The term "blastocoel" literally means "germ of the sky", and this cavity will be of great importance for the future **peritoneal cavity**.

The phenomenon of hatching results in a multitude of cells and a cavity. This cavity is always off-centre, the expression of a global polarity.

The eccentricity of this cavity leads to a concentration of cells at one pole, which is called the **animal pole** (as opposed to the vegetal pole), or assimilation pole. At this location, there is a high concentration of cells, less elsewhere.

Between this phase and the next, hatching occurs. The peripheral membrane can no longer contain the whole. A rupture occurs, and the embryo begins to leave this space.

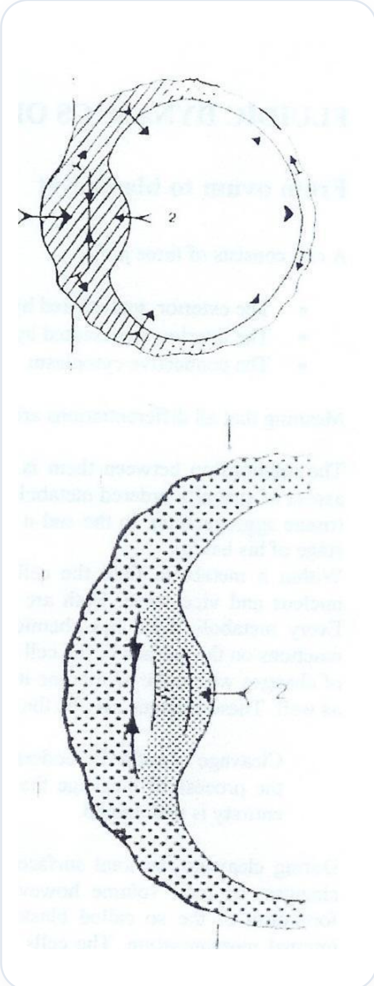


FIG. — Embryonic pinching by contraction

The passage of this first cavity containing the exudate, forming the blastocoel, will grow. The difference from the previous phase is that now, the embryo is much larger because it has emerged from the membrane.

It is crucial to remember that during the first three days, the embryo does not grow. But inside, power accumulates, until the surrounding tissue ruptures and hatching occurs.

The embryo leaves its "egg" and finds itself in the external environment. In this external environment, it very quickly organises itself into a concentrated area of cells and a cavity.

It is still the same image of the blastocoel, but this time, it will develop what is called an **embryonic pole**, with a greater concentration of cells on one side than on the other. This pole is an assimilation pole, seeking to "take in the inside".

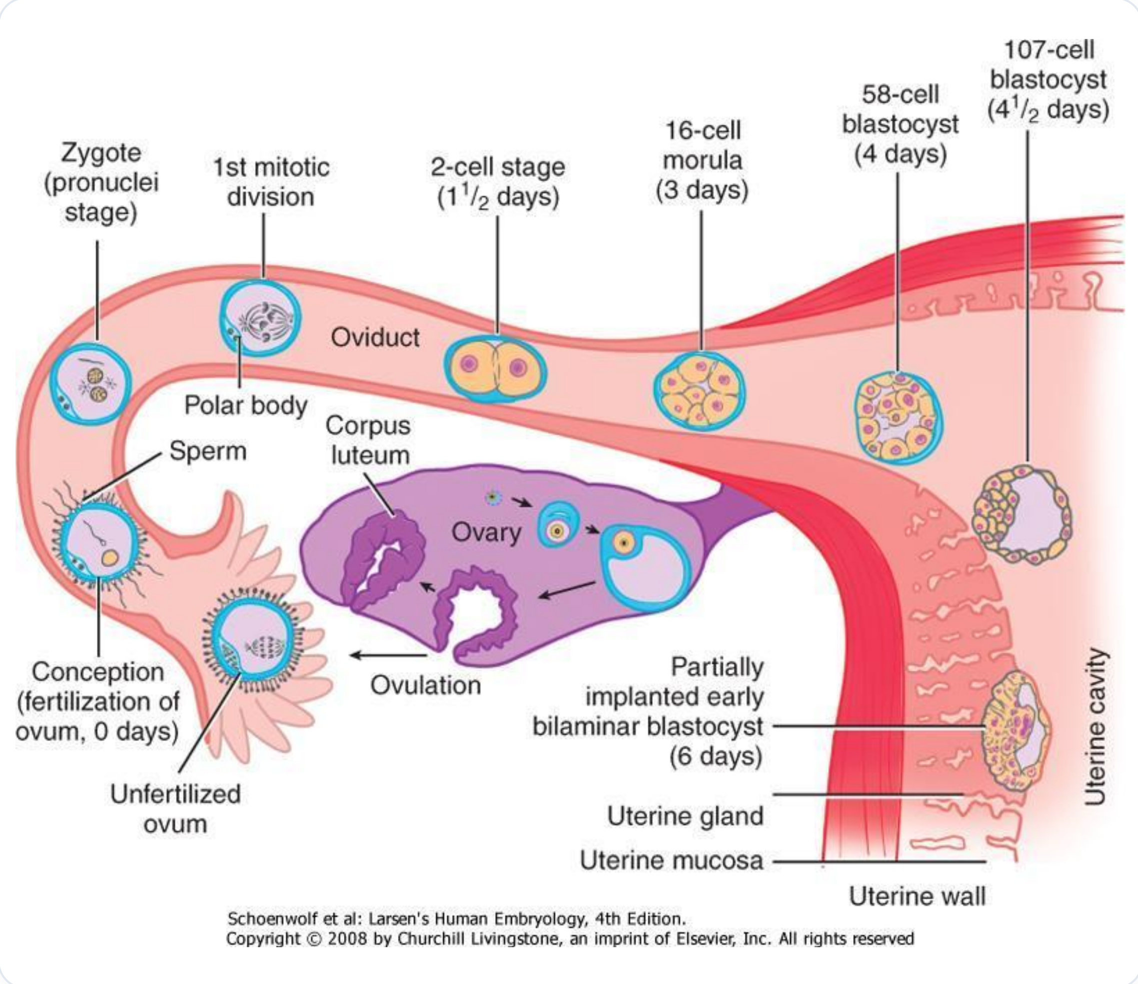


FIG. — Early embryonic development

FIG. — 2-cell embryo

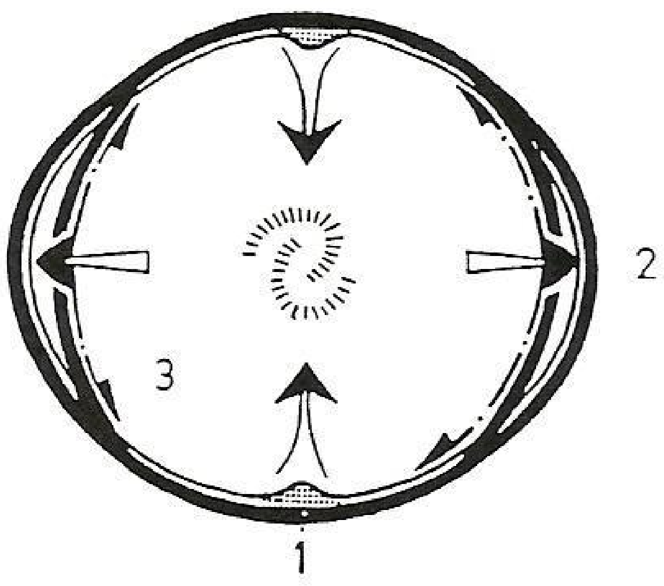


FIG. — *Deformed embryonic shell*

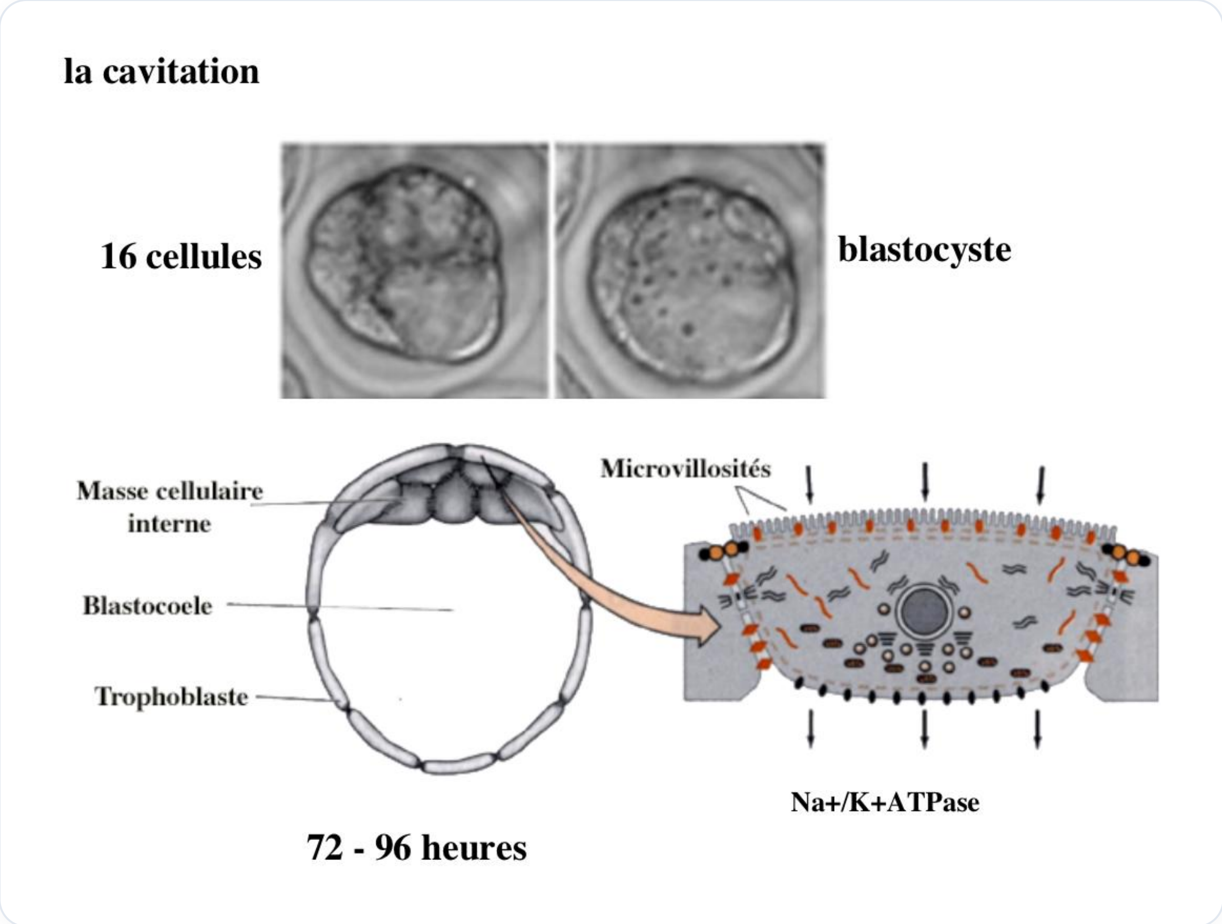
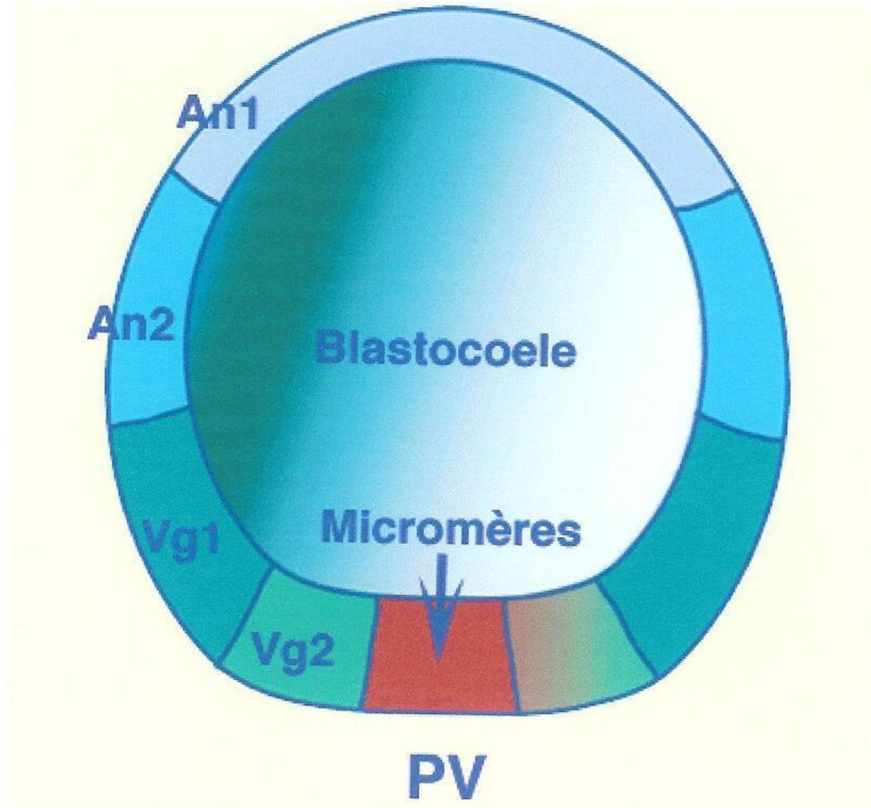


FIG. — Blastocyst cavitation NaKATPase



An1: Animal 1; An2: Animal 2; Vg1: Végétatif 1; Vg2 Végétatif 2; PA: Pôle Animal; PV: Pôle Végétatif.

FIG. — Blastula Micromeres PV

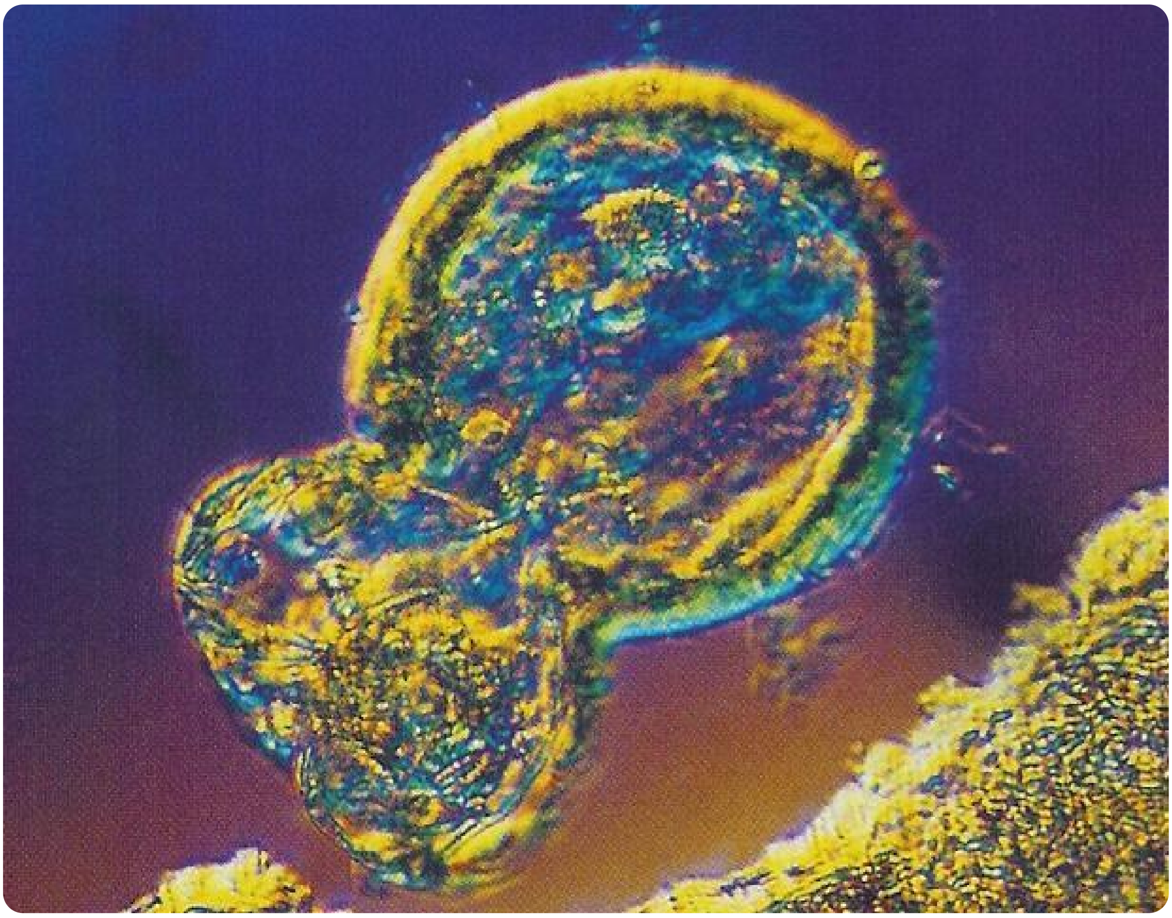


FIG. — Gastropod embryo

13. BLASTOCYST MIGRATION

DURATION : 01:12

STUDY SHEET

COURSE SUMMARY

In this video, we explore the migration of the blastocyst through the fallopian tube to its implantation in the uterus between day 4 and day 6. The process involves a search for maternal warmth, which guides the blastocyst towards optimal areas for implantation, usually the anterior part of the uterus. The uterine lining becomes welcoming, swelling and preparing to provide essential nutrients such as warmth, sugar, and oxygen, which are indispensable for the zygote in this critical phase of its development.

BLASTOCYST MIGRATION

I am now moving through the **fallopian tube**. After travelling this path, I gradually arrive, around **day 4-5**, then towards **day 6**, in the uterus.

During this time, I have grown. I still have my **nurse cells**, but now, I am in a new phase.

What do I need at this stage? I need my **mother's warmth**. That is why I will go and implant myself in a warm area. Implantation generally occurs more in the **anterior part of the uterus**, close to the **bladder**, because this area is warmer.

I will seek out the **uterine lining**. What has it done? It has become super welcoming. It has swollen, created many **glands**, and filled with **blood** to ensure it is very warm.

The **zygote** arrives in the uterus. It is there, and I, the zygote, what do I need? I need warmth, **sugar**, **electrolytic activity**, and a start of **oxygen** for future development.

We will discuss the moment of **implantation** after the break.

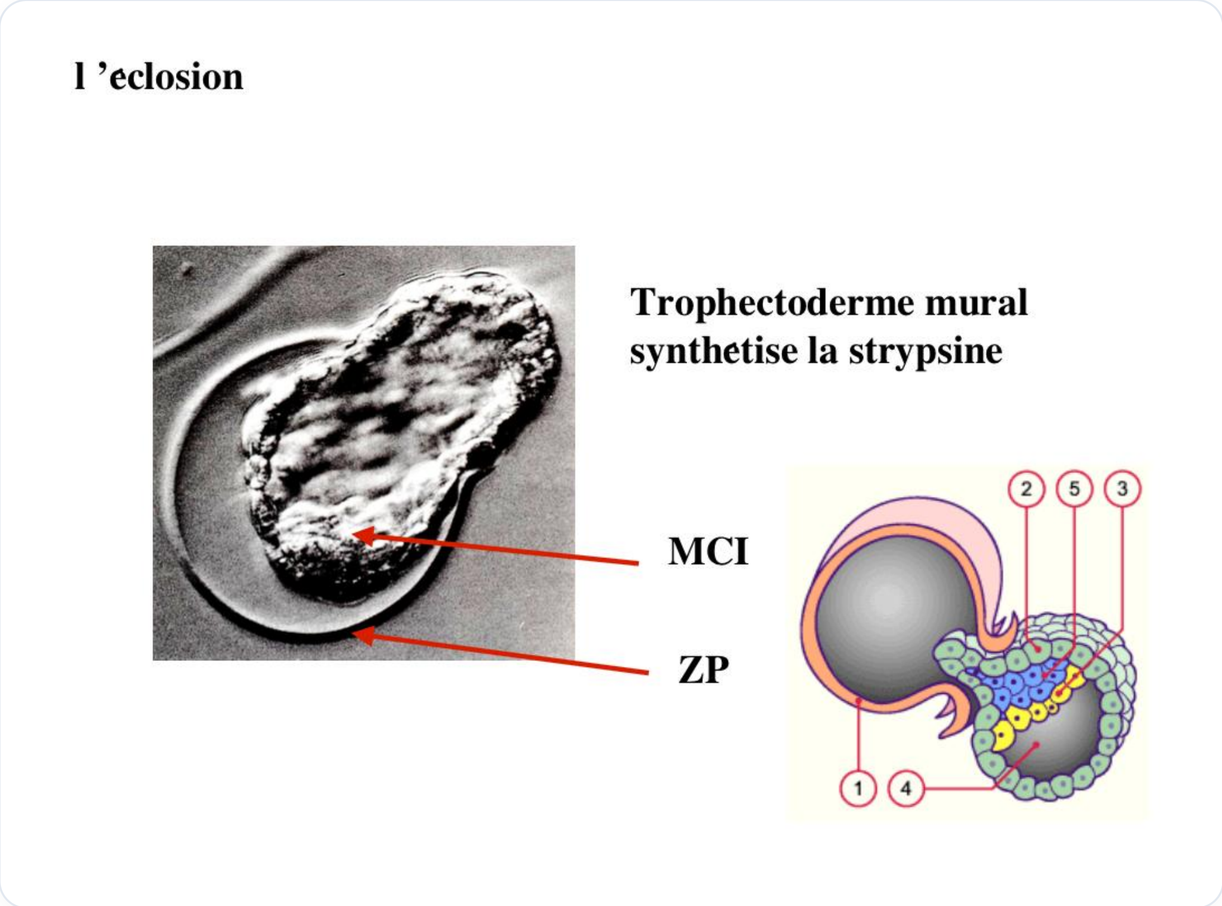


FIG. — *Blastocyst hatching*

14. IMPLANTATION AND AMNIOTIC CAVITY

DURATION : 14:52 STUDY SHEET

COURSE SUMMARY

This video explores the fascinating process of embryonic implantation and the formation of the primitive amniotic cavity. You will learn how the embryo, with its assimilative pole, penetrates the uterine lining thanks to specialised cells and aggressive syncytiotrophoblast activity, which generates micro-bleeding and prepares the ground for life. We will discuss cell differentiation between hypoblasts and epiblasts, which will respectively give rise to the digestive and neural systems. Ultimately, this session reveals the subtleties of the embryo's quest for embodiment, reinforcing an understanding of implantation challenges and their impact on successful pregnancy.

IMPLANTATION AND AMNIOTIC CAVITY

The **primitive embryo** has an **embryonic pole**, which is an assimilative pole. It always penetrates the **uterine lining** through this pole.

The uterine lining is shown here. The embryo displays **specific cells**, which capture information. This layer of cells forms what is called the **embryonic vesicle**.

The layer here is the **embryonic disc**. At the front, we have the cells of the **syncytiotrophoblast**. "Syncytio" indicates lipolytic activity, therefore an aggressive activity.

If the lining is not ready or if, and this is common, the embryo is not strong enough, **implantation** can fail. Often, during early miscarriages, the embryo lacks power, its metabolism is not strong enough. If the soul has not received this power, the embryo will not be able to enter the lining and will be naturally eliminated.

The syncytiotrophoblast releases an **acid** that attacks the lining. It's as if acid is being poured. The uterine system will then prepare to "welcome" the embryo into its "warm nest".

We will visualise this process through an animation and a more realistic film. This is a crucial moment, almost an incarnation, where the embryo agrees to "enter the flesh", into the maternal lining, into "the earth".

Very early on, there are **calcium-dependent cell junctions**. These adherent cells, called **cadherins** (calcium ions), provide cohesive strength.

When the embryo reaches the uterine lining, it has completed its journey. We observe a zone of concentrated cells, in front of which is a small layer of cells that change their name: the syncytiotrophoblast.

This aggressive lytic activity allows the cells to penetrate the lining by releasing an acid. This even causes **micro-bleeding** and a **micro-scar** in the uterus, at the implantation site. This phenomenon can sometimes lead to confusion regarding the dates of the last menstrual period.

The embryo always enters through the assimilative pole and in a perfectly straight manner. No deviation is tolerated, under penalty of implantation failure.

This penetration is made possible by a **powerful metabolic field**, an intense absorption that responds to a "will to live". It is a true quest for life, a "choice of the soul" to incarnate. This is why the embryo enters through the pole of the **micromeres**.

Let's now observe this image: we see the **blastocoel** and the cells of the embryonic disc. The cells of the syncytiotrophoblast, releasing a fluid, erode and penetrate the lining.

The uterine lining is rich, swollen, full of **blood vessels**, arteries, and small uterine glands, ready to welcome and nourish the **zygote**.

The embryo completely penetrates the lining. At the moment of its entry, a small cavity appears in the embryo: the **primitive amniotic cavity**.

Cells organise around this cavity. The cells facing the amniotic cavity are the **epiblastic cells**. The cells facing the blastocoel cavity are the future **hypoblastic cells**, of endodermal type.

In other words, some cells face the amniotic cavity and others face the blastocoel cavity. The latter will later participate in another formation, which we will explain.

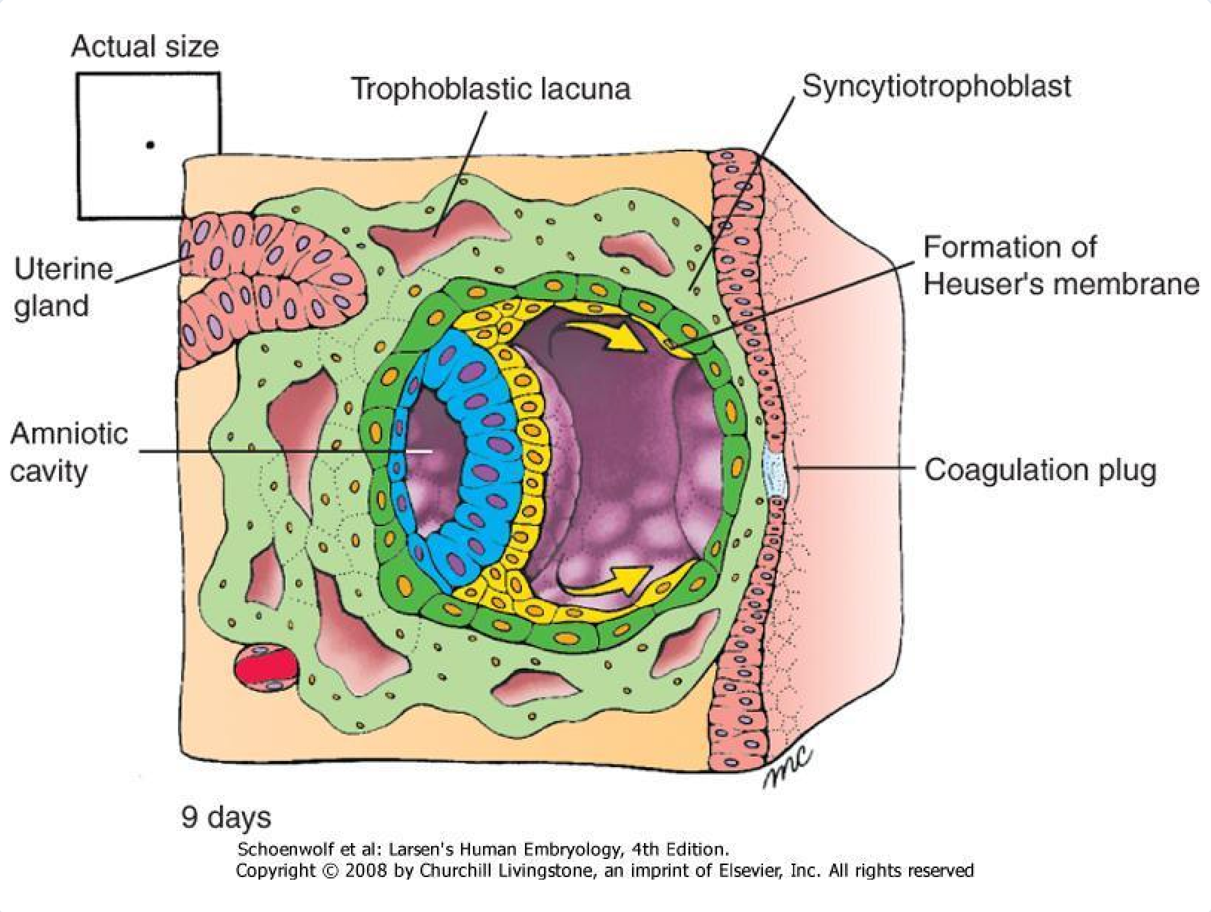


FIG. — Human embryo 9 days

It is important to imagine that at this stage, **cellular differentiation** is already appearing. Two cavities are forming:

- Everything that will be hypoblastic will give rise to the future **digestive system**.
- Everything that will be epiblastic will give rise to the future **neural system**.

Let's examine the mechanism more subtly. The embryo "is hungry", it needs nourishment. There is a continuous process of information exchange between the cells.

The cell penetrates the lining. It is constrained, it doesn't just slide. It uses acid to "push" and integrate itself.

The initial cells receive trophic information. The posterior cells are still nourished by the amniotic cavity of the blastocoel. But the central cells are compressed, lacking nourishment.

The membrane suddenly disappears, giving way to a **cellular exudate**. This exudate is the anlage of the amniotic cavity. The growth process is already underway.

What will this amniotic cavity become? It will become the "bag of waters", a primitive amniotic cavity that will completely envelop the embryo, surrounding it with a small liquid exudate.

In the end, the embryo swims and is bathed in this cavity. Cells, called **amnions**, will develop to nourish and grow this cavity.

The epiblastic cells will form the **nervous system** (the neural plate). Above this neural plate, we will later find the **primitive cerebrospinal fluid**, originating from the amniotic cavity.

This amniotic fluid, within the being, is so profound that we forget this liquid cavity within us. The **ventricular system** and this fluid at the centre of oneself are nothing other than amniotic fluid.

Outside the embryo, there is also amniotic fluid. The embryo at this stage drinks this fluid. It is completely informed by this pulsating, autocrine and paracrine fluid.

The information of **internal and external pressure** is crucial. The integration of the **coelom** (closure of the peritoneal cavity) and the closure of the **neural tube** will occur simultaneously. These are synchronicities, specific inductions.

Here, we still have a zygote. The **foetus** appears with the primitive axial process, i.e., **gastrulation**. It is then that there is a foetal form.

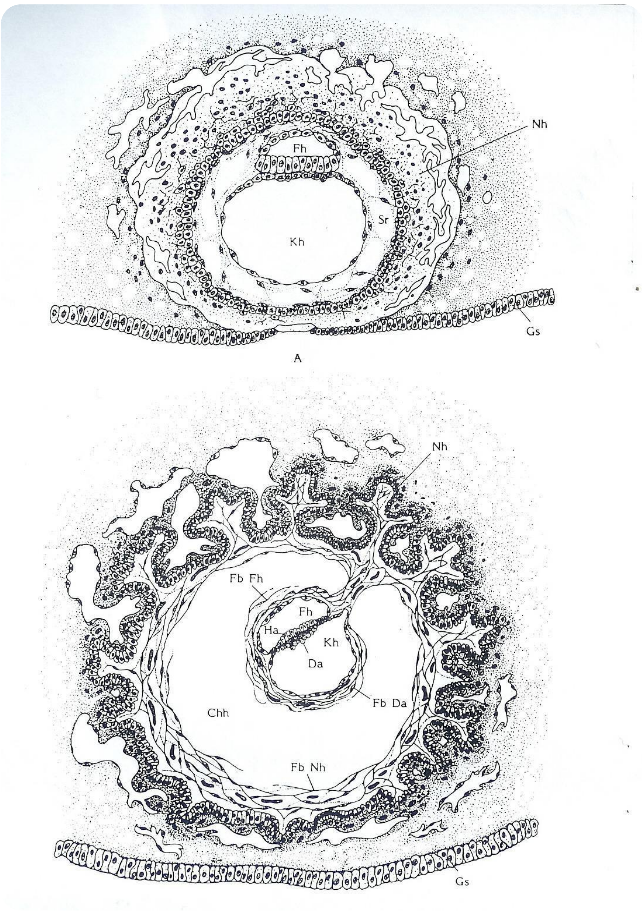


FIG. — Early embryonic development

In the meantime, the woman offers a "lunar landscape" rich in mucous membranes, glands, and nourishment. This is the moment of choice, where the embryo can penetrate the lining.

The embryo seeks to adhere, sending acid. There is very significant metabolic activity.

On the eighth day, the embryo penetrates the lining. Pressures to the right and left create the force necessary for the appearance of the amniotic cavity.

In the embryonic disc, at the moment this cavity appears, cells organise and orient themselves towards it. This is where the epiblast and hypoblast appear.

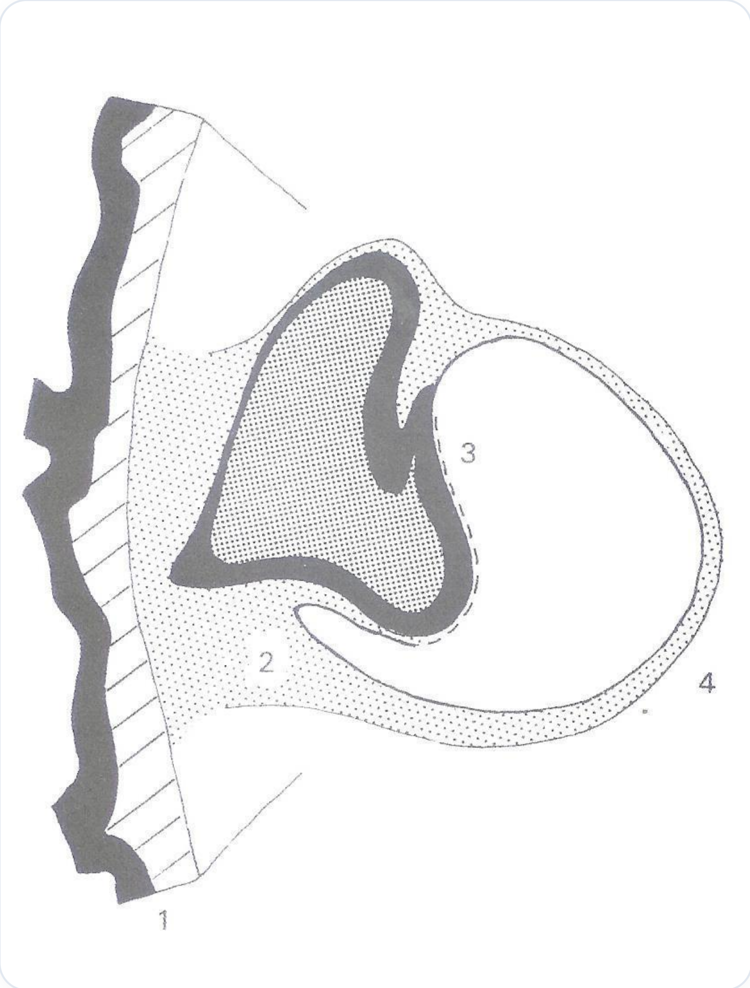


FIG. — Developing nephron

At this stage, we already have a preparation for digestive-type tissue and neurological-type tissue.

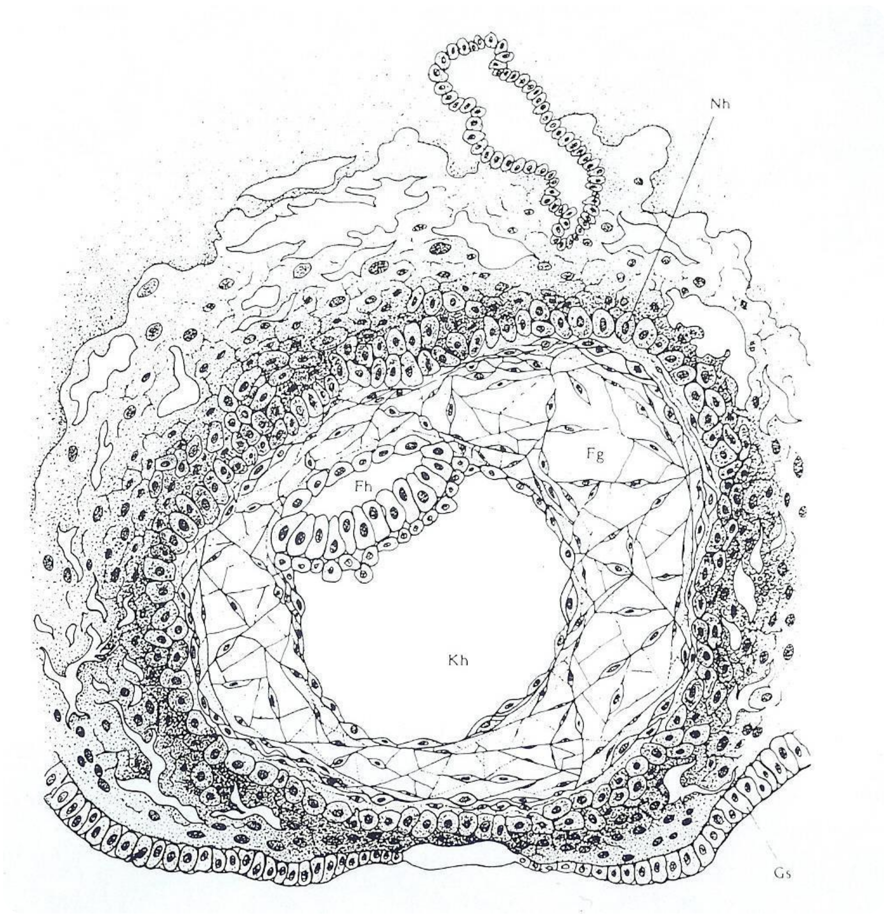


FIG. — *Neurulation</td></tr></table>*

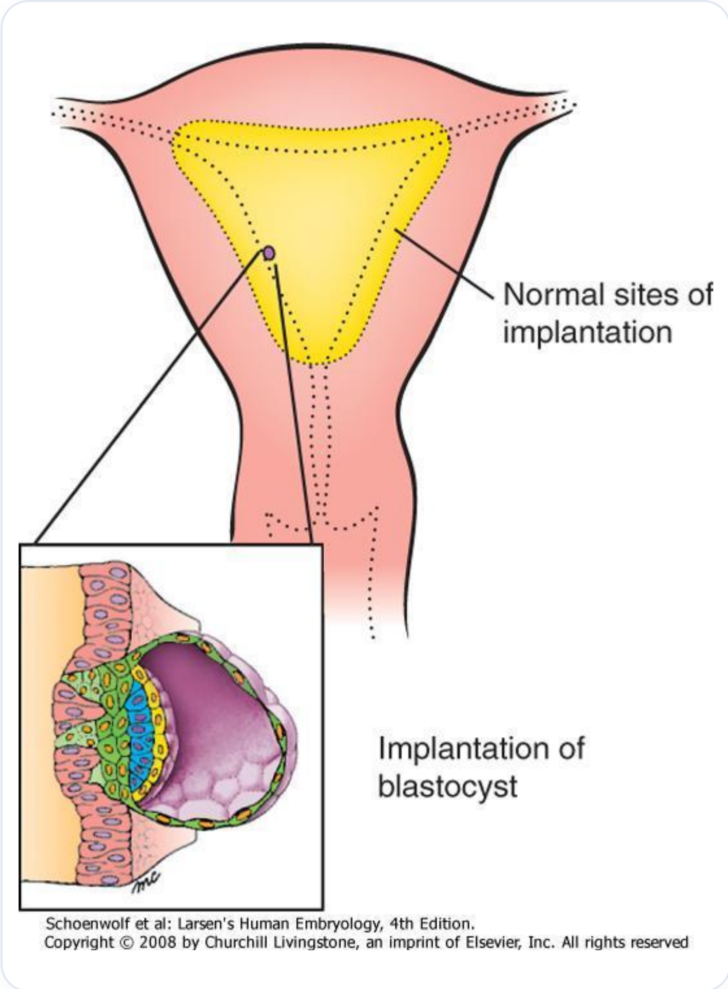


FIG. — Blastocyst implantation

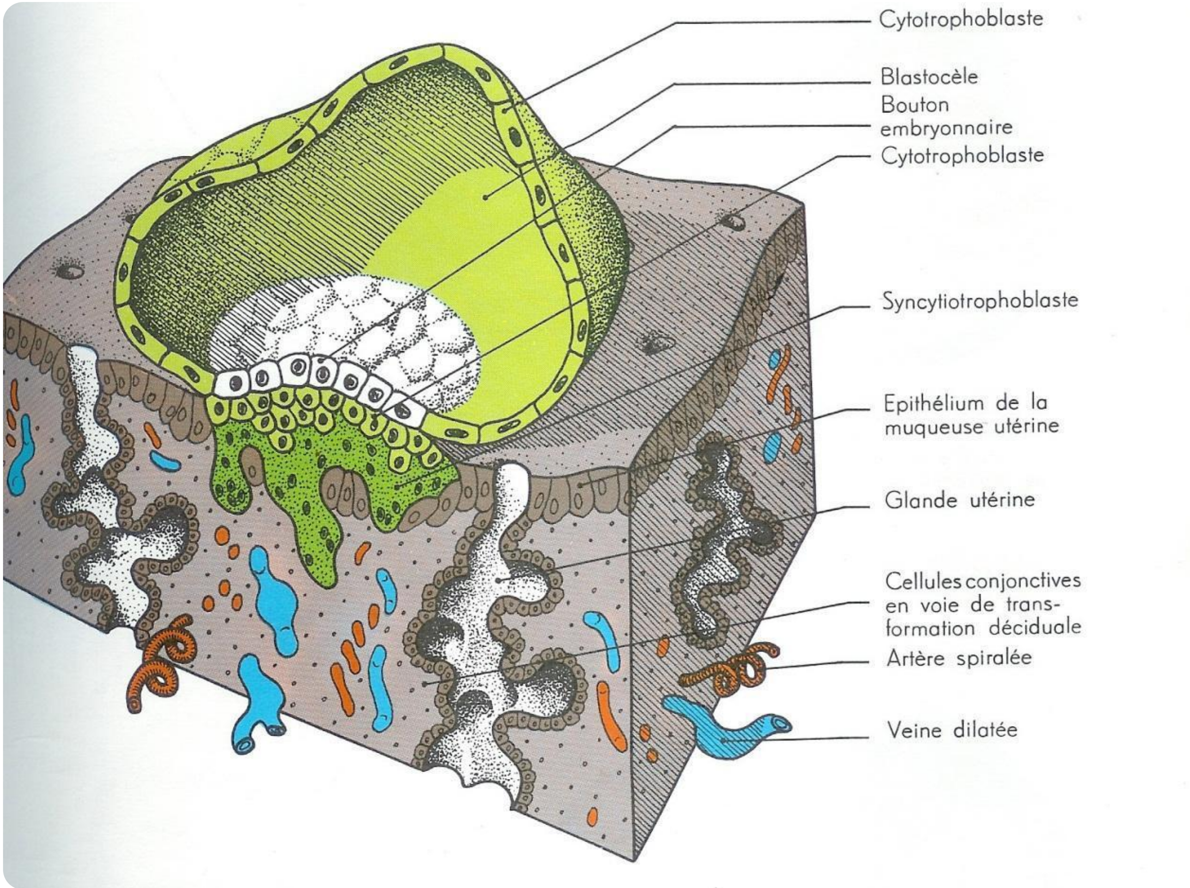


FIG. — Blastocyst implantation

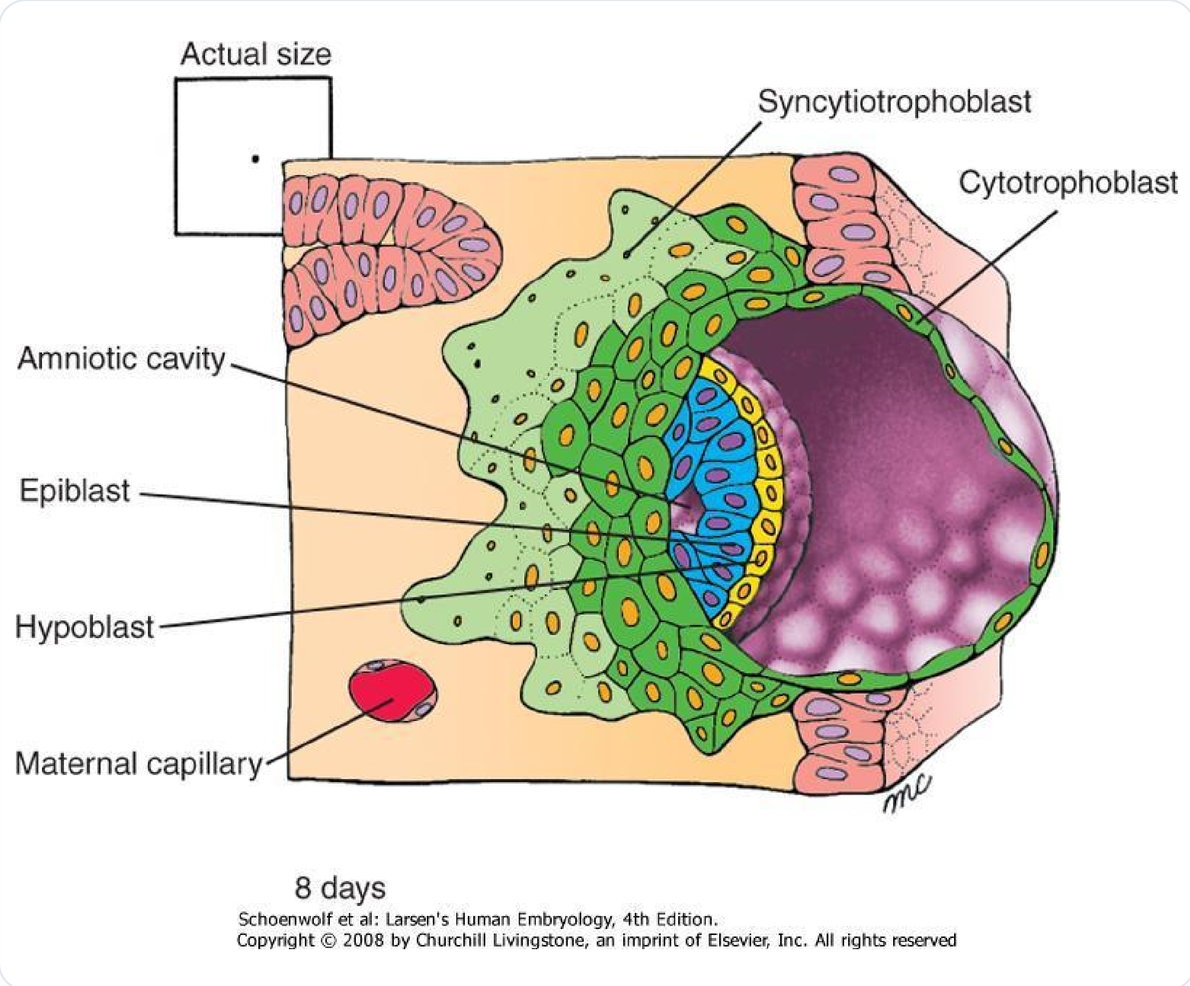


FIG. — 8th embryonic day

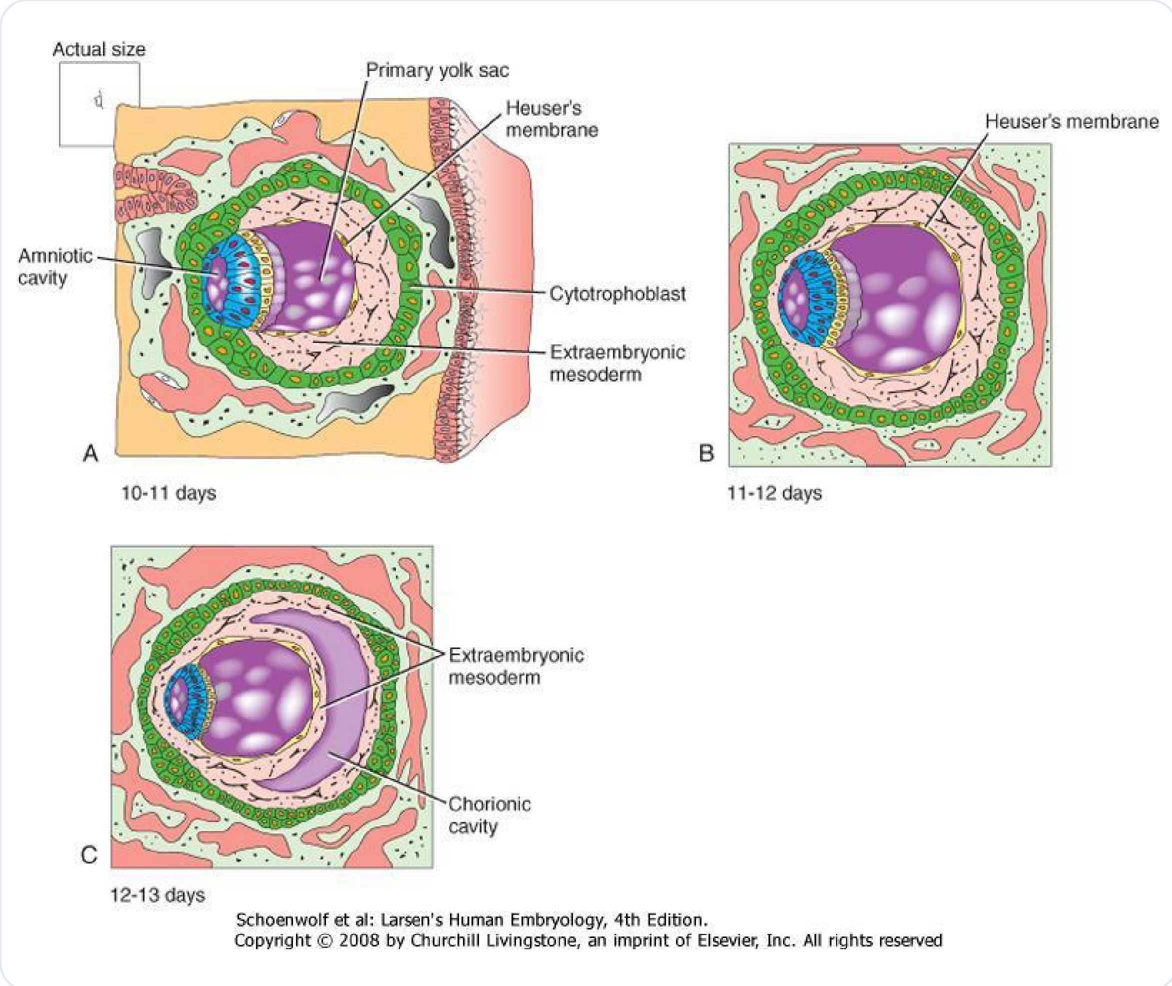


FIG. — Early embryonic development

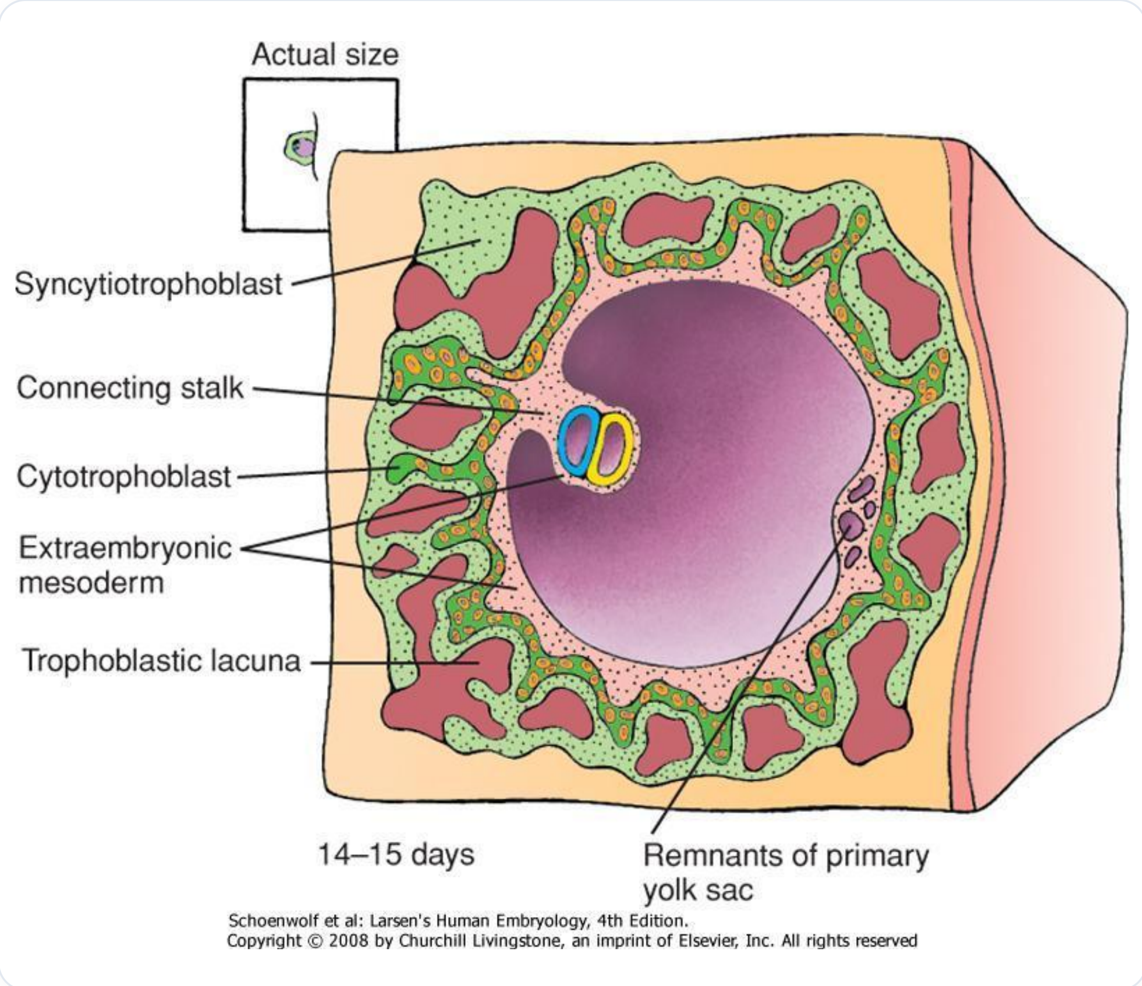


FIG. — Embryo 14-15 days

THE ECTODERM

15. Setting Up the 3rd Chamber

DURATION : 06:17

PEDAGOGICAL SHEET

COURSE SUMMARY

This video addresses the importance of the third embryonic chamber in the context of differential cell growth. It explores the role of the blastocoel, the amniotic cavity, the endocyst, and the growth algorithm that conditions the formation of different cavities: the vitelline cavity, the exocoelomic cavity, and the amniotic cavity. Emphasis is placed on fluid dynamics and the impact of cell divisions, thus merging embryological understanding with potential applications in osteopathy and biodynamics.

16. SETTING UP THE AXIAL PROCESS

DURATION : 16:49 STUDY SHEET

COURSE SUMMARY

This video teaches the mechanisms of axial process establishment in embryology, focusing on the dynamic between expanding and impanding epiblastic cells. The concepts of the zero point, notochord, and Hensen's node are detailed, illustrating their crucial role in cranio-sacral development. Students will discover how these embryonic structures interact to form a functional unit, essential for understanding all growth and development processes within the framework of biodynamic osteopathy.

ESTABLISHMENT OF THE AXIAL PROCESS

This movement is driven by a **growth force**. In the **epiblastic** tissue, some cells diverge upwards while others converge.

We observe cells in an **expansion dome** and others in an **impansion dome**. In other words, some cells are in the **convexity** and others in the **concavity**.

The algorithm intervenes: the cells at the front develop much faster than those at the back, with a fulcrum where growth is slow.

Detailing the embryo, the head (cranial, cranio-caudal) is formed from an **impulse**. Below it, another layer of cells forms.

Within this polarity, two phenomena occur in the tissue, visible in the growth rate. The **ectodermal** tissue always grows faster than the tissue below it, capturing information more quickly from the outset.

The **endoblastic** cells, which follow the movement, do not develop in the same way. They do not have the same reference, the same development, nor the same spatio-temporal and energetic information.

There is a point called the **zero point**. In a wave, this point is almost stationary.

In a growth process, it can be represented as follows: at the top is time, and at the bottom is growth. In the centre, the epiblastic tissue is initially almost flat, then quickly takes on an **S** shape.

We observe that the cells along the axis do not grow at the same rate. The upper part grows and begins its expansion, which is the zero point. This represents a rapid multiplication of cells, an accelerated **mitosis**, creating a kind of **tube** below.

This tube is the process of **gastrulation**, called the **notochord**. At this stage, Hensen's node and the tip of the notochord are identical. Hensen's node moves backwards, from the cranial to the caudal region, while the point does not move. This movement leads to the development of the **primitive cranio-sacral axis**.

This system is not yet neural; it is **notochordal**. The tip of the notochord will be the base of the skull, the space for its future development. Hensen's node, at the back, is where the **sacrum** will develop.

It is common to confuse Hensen's node with the closure of the **posterior neuropore**, which is incorrect. In adulthood, the remnant of Hensen's node is found on the sacrum, which includes vertebrae **C1**, **S1**, **S2**, and the descending notochord.

Are the zero point and Hensen's node different? Initially, they are together. Are the future base of the skull and the sacrum together? Yes. They are not really separate, but form a growth wave with a fixed point (the base of the skull) and a mobile system below.

The sacrum is a mobile **fulcrum**, while the base of the skull is a fixed fulcrum. Blais-Schmidt, in classical embryology, speaks of the gastrulation process, but prefers the term **axial process** to explain its establishment.

The zero point is below. The algorithm indicates that the part in the expansion dome will grow much faster than the part in the impansion dome.

Imagine this in a space where the **embryonic pedicle** is present, and trophic information arrives. Inside, a tube forms, thus developing a primitive axial process.

The sacral point is not yet there, but it will arrive. This movement with the zero point and the arriving dome illustrates that the zero point and Hensen's node are the same thing. At the time of this wave, it is the zero point.

When we have the S, it is the zero point. This shows that the sacrum and the occiput are, ultimately, a **functional unit** formed by a movement. The sacrum is directly connected to the base of the skull by this process.

Here we observe a remnant of the notochord, which extends to the base of the skull. This development occurs in the centre of the embryo. At this stage, the anterior part of the embryo is the head, while the small remaining part is the trunk.

The subsequent development of the embryo will change shape, but at this stage, it was in the middle. The vertebrae, the neural tube, and the notochord pass into the vertebrae. Finally, in the embryonic disc, the **nucleus pulposus** will remain.

The axis is present from the outset. To treat disc problems, it is essential to consider what is happening in the **mouth**. The base of the skull, which is the tip of the primitive **autochord**, will be organised by **occlusion**.

The hypophyseal cartilage will become the body of the **sphenoid**, while the parachordal cartilage will become the basilar part of the **occipital**. This set of elements constitutes the **base of the skull**.

It is crucial to see this in three dimensions. At the moment the notochordal process is established, a small formation called the **primitive streak** appears at the level of the embryonic pedicle. This is an energetic field of aspiration that manifests itself.

From this movement, the epiblast will transform into **ectoderm**. Through this aspiration movement, a lateral traction occurs. The great growth of the dome part leads to the appearance of the primitive streak.

This dome forms Hensen's node, which moves backwards by growth. In this phase of development, the lateral cells are drawn towards the centre, forming **bottle cells** that constrict between the tissues.

Between the two, a space opens, creating an aspiration field where cells arrive to form the **primitive mesoderm**. At the moment this third cavity appears, the notochordal process begins, creating the three tissues: **ectoderm**, **mesoderm**, **endoderm**.

The mesoderm begins to form from the primitive streak and the movement of the autochord. Cells are drawn towards the centre to fill the intermediate tissue, forming the **primitive mesoblast**.

It is the epiblastic tissue that will give rise to the mesoderm and the notochord. This process is a powerful wave. The tube created is the notochord, Hensen's node is there, and the primitive streak appears.

As soon as this process begins, the mesoderm forms simultaneously. An important observation is the **nodal flow** which breaks the symmetry of the embryo on a molecular level.

A few years ago, the question arose: why is the heart on the left or right? At the time of this wave, small cilia are observed moving. A liquid called **nodal fluid** becomes a nodal flow.

This flow is closely observed, showing that the cilia perform a circular movement, moving elements from right to left. This movement creates a preferential flow, with small morphogenetic molecules transported by this flow.

These molecules, in the form of small vesicles containing proteins, establish the asymmetry of the embryo. They inform the heart and provide the necessary information for its development.

Thus, at the time of the axial process, a left and a right side, as well as a specific axis, appear. Through this moment, all the mesodermal tissue develops, accompanied by asymmetric morphogenetic activity.

The primitive cranio-sacral axis forms between the **14th** and **21st day**.

THE ECTODERM

17. Organisation of Metabolic Fields

DURATION : 08:21

PEDAGOGICAL SHEET

COURSE SUMMARY

In this video, we explore the organisation of metabolic fields during embryonic development, starting from the embryonic knob and the formation of the blastocoel. Key concepts include the introduction of permeation, permeation, and infusion fields, which determine how trophic information is distributed and influences tissue transformation. The video also details the role of the embryonic pedicle, which guides the polarised trophic supply, as well as the dynamics that lead to the formation of the 'S' shape of the embryonic tissue. This teaching is essential for understanding the foundations of biodynamic embryology and its application in osteopathy.

PLANCHES ANATOMIQUES & SCHÉMAS (1)

FIG. 1

FIG. 1 **FIG. 1**

18. MEDITATION 1 - BREATH & CONNECTION TO THE BREATH OF LIFE

DURATION : 04:45
STUDY SHEET

COURSE SUMMARY

This video explores meditation focused on breathing and connection to the Breath of Life, emphasising observation of the patient's breathing. Participants learn to attune to a specific frequency, fostering a calm and open state of mind receptive to new perceptions. The exercise also guides participants to prolong exhalation to reach a space of inner silence, crucial for accessing embryonic potency and pre-natal breathing. Key concepts include attention to breathing, observation of respiratory changes, and the search for balance points in the primary respiration, thus paving the way for personal and therapeutic transformations.

MEDITATION 1 - BREATH & CONNECTION TO THE BREATH OF LIFE

CONNECTING TO THE EMBRYONIC POTENCY

To access **embryonic potency**, it is necessary to tune into a **specific frequency**, similar to searching for a radio station.

We will learn to observe the **patient's breathing**, starting with the first breath. Gradually, we will look for a point, a **door**, where we can place our sensation. This point is located at the end of exhalation and just before inhalation.

EXERCISE TO DISCOVER A QUALITY OF THE MIND

Let's do this exercise together to discover a **new quality of the mind**.

- Connect with my gaze.
- Let's take a **deep breath in** together.
- Then, breathe out. Follow the exhalation to the very end.
- At the end of the exhalation, voluntarily prolong the short **apnoea** slightly.
- Breathe in again. If you lose this state, breathe in again.

- Observe this state which gently suspends things, and you will notice that **thoughts slow down**.

A particular **tissue and mental quality** then appears. It is in this state that we will begin to work, because it is there that other perceptions will emerge.

ATTENTION TO BREATHING

It is essential to be very attentive to **breathing**. Simply observe the changes.

Even if inspiration occurs, we can remain in this sensation as soon as this state appears. It remains present, like a door opening at that precise moment, allowing you to go fully into it.

Let's do this exercise once more together. Connect well with the spirit of this sensation; it is constantly there.

DEEPENING RESPIRATORY OBSERVATION

We are going to make the exercise a little more complex. We will no longer perform **voluntary breathing**.

We will observe our normal breathing. Be careful to settle into this space. Keep your eyes open, fix them on this person's gaze. Observe your breathing.

Then breathe out. At the end of the exhalation, a small door opens. Settle there, like a child sitting down.

THE BREATH OF LIFE AND EMBRYONIC BREATHING

Thoracic breathing truly begins at birth. Before that, the foetus makes small **diaphragmatic** movements, but it is not yet full breathing.

This means that the **Breath of Life** exists before that moment. By entering this space at the end of exhalation, we seek to work on this **pre-birth** breath.

It is an entry point, a key to accessing this perception of the breath. We will go even further later.

We will look for **balance points** in primary respiration to find points of **stillness** where emerging problems can resurface in the pathways of transformation.

19. QUESTION D1: ORIENTATION OF THE WAVE IN THE AXIAL PROCESS

DURATION : 02:00
STUDY SHEET

COURSE SUMMARY

In this video, we explore the wave-like motion between the amniotic cavity and the vitelline cavity during phase 1 of embryonic development. This movement, where the amniotic cavity moves forwards and the vitelline cavity moves backwards, creates an S-shaped dynamic and marks the beginning of a process of infusion and permeation of essential information for the embryonic cells. The implications of this movement are crucial, particularly during nidation, where the cells of the embryonic knob receive nutrients and trophic information, allowing them to grow rapidly and develop a new polarity. This session highlights the complex interaction between movement, nutrition, and embryonic development, offering keys to a better understanding of ontological processes.

WAVE ORIENTATION IN THE AXIAL PROCESS

Between the **amniotic cavity** and the **vitelline cavity**, a **wave** movement occurs. The movement of the amniotic cavity is directed forwards, while that of the vitelline cavity is directed backwards. This movement should be considered in relation to the **pedicle**.

The question arises: does this movement align with the orientation of the country? At some point, it is an **impulse**.

The forward movement of the amniotic cavity, combined with the vitelline vesicle moving less rapidly at that moment, gives the impression of a backward movement. This phenomenon creates a wave, marking the beginning of an **S-shape**. This constitutes **phase 1**.

This movement is linked to **permeation**, to **infusion**, which are methods of information diffusion. It involves a **trophic** input, a **metabolic** substrate that arrives.

In this phase, I remain along the membrane and am in permeation. I send nutrients between the cells. From time to time, I come to nourish my apparatus.

This **nourishment** implies movement. However, initially, the force will be primarily directed towards the apparatus. This is due to its original, anterior position.

At the time of **nidation**, it can be said that when I enter the mucosa, the cells of the **embryonic knob** are the first to receive trophic information. They are already predisposed to develop more rapidly. These cells are more central and smaller, allowing them to attract more information. This represents a new **polarity**.

20. QUESTION D1: MESODERM IMPLANTATION

DURATION : 02:33 STUDY SHEET

COURSE SUMMARY

In this session, you will learn the crucial steps of mesoderm formation within the embryo, a process orchestrated by wave-like movements. The focus is on the dynamics of the primitive streak and Hensen's node, which are fundamental for cellular development. You will discover how the notochord forms in synchrony with the regression of the primitive streak, and how the epithelial cells of the epiblast contribute to the emergence of the intermediate tissue, the mesoblast, which will become the mesoderm. This understanding will enrich your practice in embryology and osteopathy, offering valuable tools for comprehending embryonic development.

MESODERM FORMATION

The **mesoderm** forms through a **wave** movement that takes shape within the embryo. This movement creates a **field of aspiration** at the primitive streak, which is a key element in embryonic development.

As the **primitive streak** regresses, it descends with **Hensen's node** until it completely disappears. This dynamic is represented by a superior view, where the **amniotic cavity** can be observed above.

The **notochord** forms in response to this movement. At this stage, a traction occurs at the **pedicle**, leading to the appearance of the primitive streak. It is important to note that the formation of the notochord and the primitive streak occurs **simultaneously**, although there is a fine chronology in this process.

The illustrated drawing shows a section highlighting the movement of the **epithelial cells** of the epiblast. These cells are drawn towards the centre, thus filling the space created between the epiblast and the hypoblast.

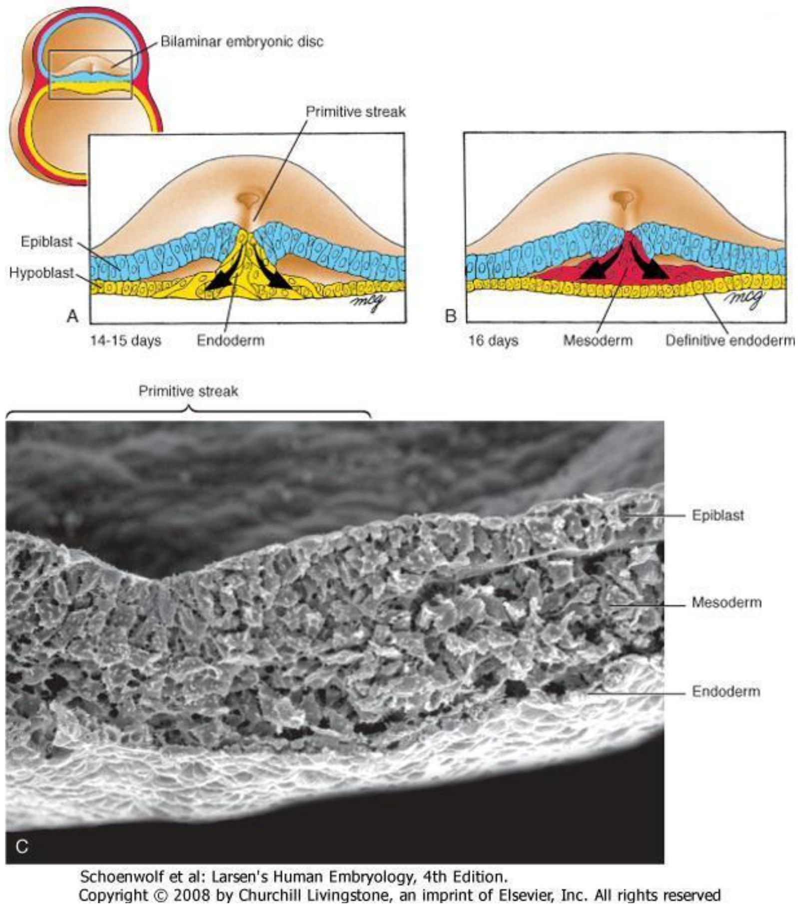


FIG. — Gastrulation

This space, which opens during the formation of the wave, is essential for the appearance of the **intermediate tissue** known as the **mesoblast**, which will become the future **mesoderm**.

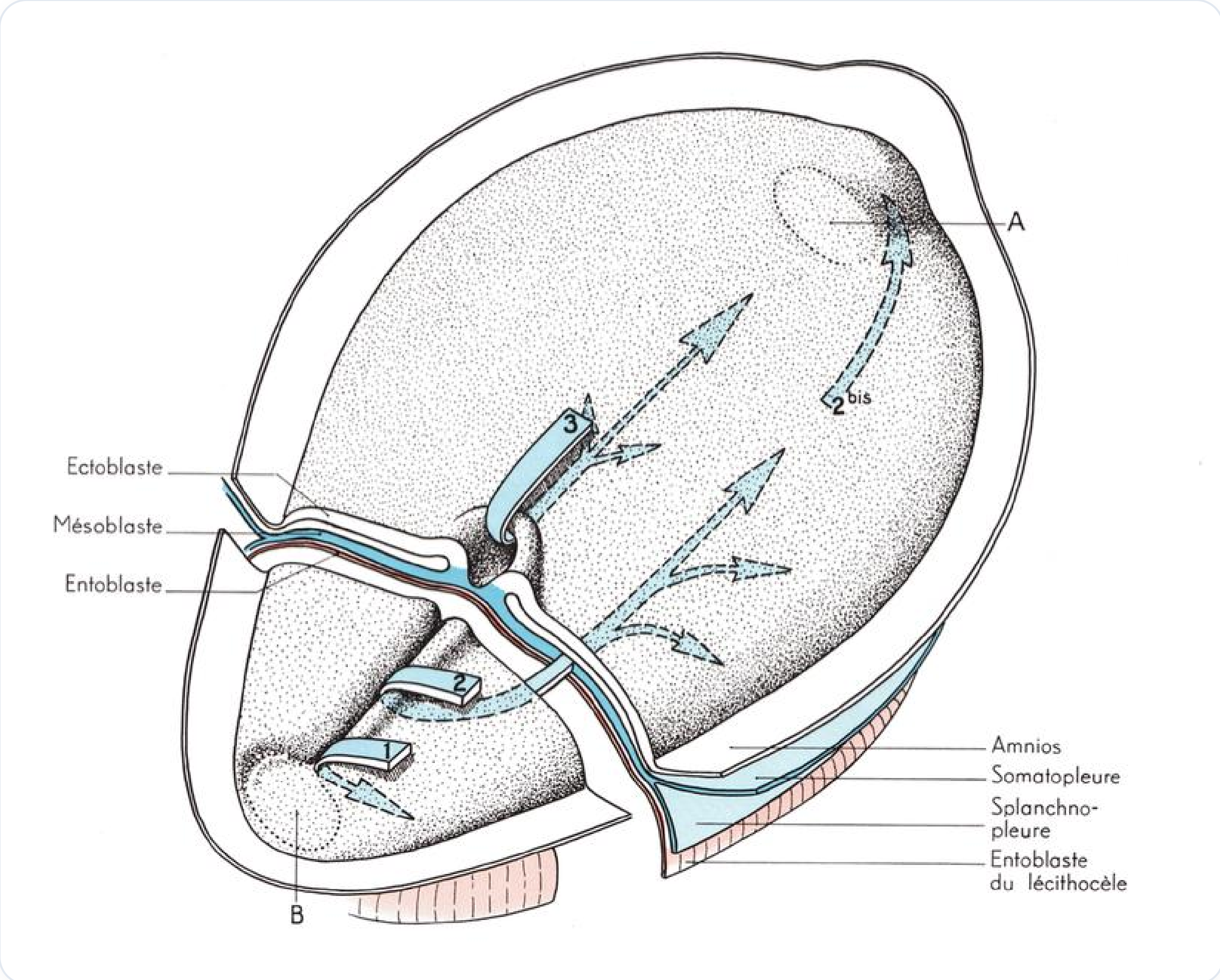


FIG. — Avian gastrulation

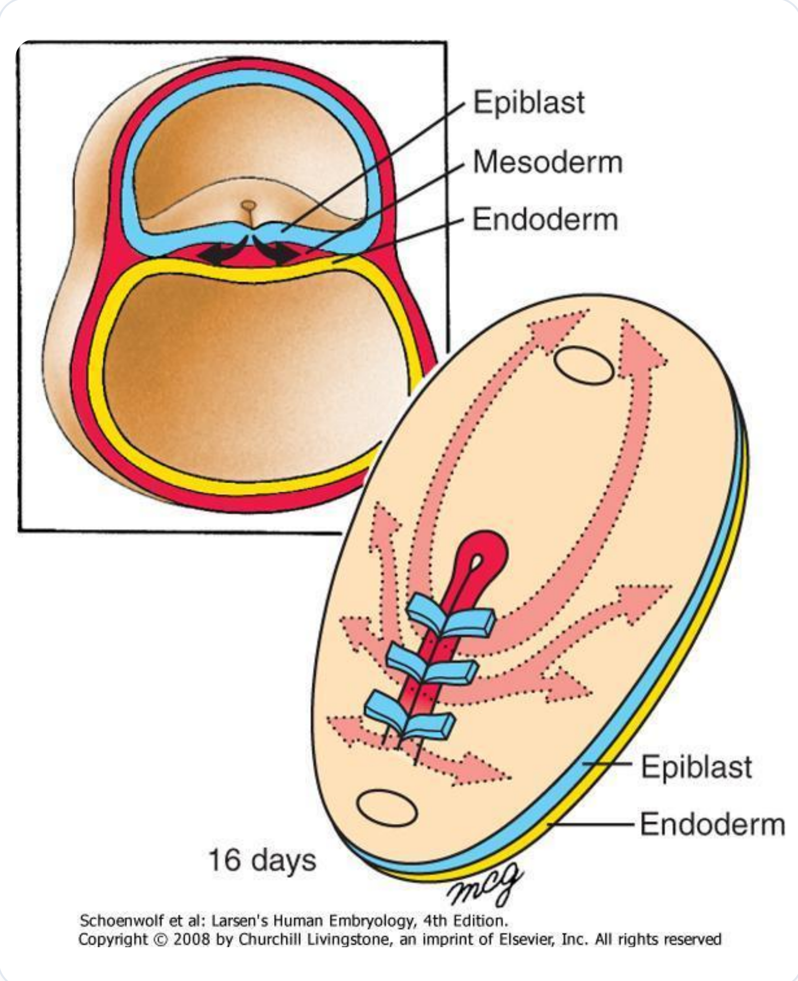


FIG. — Human gastrulation 16 days

21. QUESTION D1: PRIMITIVE STREAK

DURATION : 02:33 STUDY SHEET

COURSE SUMMARY

This session explores the emergence of the primitive streak as an aspirational force that establishes the embryonic axis. We discover the concept of the 'zero point', a zone of minimal movement where the notochord forms, essential for the development of embryonic structures. Emphasis is placed on the embryo as a functional unit, demonstrating the importance of precise connections between various parts of the body, such as the sacrum and the base of the skull, for healthy development.

21. QUESTION J1: PRIMITIVE STREAK

THE EMERGENCE OF THE PRIMITIVE STREAK: A METHODICAL SUCTION FORCE

The most plausible explanation for the appearance of the **primitive streak** is found opposite the **embryonic pedicle**, precisely at the moment when a **"wave"** occurs.

This is a huge methodical suction force that causes the establishment of an **axis** on the median lines. This phenomenon is at the origin of this extraordinary wave.

THE ZERO POINT AND THE NOTOCHORD

At the heart of this dynamic, there is a central point, described as the **"zero point"**. This is the point where movement is least perceptible.

If we observe a later embryonic section, we can already discern **cartilaginous densifications**. The tip of the **notochord**, which appears at this point, corresponds to the point where these dynamics are anchored.

THE NOTOCHORD: A CRUCIAL EMERGENCE SPACE

The tip of the notochord is not the **base of the skull** at this stage. It is the fundamental space that will allow the emergence of future structures, around which everything will be built.

In the same way, **Hensen's node** is not the sacrum, but the point around which the future sacrum will develop.

THE EMBRYO: A FUNCTIONAL UNIT IN MOTION

These mechanisms show that the embryo is not a juxtaposition of separate parts, but a **functional unit** driven by continuous movement.

It is as if we were holding the sacrum and had to connect, with a precision comparable to a **laser beam**, this structure to our sphenoid or to the base of the skull. It is essential to find this link, this **subtle connection**.

22. QUESTION D1: FORMATION OF THE EMBRYONIC PEDICLE

DURATION : 02:33

STUDY SHEET

COURSE SUMMARY

In this captivating video, we explore the formation of the embryonic stalk, a crucial moment in embryogenesis. We discover the dynamics of differential growth between the periphery and the centre of the embryo, and how this tearing movement contributes to the orientation and structuring of the embryonic stalk. The emergence of the latter is linked to mechanisms such as the growth of the analysis and the external coelom, influencing the creation of internal spaces. Finally, we address the transition from the embryonic stalk to the umbilical cord, highlighting the importance of the yolk sac in this process. This essential lesson offers an enriching perspective on biodynamic embryology and its clinical implications.

FORMATION OF THE EMBRYONIC PEDICLE

The establishment of the **third chamber** of the embryo is marked by a **differential growth rate**.

This rate is observed between the outer part of the embryo, which receives numerous trophic levels, and the centre, whose growth slows down compared to the outside. This phenomenon creates a movement, similar to a **tearing**, where the faster growth forms a layer of cells that appears on the edge.

THE CONCEPT OF "TEARING"

The term "tearing" is a **metaphor**. It is not a true tearing, but an image that evokes the dynamic movement within the embryo.

It is at the centre of the embryo that this movement occurs, and it is in this developmental process that the **embryonic pedicle** begins to form. Previously, it was uniformly distributed, but thanks to this movement, it orientates towards a central point.

ORIGIN OF THE EMBRYONIC PEDICLE

The phenomenon of embryonic pedicle formation is directly linked to the **differential growth rate**:

- Between the peripheral part of the embryo.
- Compared to the slowed rate of the centre.

This process occurs when the **astrocoele** begins to transform into a cavity. The ground is in a reaction position, which influences the growth system of the analysis. The latter, associated with the **growth of light**, also contributes to the formation of the body.

What is called the **growth of the analysis** and the **external coelom** ultimately gives rise to this space, creating a convergence of forces that leads to the formation of the embryonic pedicle.

FROM EMBRYONIC PEDICLE TO UMBILICAL CORD

It is important to note that the embryonic pedicle is not yet the **umbilical cord**. At a certain point, due to the excessive growth of the **amniotic cavity**, it will push the **vitelline vesicle** onto the embryonic pedicle.

The meeting between the embryonic pedicle and the vitelline vesicle is what will form the umbilical cord.

23. REVISION D1: NOTOCHORDAL PROCESS

DURATION : 03:09 STUDY SHEET

COURSE SUMMARY

In this session, we explore the notochordal process, which is essential for embryonic formation. We study the dome structure, where the ectodermal tissue loses its polarity and becomes a fulcrum for development. The aspiration field generated by the primitive streak draws epithelial cells towards the centre, leading to the formation of the mesoderm. The mechanisms involved, such as the release of hyaluronic acid and the dynamics of cell growth, are examined in detail, highlighting the crucial importance of these transformations in early embryonic development.

DAY 1 REVISION: NOTOCHORDAL PROCESS

We will examine the **notochordal process** to deepen your understanding.

Imagine a **dome-shaped** structure. The **embryonic pedicle** is not present here, and therefore the trophic source is located at this point.

The **ectodermal tissue** above is no longer polarised and no longer absorbs. This becomes a fulcrum. It is important to visualise that everything is **growing**. Everything organises around and creates itself.

This becomes **Hensen's node**. Simultaneously, everything acts as a **suction field**. A space will open between the ectoderm and the endoderm, between the epiblast and the endoderm.

This space generates, at the level of the **primitive streak**, this suction field. The cells, which are **epithelial** cells, are drawn towards the centre.

They are pulled towards the centre, compress, become **bottle cells**, pass underneath, and release an acid, **hyaluronic acid**. This acid will be very important later in protein and structural formations, but that is not the main point for now.

These cells pass underneath and fill the space. This is how the **mesoderm** develops.

We have this wave, this suction field that creates a primitive streak. This movement pulls the cells towards the centre. Meanwhile, Hensen's node progresses.

I am always at the shortest point. I do not move. What am I? The **zero point**. And I observe all of this.

In parallel, in this same development, there is **nodal fluid** inside. Since it is in the notochord, it is called nodal fluid.

Why do the cells in the dome develop faster? It has been observed that cells in this expanded process, as if we were no longer in an expansion dome, multiply much faster than those on the same line, but which are converging.

These latter develop less quickly.

24. DAY 1 REVISION: THE NODAL FLOW

DURATION : 03:43

STUDY SHEET

COURSE SUMMARY

This session addresses the origin of internal asymmetry through Hensen's node and nodal flow. Rotational cilia create a movement that allows the displacement of molecules and vesicles, thus generating morphogens that influence the asymmetrical development of organs, such as the heart and liver. The genetic switches involved, such as Sonic Hedgehog and fibroblast growth factors (FGF), play a key role in the organisation of internal structures.

The embryonic axis is presented as an organising centre, with an emphasis on the importance of the intra-embryonic mesoderm. The impact of these concepts on osteopathic practice is also highlighted, by showing how understanding these processes is crucial for patient care, thanks to an appropriate mental image of these internal dynamics.

24. DAY 1 REVISION: THE NODAL FLOW

THE ORIGIN OF INTERNAL ASYMMETRY: HENSEN'S NODE

At the origin of **internal asymmetry**, at **Hensen's node**, small **cilia** are observed. These cilia do not perform a simple movement, but a **rotational movement**.

They accelerate when they are higher relative to their base and bring small **molecules**. These molecules come from the "roof" (from inside our body) towards the inside of the node.

It's a big "mess" where cilia are present. Above, small **vesicles** fall and concentrate towards a specific side.

VESICLES AND MORPHOGENS

These small vesicles collide due to "excess" and release a large number of **morphogens** onto the ground below. There are **activation genes**, which amplify the phenomena, and **inhibition genes**.

It is this genetic mechanism that will create the structures of **asymmetry**. Later, this will determine that the **heart** will develop on the left, the **liver** more to the right, and that the internal organs will be asymmetrical.

We do not have perfect **symmetry**, but rather **functional asymmetry** which will give us long-term movements. All of this is organised down to the **cerebral connections**.

THE STARTING POINT OF ASYMMETRY AND NODAL FLOW

Nodal flow is the starting point of asymmetry. It is this small movement of the cilia that moves the **nodal fluid** and concentrates the vesicles more on one side than the other.

The vesicles are of **epiblastic** origin, coming from the roof of the epiblast. It is here that the entire cascade, left and right, will occur, with this concentration towards the right.

This will trigger cascades involving important genes such as **Sonic Hedgehog** and genes from the **FGF** (Fibroblast Growth Factors) family. These genes will be involved up to Track 2 and will be important later for the creation of new structures, new **proteins**, and for the organisation of asymmetry.

THE AXIS AS A CENTRE OF ORGANISATION

This organisation is established very early. This **axis** is a true centre of organisation.

The epithelial cells called "**bottle cells**" and the **mesenchymal** cells will form the **mesoderm**. There is **extra-embryonic** and **intra-embryonic** mesoderm, but it is mainly the intra-embryonic mesoderm that interests us here.

It essentially originates from the epiblastic epithelium. This is why, initially, it is first **electricity** that induces and organises the fluid, and then the fluid that organises matter.

THE IMPORTANCE IN OSTEOPATHY

This is crucial in our **osteopathic** thinking, in our work. We work with our **mental image**.

What is our mental representation of these processes? The body uses it, whether we are aware of it or not.

25. NEURAL TUBE CLOSURE

DURATION : 03:10

STUDY SHEET

COURSE SUMMARY

This session explores in detail the closure of the neural tube, a crucial moment in embryonic development that begins with the formation of an elongated structure from the epiblast. This process involves the fascinating interaction between the hypoblast and the ectoderm, leading to the emergence of the notochord and its transformation into a chordal rod. Students will learn how this dynamic influences vertebral formation and integrates essential information for embryonic development. By understanding these interactions, therapists can better grasp the embryological foundations related to their practices.

NEURAL TUBE CLOSURE

Neural tube closure is a crucial process in embryonic development. This tube, by its nature, is hollow and forms from the epiblast.

Initially, the tissue deposits on the epiblast, then closes to form an elongated structure. Beneath this structure lies the **hypoblast**, which can be considered the "mouths" of the **endoderm**.

As one moves towards the **endocoel**, the shape of the tube becomes more defined. This growth process leads to a flattening on the hypoblast, followed by a regaining of height and a restructuring into a cord. It is essential to visualise this formation as a tube.

Observing a cross-section, one can see the inside of this structure after removing the epiblast. At this stage, the **ectoderm** and **endocord** can be placed above, while the structure flattens against the hypoblast.

At some point, this structure completely inserts itself between the other tissues and becomes a cord without an internal lumen. It is at this stage that it acquires **ecto-mesodermal** characteristics, but this primarily concerns the **notochord**, and not the **mesoblast**.

The notochord, at this stage, is considered a chordal tube, subsequently evolving into a chordal cord. Although it is epiblastic, during its closure, it integrates hypoblastic information, but remains essentially epiblastic.

This dynamic is important: the notochord is epiblastic due to its development. When it becomes chordal, it retains hypoblastic information.

This information is crucial, as it constitutes the **neural tube**, around which the vertebra is outlined. One can already observe the vertebra forming around the notochord, which then leaves a vestige of its existence.

26. VENTRAL MIDLINE REFERENCE

DURATION : 05:28

STUDY SHEET

COURSE SUMMARY

In this session, we explore the key role of the ventral midline in embryogenesis, where it serves as a vital reference that guides embryonic development. The notochord, as a central element, induces the formation of the neural tube and influences the ascent of the sacrum. We highlight the importance of this reference for metabolism, the structure of body systems, and its link to health issues such as osteoporosis. Embryonic coilings and the structuring of the midlines are discussed, as is their integration with the diaphragms and the formation of the body's various systems, thus emphasising the complexity and harmony of the developing embryo.

VENTRAL MIDLINE REFERENCE

The **ventral midline centre** is a fundamental element of the embryo. It is a key centre that allows the **genome** to execute its programme and organise the entire embryo.

Without this axis, there is no **vital reference**. In a therapeutic approach, the aim is to allow the body to regain this reference.

A growing child, even a baby, has a fundamental need for this reference, for example for **calcium metabolism**. The loss of this reference hinders the ability to restructure or integrate calcium, leading to degradation and, ultimately, to degenerative processes such as **osteoporosis**.

This axis is crucial for the **ascension of the sacrum** and its development. It generates a compressive force on the sacrum, linked to **sphenobasilar synchronisation (SBS)**, which influences balance, including at the level of the sternum.

The **notochord** is the basis of this ventral midline. It induces the formation of the dorsal midline, which will become the **neural tube**.

The neural tube is built from the notochord, which means that the notochord is the inducer of the formation of the **posterior nervous system**. This posterior nervous system will grow and converge towards the centre, forming the anterior midline.

All this movement integrates the **cranio-sacral axis**, the repercussions of which manifest on the anterior part, at the sternal level. The sternum, in this lineage, is symbolic of a **"stretching star"**.

THE EMBRYO AND ITS STRUCTURING

The embryo initially consists of an **endodermal layer**, a notochord with **lateral mesoblast**.

If one visualises the embryo by turning it over, from a lateral point of view, one observes a **neural plate** above the notochord. This neural plate will evolve into a **neural groove**.

The neural groove will reorganise the fluid structure of the mesoderm. The appearance of a **primitive lateral aorta** is a key step.

The internal mesoderm does not grow at the same rate as the neural plate, which causes **flexion of the embryo**. During this flexion, the neural tube forms to become the **posterior neural axis**.

COILING AND MIDLINE FORMATION

The restructuring of the internal fluid system leads to the appearance and organisation of the **lateral aortae**. The differential growth of these structures brakes the embryo forwards while pushing it from behind.

- This induces a **coiling of the embryo**, orchestrated by the aortae and the amniotic cavity.
- The **yolk sac** deposits on the embryonic pedicle, forming the **umbilical cord**.
- At the same time, the anterior part is drawn forwards, giving rise to the future sternum and heart.

This is how the three midlines are built:

- The **notochordal** (ventral midline).
- The **posterior midline** (that of the neural tube).
- The **anterior midline** (that of the positive tube).

This embryonic coiling, a complex process, is at the heart of embryonic organisation. All systems are organised around their **primordial fluid system**.

This precise synchronisation and these inductions delimit the embryo:

- The ventral midline induces the development of the dorsal midline.
- The dorsal midline induces the development of the medial midline.

INTEGRATION AND DEVELOPMENT OF SYSTEMS

These midlines are integrated into the different **diaphragms** (thoracic, cervical, etc.). There are at least seven diaphragms.

These structures are fundamental for the development of the following systems:

- **Urogenital**
- **Metabolic digestive**

- **Rhythmic**
- **Neurosensory** (or mineral, vegetal, animal)

27. DEMO

DURATION : 16:33

STUDY SHEET

COURSE SUMMARY

This video offers an osteopathic demonstration focused on working with the sacrum and the notochord. The instructor highlights the importance of observing breathing and listening to the patient's sensations, while navigating the body's systems. Key concepts such as the zero point, polarity, and asymmetry are discussed, as well as the need to remain attentive to stillness and energetic flows. The instructor emphasises the importance of intuition and connection to the patient's life wave to facilitate the healing process.

OSTEOPATHIC DEMONSTRATION

In this demonstration, I take the **sacrum** quite low. My fingers are placed on the highest part you can feel. I don't grasp the entire sacrum, but low enough to reach the tip of the **notochord**.

I bend and rock, all while observing. I use my elbow to get a good fulcrum, a good **"full crew"**. It's very important not to create torsion, as this can be dangerous if the hand is placed too far, leading to rotation of the hand and pelvis.

I position myself well in this axis, with a mental image of this **autobody**. I wait a moment and observe its breathing. I will gently enter another system, the one you described this morning.

I feel that it moves slightly, deviated to the left. There is a correction happening very gently. I reconnect and wait, observing its breathing.

The door opens, I enter the **aptism** and wait for a small ascent. I try to reconnect with this base, like with a laser. I talk too much and lose my train of thought.

As we have seen, this is a reference line. We discussed references for the **genome**, for the entire body, and for the formation of my posterior and anterior midline. This includes the entire **calcium** side, the strength at the level of **collagen** and **elastin**.

We also saw the power of the organisation of **asymmetry** and **polarity**. I try to connect with this wave of life, and at some point, it works gently.

I observe her eyes, she seems to be looking for images in her head. It's working, but the wave isn't there yet. I am on a movement, a wave. Something is falling, and I feel that I have reached my hand.

Her eyes show that it's working. She might be releasing something related to a memory. I won't push too hard, as there is a reaction in the tissue. I go back down and wait.

Again, something is working. It has moved above the left breast, and her breathing has just changed, indicating a gentle oxidative change.

SHARING SENSATIONS

I ask the patient to share her sensations. She feels increased intensity. I will try to continue this video, because it takes time, sometimes it's fast, sometimes it's not. The idea is to have a feeling of **homogeneity**.

For now, something is working. I will give her a zero point so she knows what to do. I move to the right, remaining a welcoming point while being mobile to allow for a lightening.

I remain attentive. At some point, something leaves, then I am there. This is the organism system.

For now, it made a small movement, then it came back. It's starting to work at the head level. When I say I haven't finished the process, I mean I shouldn't calm down regarding the breathing.

THE ZERO POINT

The zero point corresponds to the base of the skull, at the articulation between the **occiput** and the **sphenoid**. I take the occiput and my fingers go to the greater wings of the sphenoid, looking for the **SSB** (sphenobasilar synchondrosis).

I am at the centre of the SSB, below, above, throughout the SSB. I seek stillness, because it is a point created in stillness. I am not looking for a goal, but I always observe the breathing, the door, and the sensation behind.

It is important not to close your eyes when working, as this makes you lose 30% of information. I remain attentive to her gaze up to the limit of the field.

I am careful not to say "I see". I remain outside, just in connection with the vibration. I look for a fulcrum for a reference, by putting my attention on the base of the SSB.

I am looking for **silence**. My touch is hyper light, an attentive listening. I gently move to the level of the skull to look for this point of dynamic stillness. This takes time, because the body needs to organise itself.

I trust the principle of **osteopathy**. I must be neutral, and all of a sudden, I will be attracted by something. I don't enter the lesion, I let it pass.

I stay on health, on the **fulcrum** of health. It is essential to put 30-40% of attention outside of oneself and 70% inside. Will can sometimes be an obstacle, because it is often linked to our ego.

It is crucial to leave this notion of wanting to do well to move towards this reference. The power of life is connected to this embryonic force. We are here to provide a fulcrum, and if necessary, it will take it or not.

This is not a technique I am teaching you, but a connection to the frequency.

28. MEDITATION

DURATION : 02:05

STUDY SHEET

COURSE SUMMARY

This meditation session guides us through an experience of deep connection with our breath. By stabilising our gaze and becoming aware of the eye as an expansion of the brain, we learn to observe the present moment, particularly the end of the exhalation. Through a benevolent thought exercise, we are encouraged to send love and tenderness to a person who comes to mind. This process not only promotes our own well-being but also enriches our relationships with others.

MEDITATION

Connect with your **breathing**.

Stabilise your eyes. The **eye** is an **expansion of the brain** and the **third ventricle**. Breathe.

Observe the end of the **exhalation**.

In this moment, say goodbye to a **bubble** you have filled.

Try to finish thinking about something. Think of someone who comes to mind, to whom you wish to send **love** or **goodness**.

Imagine that, through this thought, you can transmit **well-being**, **tenderness**, **love**, and **joy** to them.

Do this.

29. EMBRYONIC FOLDING

DURATION : 59:59

STUDY SHEET

COURSE SUMMARY

In this video, we delve into the crucial process of embryonic folding, revealing how the notochord influences the structural development of the embryo. We explore the central role of the notochord in establishing polarity and forming the neural tube, while also discovering the dynamics of differential growth between the ectoderm and mesoderm. This process leads to the creation of the neural groove and pre-aortic vessels, essential for embryonic blood circulation.

Furthermore, we analyse how the flexion of the embryo around its vascular system, using the heart as a fulcrum, is similar to natural and comfortable movements. This complex interaction between different embryonic tissues and structures is key to understanding joint formation and body organisation, offering essential insights for students and practitioners interested in biodynamic embryology and osteopathy.

29. EMBRYONIC FOLDING

We will continue with the movement of the embryo and address **folding**. This concept was a revelation for understanding what an **articulation** is. It is the algorithm that generates this **differential growth rate**, leading to the flexion of the embryo.

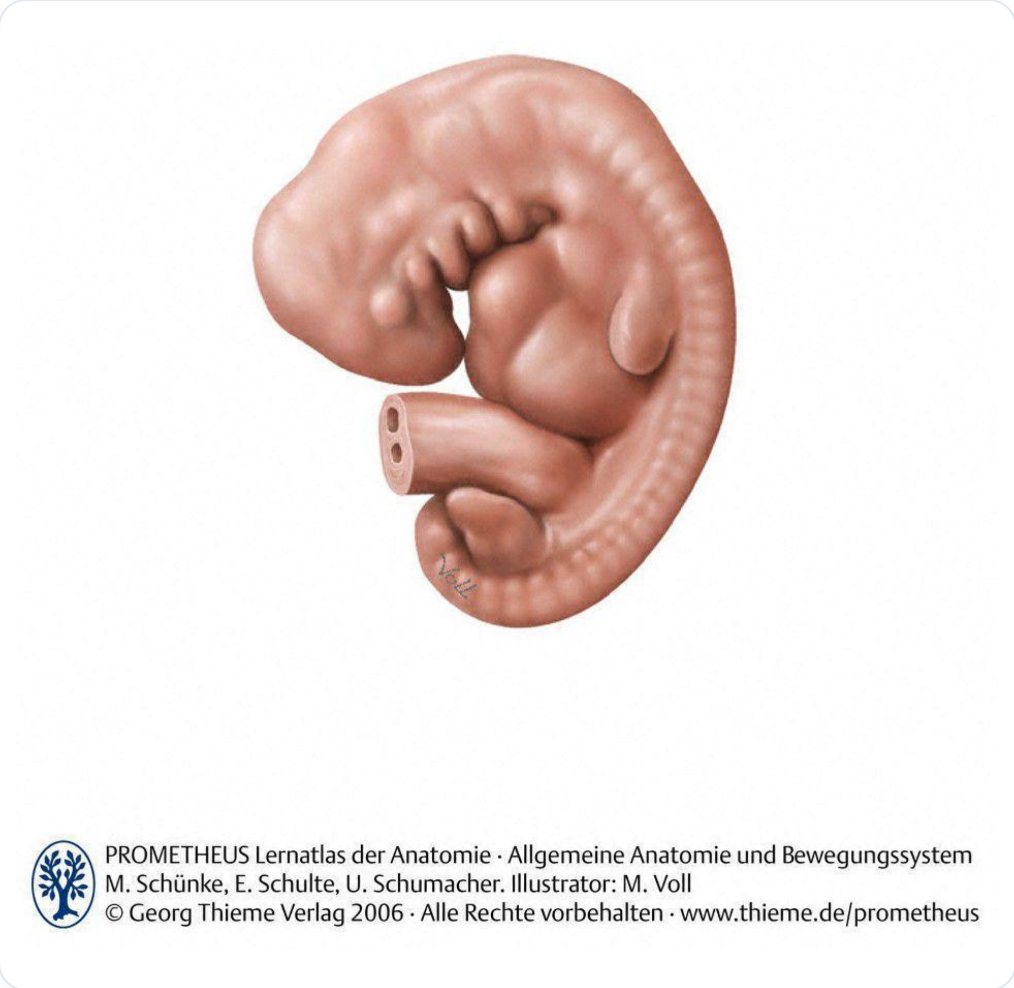


FIG. — Human embryo, early stage

A **fluid dynamic** organises the embryo at this level. The **notochord**, essential for its development, will now influence a structure above it: the **neural tube**, or rather the future neural tube.

THE INFLUENCE OF THE NOTOCHORD

We are between the **eighteenth** and **twentieth day** of development. We have the **axial process**, i.e., the notochord, and above it, the **ectodermal** tissue. We are progressing in the study of embryonic stages.

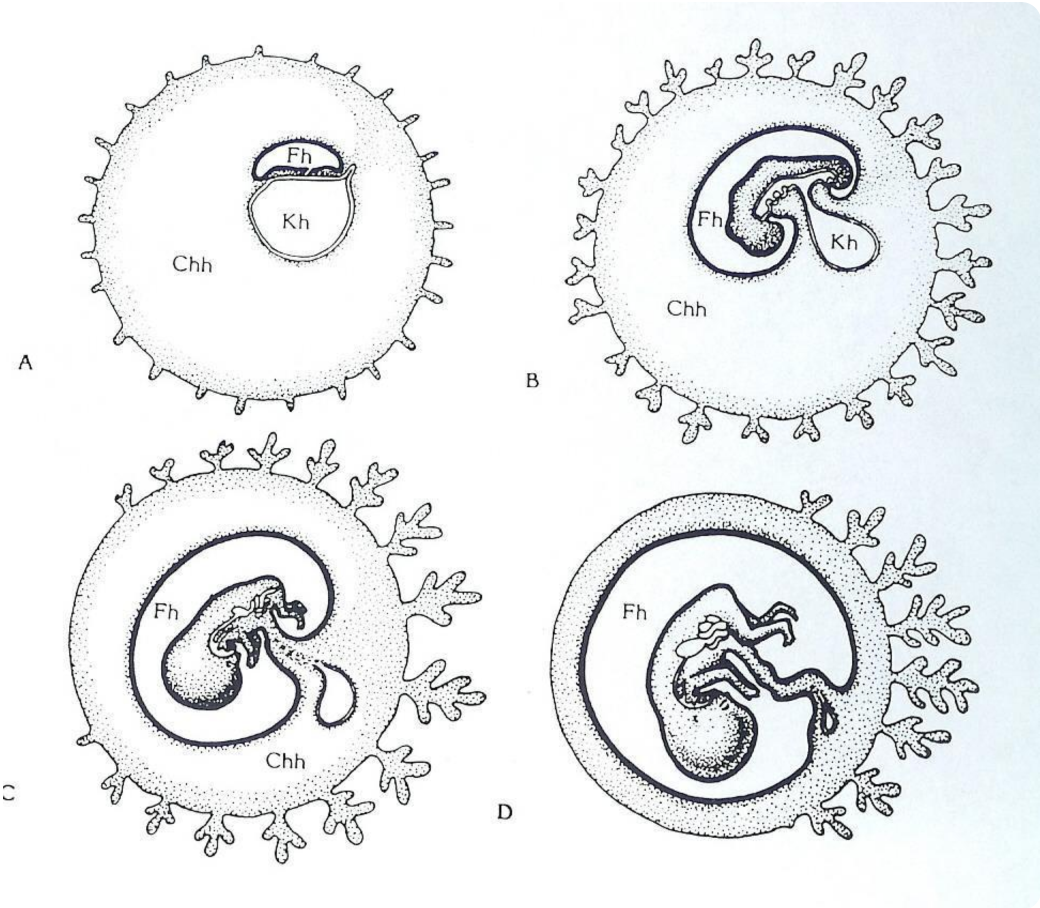


FIG. — Bird embryonic development

The notochord, which has established **polarity** and created a future space for the **sacrum** and the **base of the skull**, influences the development and structural organisation of the ectoderm. Above the notochord, we observe a **neural plate**.

Let's take a cross-section. The ectodermal cells directly above the notochord receive **inhibitory genetic information**, leading to a slowing of their growth.

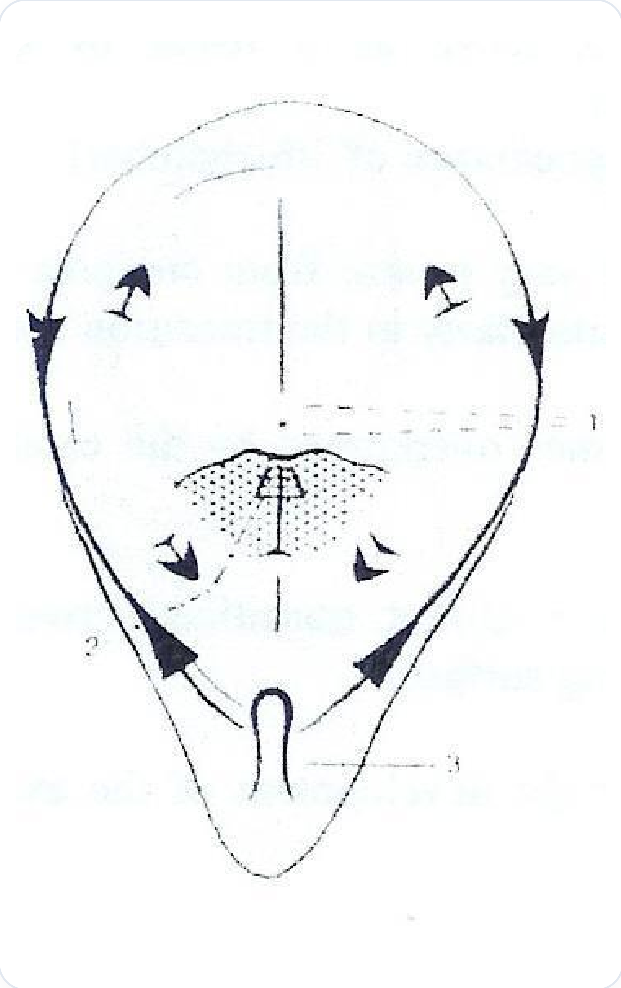


FIG. — Gastrulation

- The central cells at the level of the notochord are inhibited in their growth.
- The lateral ectodermal cells, however, do not have this brake and even develop faster.

This process leads to a **change in the shape** of the ectoderm.

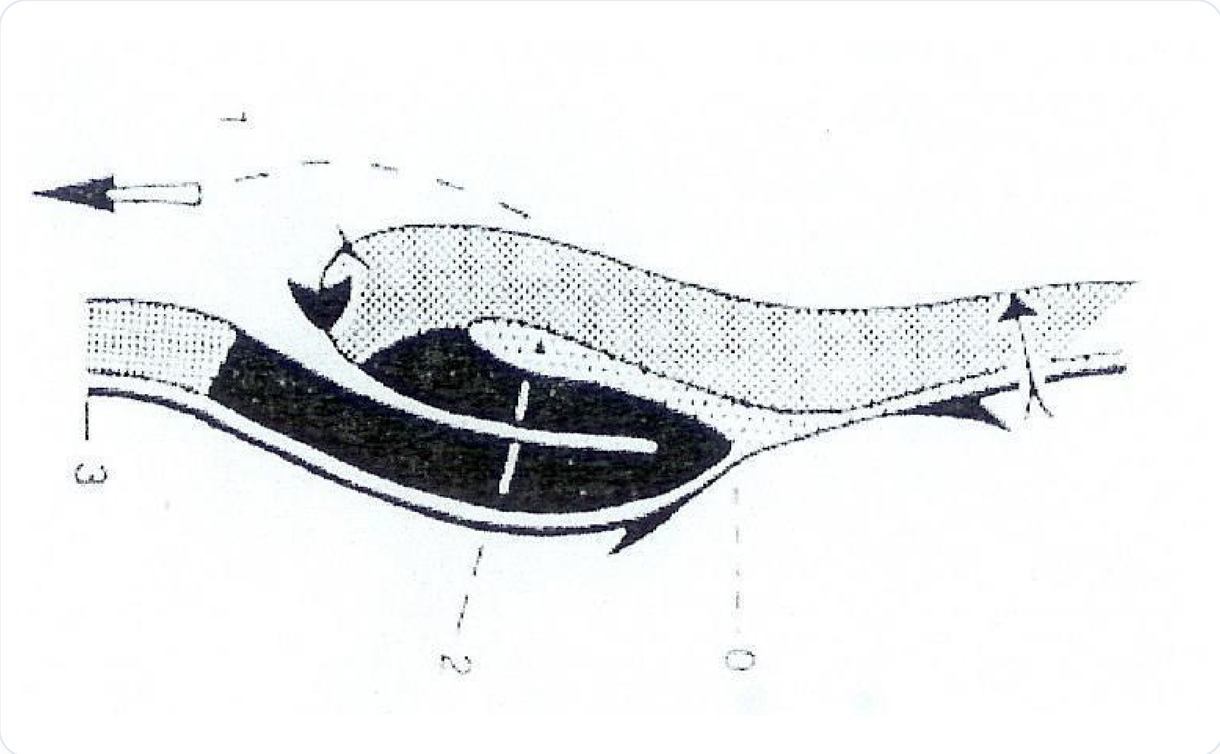


FIG. — Amphibian gastrulation

FORMATION OF THE NEURAL GROOVE AND PRE-AORTIC VESSELS

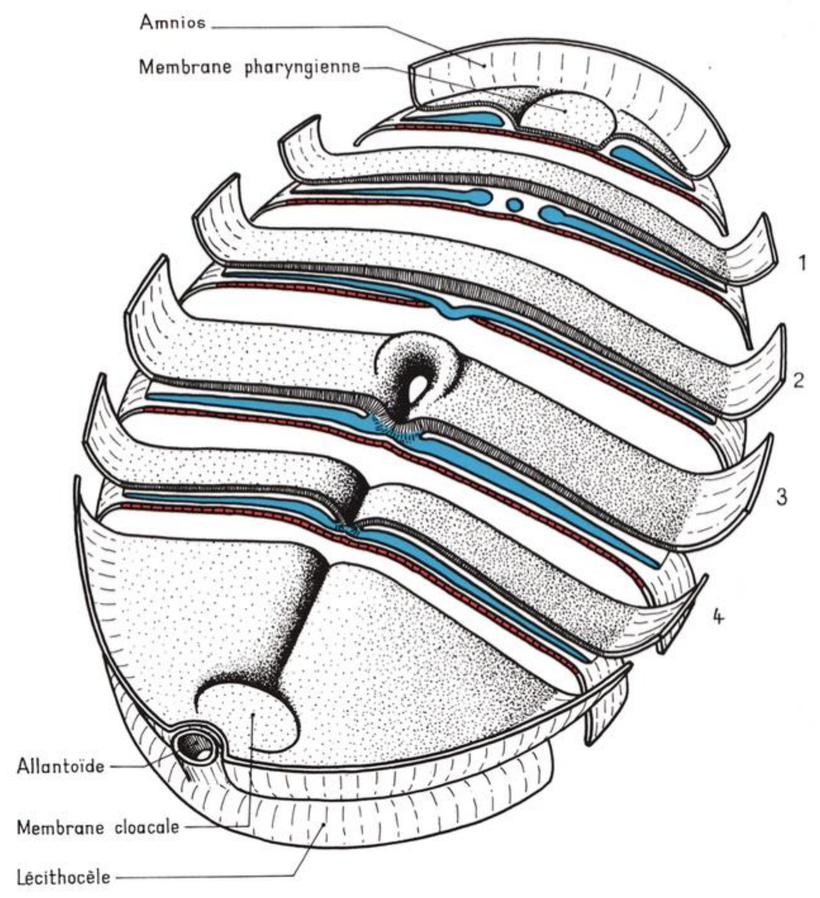


FIG. — Embryo in sagittal section

This differential development causes the formation of a **groove**, called the **neural groove**. The lateral cells of the neural plate develop more rapidly, creating this excavation in the ectoderm.

Simultaneously, the surrounding **mesoderm**, little modified and quite liquefied, undergoes reorganisation. We observe the formation of fine **pre-aortic capillaries** in the form of vesicles, which require a specific trajectory and metabolic concentration.

These vessels organise into two mesodermal fluid conduits. The differentiated growth rate between the ectoderm and the mesoderm is crucial. The ectoderm, a large consumer of information, grows very rapidly.

EMBRYONIC FLEXION AND THE ANGELIC ZONE

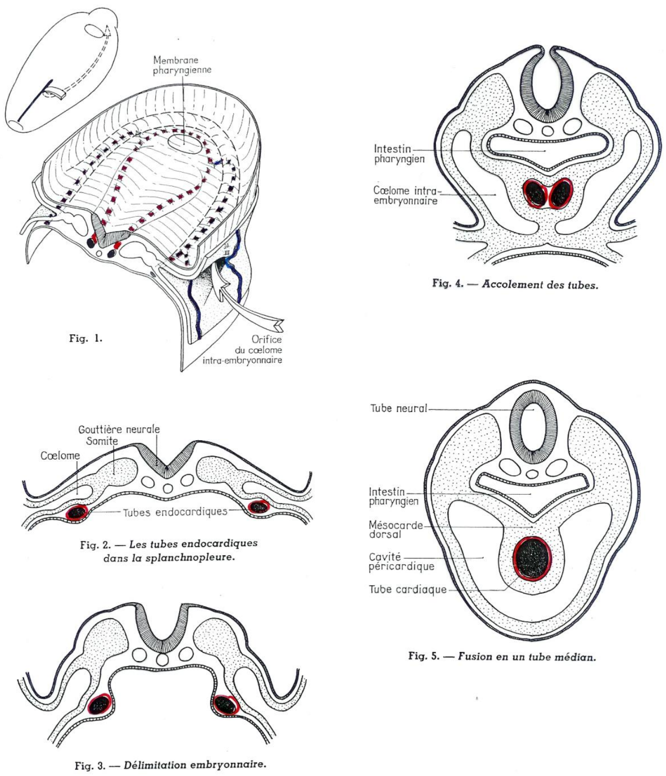


FIG. — Heart tube development

At some point, the embryo will perform a **flexion movement** forwards. This movement is fundamental.

The neural plate begins to organise. On either side, the pre-capillary aortic mesodermal vessels appear. They create a **congestive zone** above the neural plate, a small "pond" in front, supplied by two small lateral vessels. This is called the **superior angelic zone**.

This angelic zone will undergo many modifications. It moves towards the midline to form a tube due to the growth of the embryo. The ectoderm (neural plate) is the engine of this growth, while the zone of the aortic capillaries and the pre-cardiac zone act as a **relative brake**.

THE ROLE OF THE HEART AS A FULCRUM

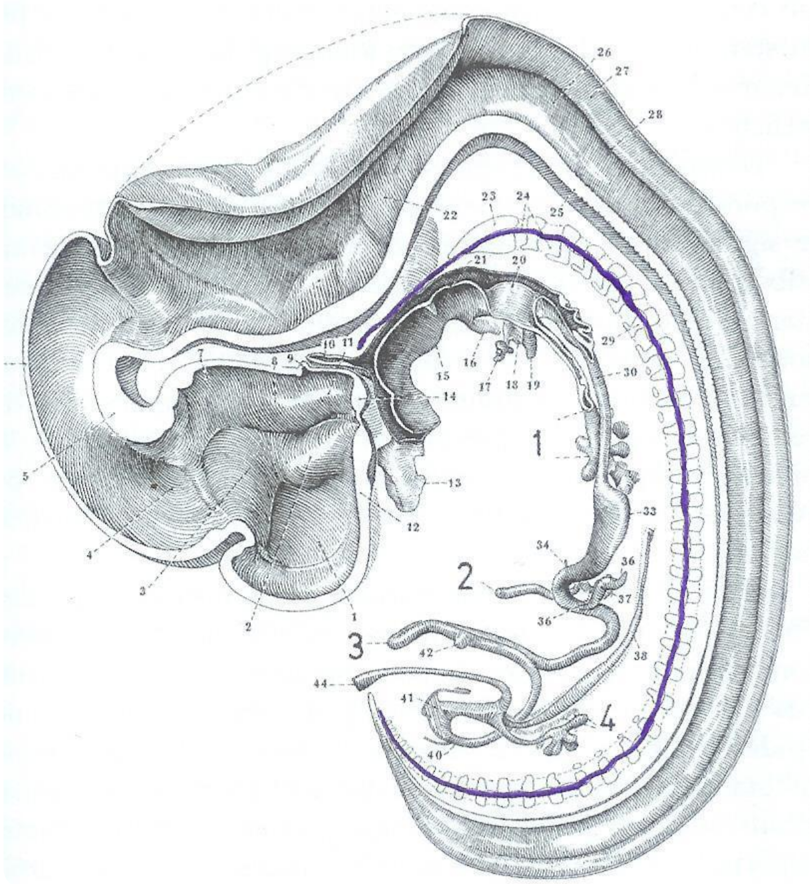


FIG. — Human embryo sagittal

The embryo grows, but it encounters a brake imposed by this pre-cardiac zone. What becomes a fulcrum? The **heart**.

The embryo flexes, coiling around its vascular system, which acts as a **pivot**. This movement is often presented as flexion or folding, but it is actually an **expansion** around this vascular anchor point.

Our true initial postural axis is a **vascular axis**. We coil around it. This principle organises all the articulations of the body. If a path is blocked (no developed artery), growth adapts and takes another route.

This movement is similar to that of a baby coiling in on itself, around its vascular system. It is a position of well-being, reproduced in adulthood in the foetal position in case of discomfort.

ENGINE OF GROWTH AND TROPHIC INFORMATION

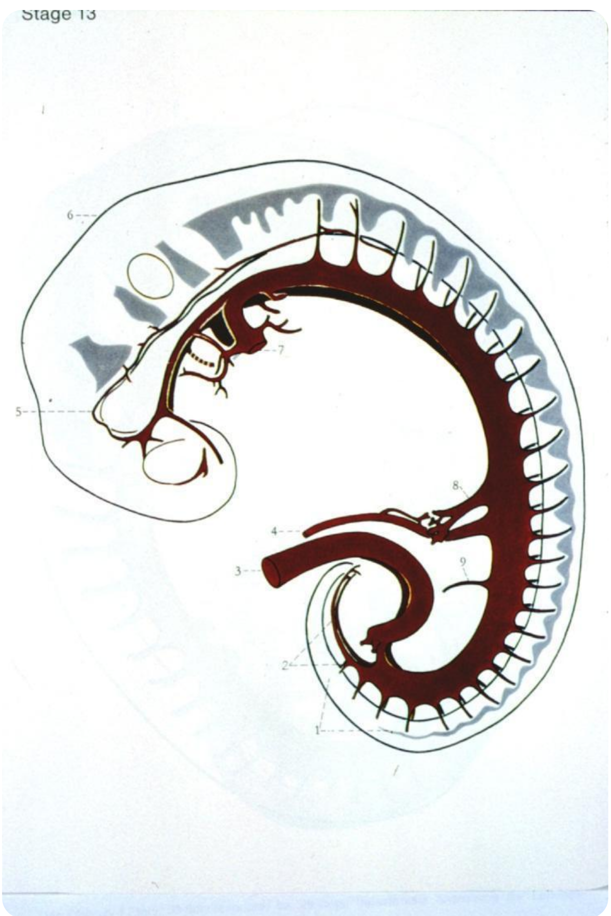


FIG. — Vascularized human embryo

The ectoderm is the **engine of growth**. It is supplied by the **fluid mesoderm**.

- **Engine of growth:** The ectoderm.
- **Trophic information:** It comes from the endoderm, via the mesoderm, and from the mother via the cord and placenta.

The overall coiling of the embryo at this stage will define specific aspects of development.

THE CEPHALISATION CASCADE

Cephalisation, i.e., the development of the head, leads to a series of processes:

1. **Cephalisation**: The process of cerebralisation of the cerebral vesicle coils and triggers the sequence.

2. **Cardialisation**: Cerebralisation initiates cardialisation, the flexion movement positioning the heart. The heart, initially higher, descends to this location due to growth.

3. **Diaphragmatisation**: The heart presses on the supero-internal part of the vitelline vesicle, creating pressure that forms the **septum transversum**, the primitive anlage of the diaphragm.

4. **Hepatisation**: The septum transversum leads to a phase of sub-diaphragmatic congestion, forming the mesodermal anlage of the hepatic zone.

5. **Adrenalisation and Renalisation**: Later, during straightening, absorption spaces create the adrenalisation and renalisation zones.

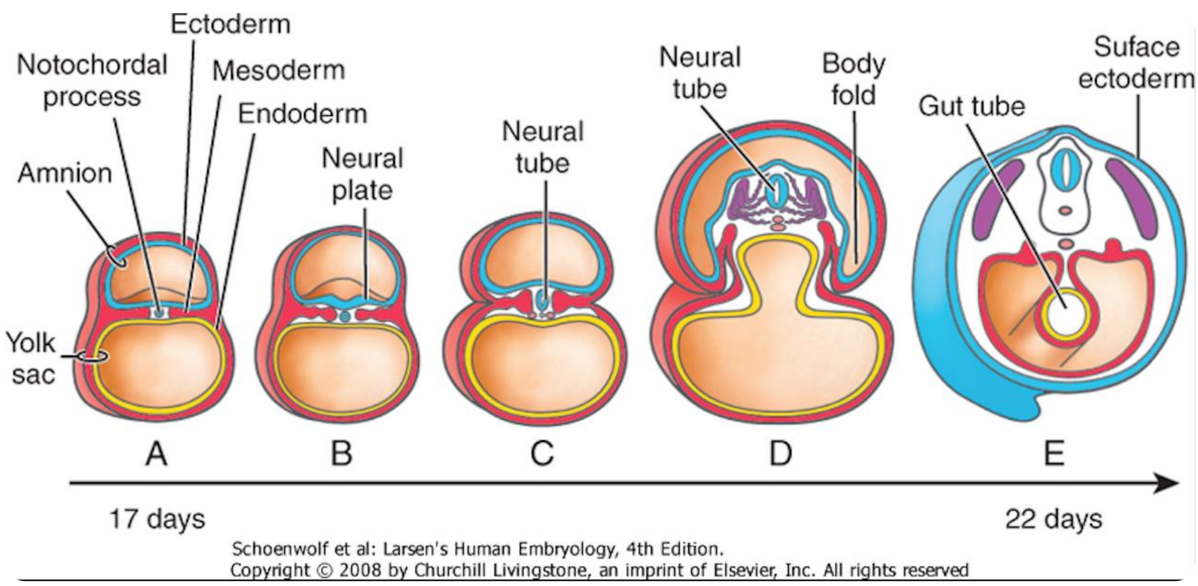


FIG. — Early embryonic development

THE LIVER AND CONGESTION

Congestion represents the continuous flow of information from the **embryonic pedicle**. The liver is initially formed only by the waste and exudates of the embryo. A global accumulation of these exudates creates the first impulse of congestion.

The continuous flow of nutritive information from the embryonic pedicle leads to a more significant phase of congestion under the diaphragm, impelling the development of the liver.

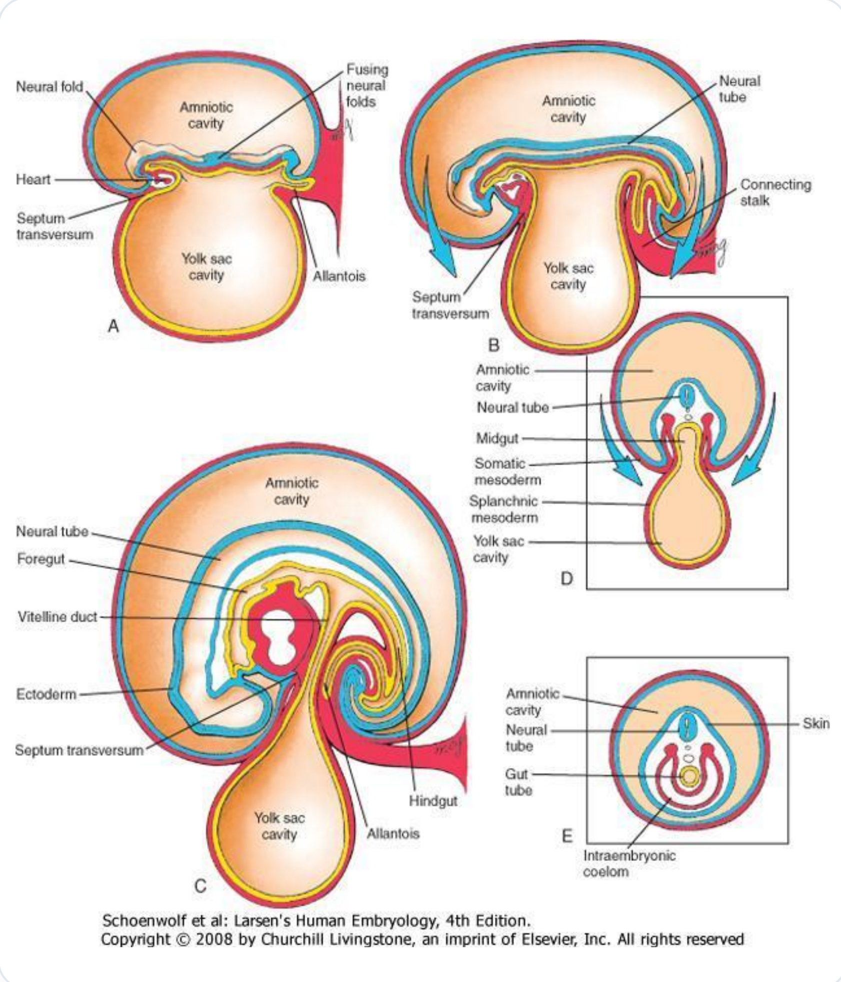


FIG. — Lateral embryonic folding

The liver is composed of **mesoderm** (for its vascular plan) and **endoderm** (for its digestive plan). The meeting of these two tissues forms an island. This primitive sub-diaphragmatic congestion is the first vascular impulse.

VASCULAR DEVELOPMENT

The embryonic pedicle and vitelline systems provide trophic information from the mother via the placenta.

Initially, **primitive aortic vessels** are distributed homogeneously. Then, the embryo segments via the venous system, through a process of assimilation and dissimilation.

Explanatory diagram

FIG. — Explanatory diagram

A venous network, composed of the **cardinal**, **subcardinal**, and **supracardinal veins**, forms a vast central vascular network (vena cava, portal veins, etc.).

At the aortic level, the flexion movement of the embryo fuses the two primitive aortas into one.

Two large vascular networks develop almost simultaneously:

- An aortic network.
- A venous network.

It is essential to understand that initially, the embryo coils around these **primitive mesodermal vascular axes**.

THE CRANIO-SACRAL-STERNAL COMPLEX

This mechanism is fundamental for **occipital** and **sacral** development. The heart is a central point around which everything coils. This story is repeated at the level of the sternum.

This is why we speak of **cranio-sacral-sternal treatment**. It is a movement of flexion, internal rotation, and adduction. It is a developmental movement towards the centre, resulting from differential growth.

The occipital system, the sacral system, and all lines of force reverberate at the level of the sternum. The sternum is a very **somato-emotional** zone.

EVOLUTION OF THE STERNUM AND DIAPHRAGM

Initially, there are two **sternal bars** that meet at the manubrium. This junction forms the **angle of Louis**, which appears when the mediastinum reaches equilibrium.

Later, the **xiphoid process** forms. It expresses the equilibrium of the peritoneum when the abdominal cavity is completely integrated and the intestine has completed its rotation.

The xiphoid process therefore symbolises the equilibrium of the peritoneum and the entire history of its development.

A point of equilibrium for the mediastinum is often located around **T3-T4**, as these vertebrae share a common mesentery with the aorta and part of the digestive system. For the peritoneum, **T12** is the axis of rotation of the digestive system.

The body is a complex knit. In **biodynamics**, we work on the eight fields. When the occiput descends and the sacrum descends (flexion), the sternum ascends. This **"whiplash"** movement is often associated with a loss of decision-making power.

If the sacrum ascends when it should descend in flexion, this creates a misalignment in the movement.

30. THE DIAPHRAGM

DURATION : 14:51

STUDY SHEET

COURSE SUMMARY

This video explores the formation of the diaphragm through four essential embryonic structures: the septum transversum, the pleuroperitoneal membranes, the paraxial mesoblast, and the oesophageal mesenchyme. Key concepts include the dynamics of embryonic movement, the functional connection between the diaphragm and the pericardium, and the importance of embryonic flexion for the organisation of internal structures. The interaction between these elements is crucial for cardiac rhythmicity and for the growth of neighbouring organs such as the lungs.

30. THE DIAPHRAGM

The process of **embryonic flexion** is fundamental. It not only organises the establishment of the joints but also initiates a phase of embryonic delimitation. This process involves four main embryonic structures.

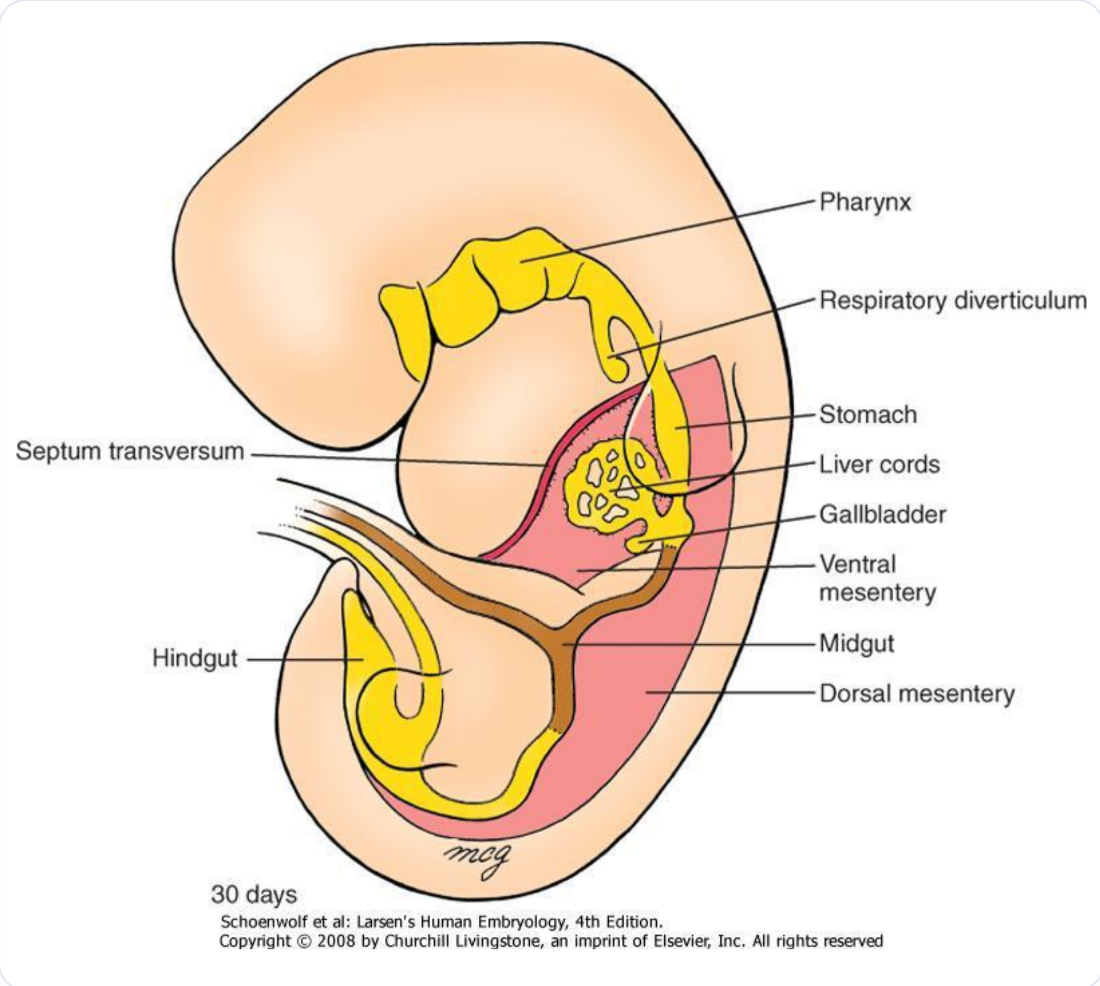


FIG. — Digestive primordia 30 days

THE FOUR EMBRYONIC STRUCTURES OF THE DIAPHRAGM

The four embryonic structures that contribute to the formation of the diaphragm are:

- **The septum transversum**
- **The pleuroperitoneal membranes**
- **The para-axial mesoblast** of the trunk wall
- **The oesophageal mesenchyme**

ORIGIN OF THE PERICARDIUM AND SEPTUM TRANSVERSUM

Beyond the precordial plate, in the cephalic region, the **mesenchymal** cells of the embryonic disc give rise to the pericardium and the septum transversum. It is important to note that your heart is already developing above your head at this stage. This is due to the embryonic flexion movement, which brings these structures into place very early.

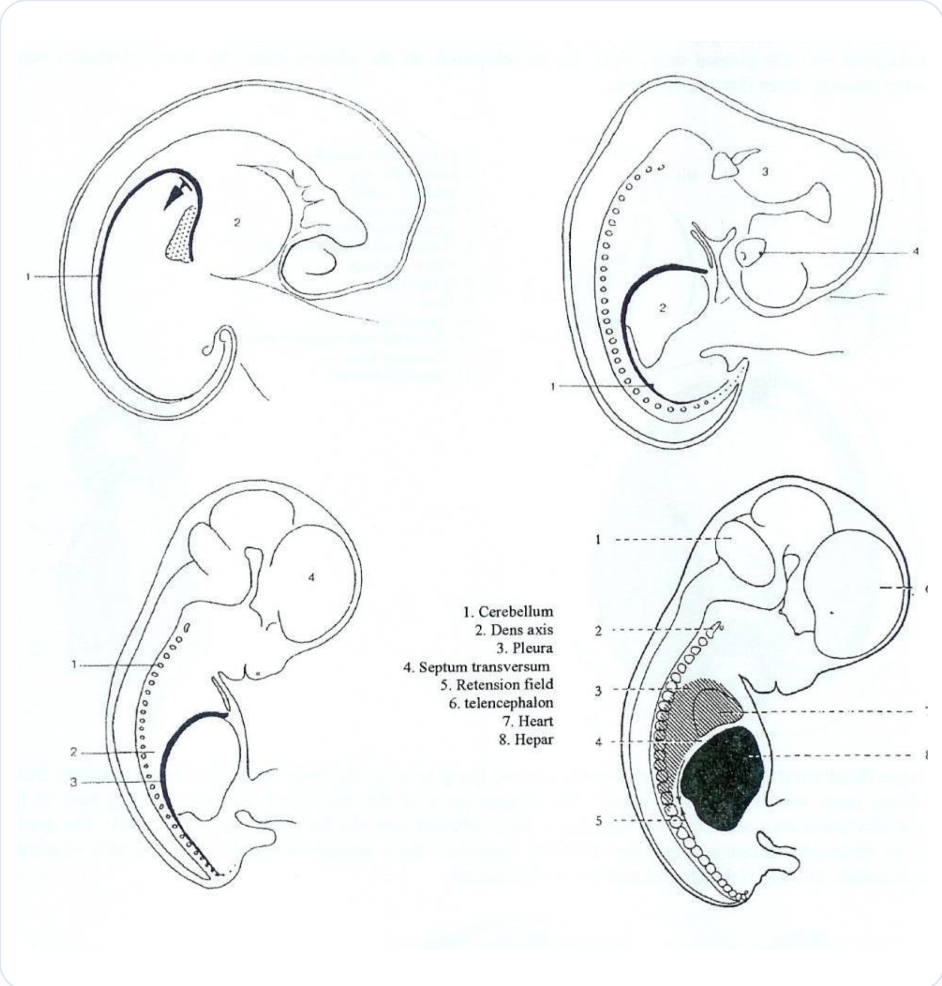


FIG. — Human embryonic development

THE DYNAMICS OF EMBRYONIC MOVEMENT

Contrary to classical embryology, which describes a "descent" of the heart, it is actually the rest of the body that "ascends" around the structures that are already in place. Similarly, for the kidneys, we speak of kidney ascension, but in reality, the kidney remains in place and it is the rest that descends.

THE DIAPHRAGM AND THE PERICARDIUM: A FUNCTIONAL UNIT

The pericardium and the diaphragm form an inseparable **functional unit**. The rhythmicity of the heart is supported by the pericardium. The latter does not merely contain the heart; it actively participates in its function. The closed oscillation system of the pericardium is essential for supporting cardiac rhythmicity.

This functional unit is completely connected between the heart and the diaphragm. They are not two separate entities, but rather structures originating from the same cellular source.

INTEGRATION OF MESENCHYMAL CELLS

The flexion of the embryo, induced by the **notochord**, leads to a specific movement at the level of the neural tube. This neural groove movement organises the lateral structure into small vessels, bringing nutrients but limiting growth. The surrounding ectoderm rolls up, and it is in this movement that mesenchymal cells integrate to form the future pericardial tissue and the diaphragmatic openings.

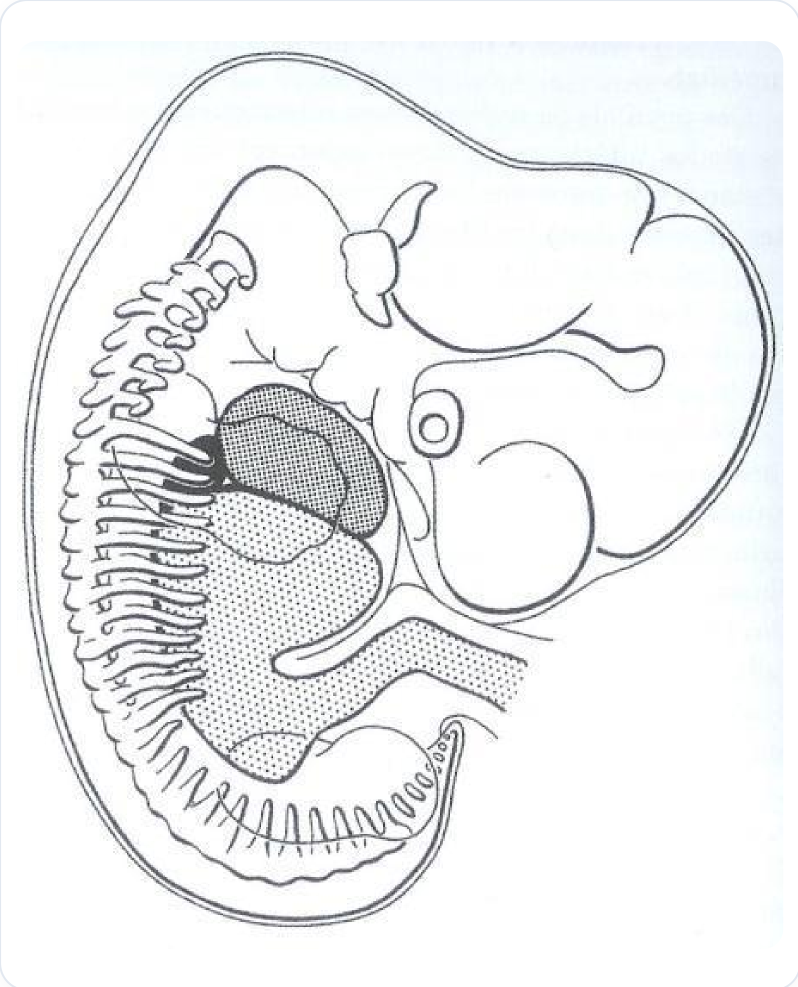


FIG. — Embryo 4 weeks

EVOLUTION OF THE YOLK SAC

As the embryo grows, the **yolk sac** ceases its growth and becomes a mere remnant. It inserts with the vitelline vessels at what is called the umbilical connection. The yolk sac integrates into the pedicle to form the umbilical cord. The pericardium integrates into the septum transversum, whose origin is beyond the precordial plate.

GROWTH OF THE SEPTUM TRANSVERSUM

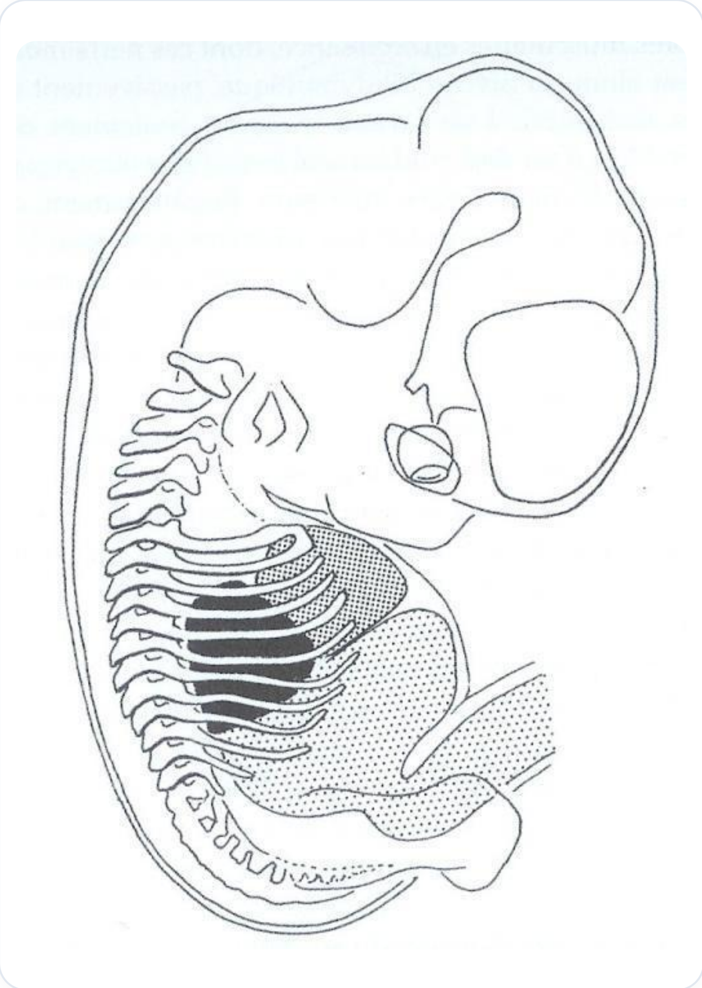


FIG. — Embryo, internal organs

The septum transversum, an important tissue, undergoes significant growth and transformation. In the fourth week, it is initially positioned. In the fifth week, it horizontalises and then begins its "descent". However, this "descent" is actually a consequence of the growth of the rest of the embryo.

THE ORIGIN OF THE PHRENIC NERVE AND THE MOVEMENT OF THE DIAPHRAGM

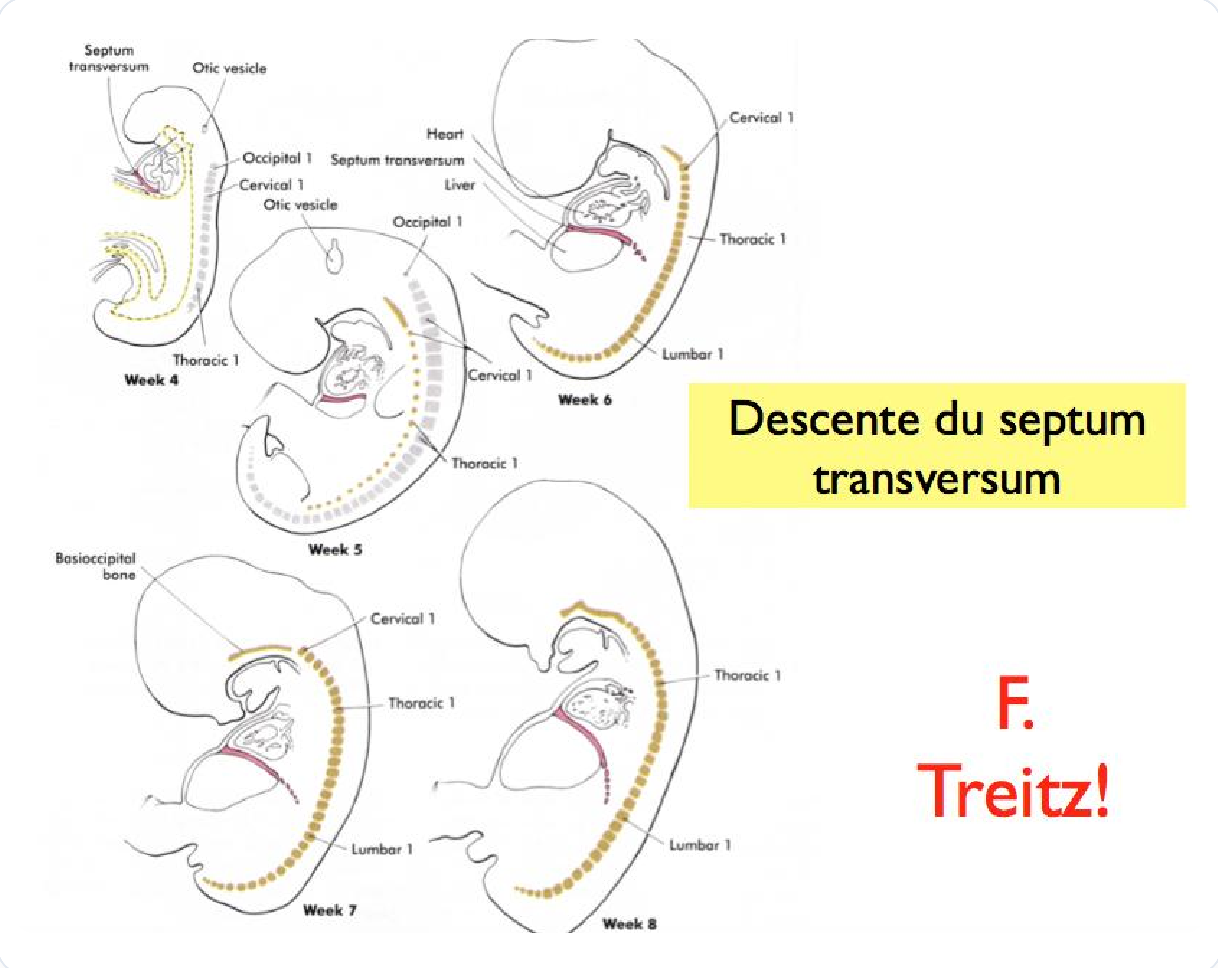


FIG. — Transverse septum descent

The origin of the septum transversum, when the embryo is 2 mm long, is at the very high cervical level (C3, C4), which is precisely the origin of the **phrenic nerve**. The diaphragm eventually ends up at the lumbar level. The idea of "descent" is false in biodynamics. The diaphragm settles, positions itself, and it is the enormous growth of the surrounding structures that modifies it.

CREATION OF SPACES AND ATTRACTION OF THE LUNGS

The growth of the posterior part of the embryo creates an absorption space. This space opens and attracts the lungs, which are a suction field. The lungs are, in a way, drawn into this forming space. The diaphragmatic origin is very high, at the level of the second and third cervical vertebrae.

THE STRAIGHTENING OF THE EMBRYO AND THE POSITIONING OF THE DIAPHRAGM

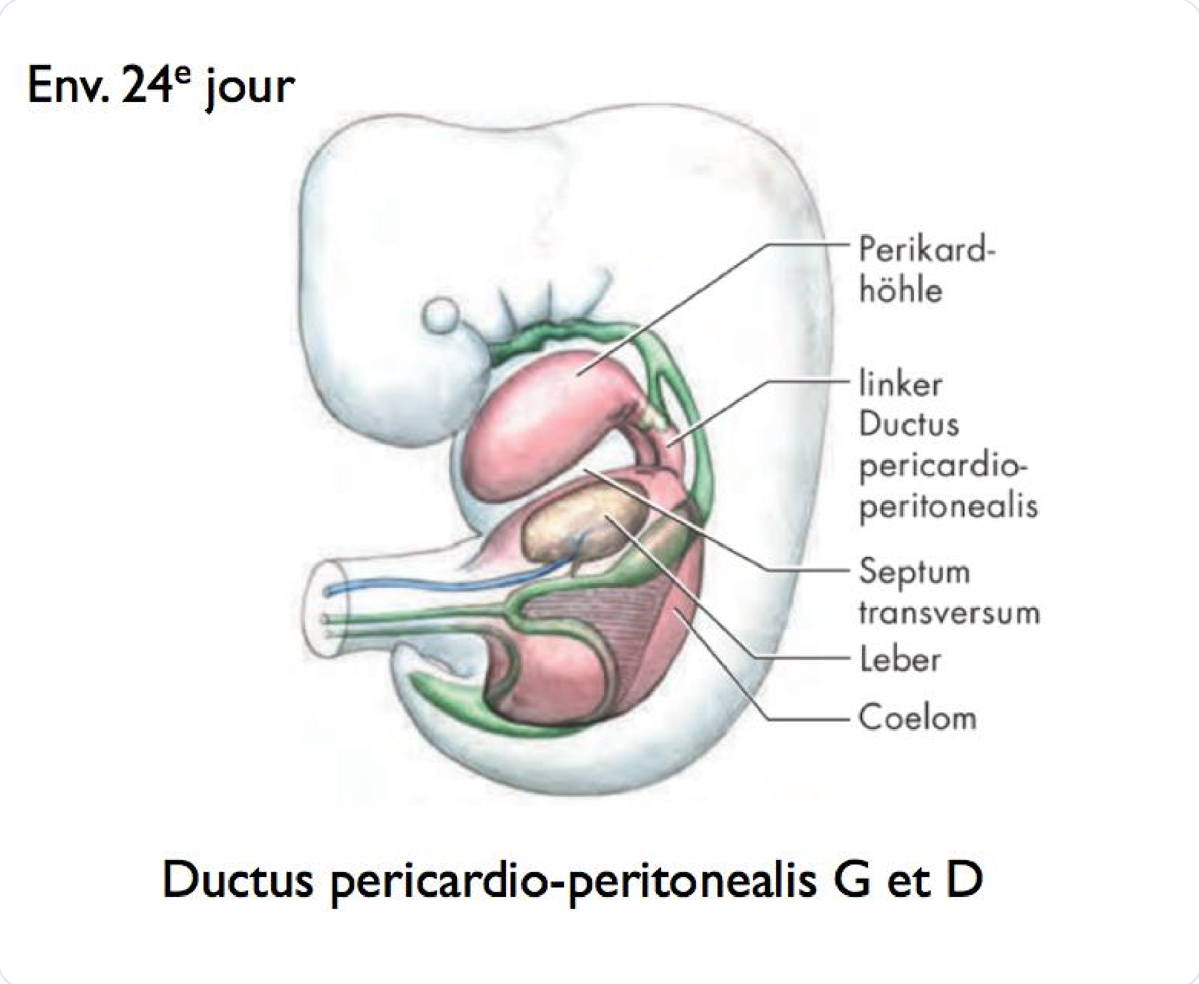


FIG. — Ductus pericardio-peritonealis

The phrenic nerve is not "pulled" but rather "accompanied" by this growth. The positioning of the diaphragm is linked to excessive growth. At a certain point, the embryo stops flexing and straightens up. This straightening is a matter of fluids on the lateral sclerotome, with the ascent of the telencephalon, the neural tube, and the descent of the digestive system. This movement is crucial for integration and the creation of spaces.

THE DIFFERENT TISSUES OF THE DIAPHRAGM

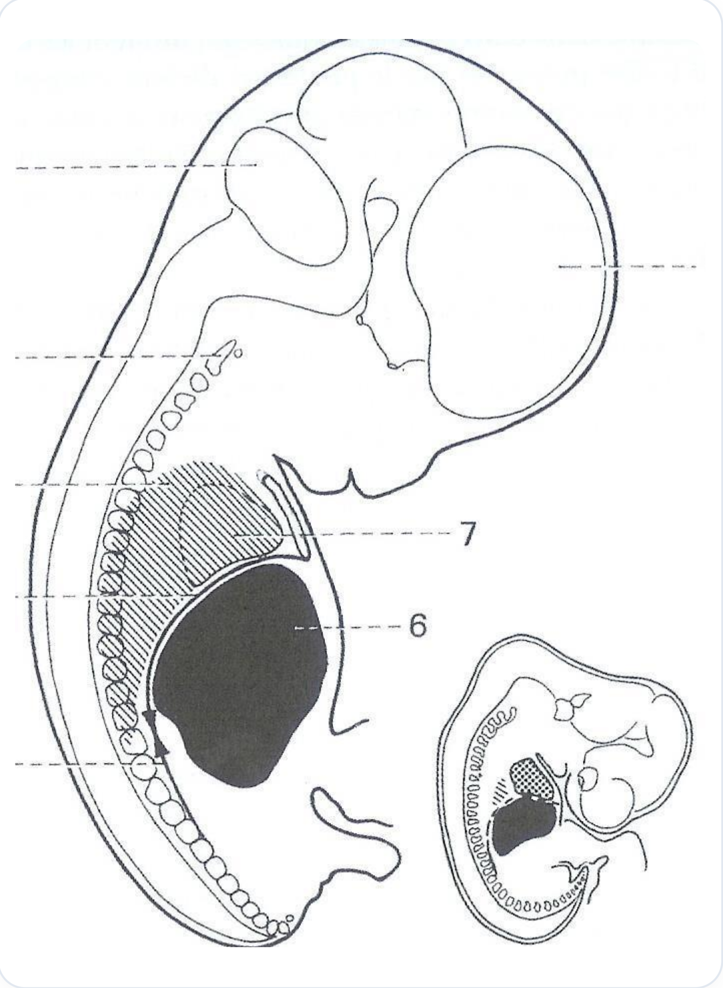


FIG. — Human embryo 6 weeks

On a section, one can observe the different tissues of the diaphragm:

- **Septum transversum**
- **Pleuroperitoneal membranes**
- **Oesophageal mesentery**
- **Para-axial membranes**

Working on a diaphragm is not limited to these spaces, but also to the release of the aorta, the vena cava, the oesophagus, and especially the **oesophageal junction**.

THE IMPORTANCE OF THE OESOPHAGEAL JUNCTION

The oesophageal junction is a very important area for releasing a diaphragm. Oesophageal and peptic disorders are common and often linked to a blockage in this area. It is essential to release the peri-oesophageal space and associated muscles. This small space can block the entire digestive system, suspended from the carotid foramina, the pre-jugular system, and the pterygoid attachments at the base of the skull.

FUNCTIONAL CONTINUITY AND THE LIGAMENT OF TREITZ

There is a functional continuity with a small muscle, the **ligament of Treitz** (or suspensory muscle of the duodenum), which is vital for the opening of sphincters and the organisation of the metabolic and digestive system. This muscle contains two types of fibres, muscular and non-muscular, offering different information. It can open or close the air zones.

CONNECTIONS AND DERIVATIONS OF THE SEPTUM TRANSVERSUM

The **falciform ligament** and the **greater omentum** are elements derived from the septum transversum. There is therefore a continuity from the mesenchymal cells, via the pericardial system and the septum transversum, and from the latter derive the falciform ligament and the greater omentum.

The greater omentum is an integral part of diaphragmatic function. It ascends towards the bronchial area, orientates towards the spleen, then descends above the colon to form a coalescence of four layers, creating a large **mesangiolar** omentum (lymphatic B, T and M). Its motility is related to the diaphragmatic system.

RELATIONS WITH THE ARCUATE LIGAMENTS

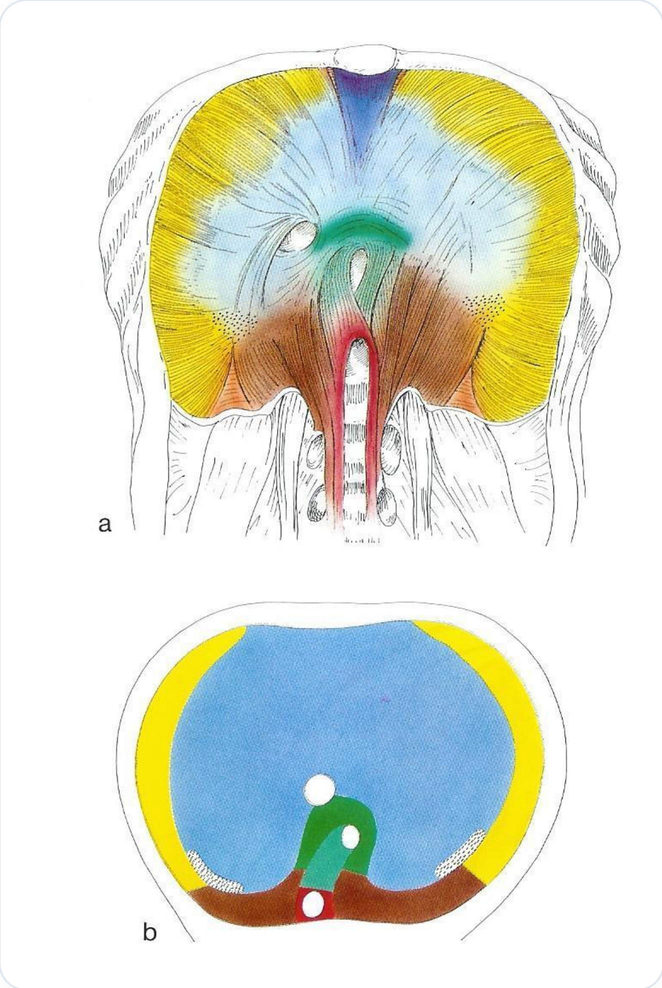


FIG. — Diaphragm skeletal muscle

The **arcuate ligaments** play a crucial role in the diaphragm's relationship with surrounding structures:

- The medial arcuate ligament attaches to the psoas arch.
- The lateral arcuate ligament attaches to the quadratus lumborum.
- The median arcuate ligament attaches to the aortic hiatus.

This continuity extends to the ligament of Treitz. The diaphragm continues indirectly through the common masses, the quadratus lumborum, and the psoas muscles. This is a fascial continuity of diaphragmatic organisation.

THE DIAPHRAGM: AN ESSENTIAL REGULATOR

An oesophageal issue can disrupt the movement of a knee, for example, by affecting overall mechanics. Each diaphragm must be considered in relation to the others. The three main diaphragms are:

1. **The cervical diaphragm**

2. **The thoracic diaphragm**

3. **The pelvic diaphragm**

An imbalance in one disrupts the other. The thoracic diaphragm is a pressure, tension, hydraulic, and thermal regulator.

MULTIPLE FUNCTIONS OF THE DIAPHRAGM

The diaphragm is a **lymphatic pump**, essential for the digestive, vascular, and lymphatic systems. Its relationship with the oesophagus, the cardiac system (especially the cardiac relationship with "phoepactis"), and the greater omentum is fundamental. Posteriorly, it is related to the quadratus lumborum and the psoas muscles. This continuity extends from the heart to the oesophagus, up to the base of the skull, in connection with the craniosacral axis.

THE COSTODIAPHRAGMATIC RECESSES

The **costodiaphragmatic recesses** are accumulation sacs for exudates and waste, whether transperitoneal or transpleural. During an infection such as bronchitis, microbes and mucus are eliminated and can accumulate in these sacs. This is why, after an infection, the lung bases can be blocked by exudates. These accumulations require space to be managed by the body.

THE CELLULAR AND TISSUE UNITY OF THE BODY

The human body is a **cellular and tissue unit**. Its defence and integration reactions can manifest at any level thanks to the tissue continuity of the splanchnopleure and somatopleure, as we have seen with the diaphragm.

FIG. — Thoracic abdominal sagittal section

31. DIAPHRAGM PRACTICE - 3 TECHNIQUES

DURATION : 16:09   STUDY SHEET

COURSE SUMMARY

This video presents two techniques for diaphragm release, essential for improving respiratory function and the general condition of patients, particularly children. The first technique focuses on the oesophageal and diaphragmatic junction, utilising spiralling movements and tissue listening to relieve pressure in the diaphragm. The detailed practical protocol includes steps for neck assessment, release, and mobilisation, aiming to restore freedom of movement. The second technique targets the omental bursa and oesophageal insertions, highlighting the importance of the interaction between the ribs and the diaphragm. These approaches, rooted in biodynamic practice, aim for the patient's overall well-being.

DIAPHRAGM PRACTICE - 3 TECHNIQUES

INTRODUCTION TO DIAPHRAGMATIC TECHNIQUES

We will explore two main techniques. The first is a **mechanical** approach to release the diaphragmatic space. The second is a **global** approach to the diaphragm, particularly effective in children and babies, to restore **optimal expansion**. This technique falls within a **biodynamic** practice.

TECHNIQUE 1: RELEASE OF THE OESOPHAGEAL AND DIAPHRAGMATIC JUNCTION

We will work specifically on the **junction** between the oesophagus and the diaphragm. This region is rich in **nerve networks**, particularly with the trunk of the **vagus nerve**. The objective is to gently release this area, which will have the effect of unloading the entire surrounding vagal area.

KEY ANATOMY

- We will observe the right and left diaphragmatic portions.
- The **pleura** is also present here.
- There is an anatomical remnant, formerly a vein, called the **vein of Treitz and Lehmer**.

- A small muscle inserts onto the diaphragm and peritoneum, forming the **peri-oesophageal** space. This space has a function of adaptability and rebalancing in relation to the diaphragmatic junction.
- This structure extends with the **muscle of Treitz**, which is the suspensory ligament of the duodenum.

THE BODY'S SPIRALLING MOVEMENT

The body is often organised in **spirals**, a powerful movement found everywhere. The embryo itself develops according to spiralling movements. We use spirals in energetic work to target precise points. Here, we will apply this principle of spiralling movement.

PRACTICAL PROTOCOL

1. **Location:** Position yourself at the junction, where you can access the highest part of the stomach.

2. **Preliminary Neck Test:**

- Turn the head to the left and feel the sensation.

- Return to the centre, then turn the head to the right.

- Return to the centre, then turn the head to the left again. Note if the rotation is limited. Very often, this diaphragmatic area prevents good neck rotation.

3. **Positioning and Release:**

- Position yourself and let the person fall back towards you.

- Ask them to relax completely. Support the person.

- Maintain a point of support with your thumb.

4. **Descent and Tissue Listening:**

- Start to descend gently, deeply. Feel for the tissues.

- Listen to the tissues. Observe the person's breathing. On exhalation, descend and meet a slight resistance, a certain density.

5. **Stabilisation and Progressive Mobilisation:**

- Position yourself gently, taking a good posterior support point, straight, to be stable.

- Check that the person does not feel any pain.

- Descend further, leaving a small margin due to clothing.

- Ask the person to sit up very gently. You can feel the "hop" come up with you.

- Repeat the movement: gentle ascent, then release. Gain ground.

- Sometimes, feel a "brake" and search a little further, ascending and descending again.

6. **Posterior Hold and Sensation:**

- Then take the posterior part between your two hands.

- Ask the person to relax completely. Check for the absence of excessive pain.

- A sensation may be felt, then "opens at the gallbladder level".

7. **Post-Technique Verification:**

- Seek posterior support and return.

- Ask the person to turn their head to the left again to check mobility.

TECHNIQUE 2: RELEASE OF THE LESSER SAC AND STOMACH

This technique focuses on the lesser sac and oesophageal insertions. Do not forget the importance of the **ribs** in relation to the diaphragm, where there is a significant concentration.

PRACTICAL PROTOCOL

1. **Location:** Place yourself where the "hole" is made.

2. **Hand Positioning:** Place your hand flat towards the junction, at the level of the 7th, 8th, 9th ribs.

3. **Sliding and Release:** Slide your hand underneath and lay it flat. Then, come with the other hand.

4. **Leverage:** Release and create a small lever. This area is an important crossroads of tensions.

5. **Effects:**

- This will release the entire stomach-cardia junction.

- This will relax the muscular area of Treitz and associated structures.

- This acts on this vital relationship.

GLOBAL SENSATION AND RETROGASTRIC POUCH

- Focus on a more global sensation. Touch the whole body.
- The **retrogastric pouch** is very important to mobilise. Ulcers are often posterior, and this pouch needs mobility for good peristalsis and function. It must be free.
- If it is not well free, it is not released in relation to the diaphragm.

ESSENTIAL ANATOMICAL LINKS

- The oesophagus and stomach are the first organs in relation to the base of the skull.

- They have a relationship with the **posterior mesogastrium**, the horn, and the left main bronchi. You can feel the warmth rise, then descend.

BIODYNAMIC APPROACH AND EMBRYONIC DEVELOPMENT

I am now exploring another level: the strength of **embryonic development**, support, power. I observe the respiration and perceive a fascial work that aims to create more space between the diaphragm and the posterior pulmonary area.

THE PERITONEAL PLANE

- On the peritoneal plane, the **left colic flexure** is the first fixed point of the global development of the peritoneum. It is the first point of support.
- Focus on this point of support, then on the point of mobility.
- The most mobile point is the **caecum**, the last to be put in place. Assess how it is.
- Between your two hands, you have the entire axis of rotation of the colon frame. The diaphragm works on these structures.
- In this space, you have all the ligaments, such as the **falciform ligament**.

THERAPEUTIC CHOICE

The practitioner can choose what to work on, but often the patient's body guides this choice.

- For example, a tension on the falciform ligament, or on its leaflet.
- Sometimes, there is the expression of a "gentle anger", as if there was a lack of posterior support, or a need for release here.
- This work is done in synchronicity with the tentorium cerebelli, the falx cerebri, at the level of the **crista galli**. We will work on the sinus.

EXPRESSION OF ANGER IN BIODYNAMICS

Anger can be expressed as an inability at a given moment to express oneself. This can date back a long time, linked to what could not be said. In biodynamics, I grasp the protocol by placing myself on the globality. I leave the thoracic respiration and enter the space, leaving it open. The body then guides me to points of support. "Flashes" may appear, indicating tension.

THE KEY TO THE BIODYNAMIC APPROACH (INITIATORY)

- Dive into the water (metaphorically).
- Listen to the movement.
- Feel the movement.
- Seek the **balance** (the point of equilibrium).
- Once you have found the balance point, listen to the silence behind it.

When the cells "respond" and open to reception, it is a cellular **"ignition"**.

PRECAUTIONS AND HAND POSITIONING

When approaching the diaphragm, it is about giving space to the diaphragm.

MISTAKES TO AVOID

- **Pressing on the head:** You must have an equitable sensation between your hands. A major mistake is to press on the person's head, which is very uncomfortable in the long run.
- **Poor posture:** Have good support on your elbows.

POSITIONING TIPS

Imagine you are eating garlic and taking up the right space. It is crucial to find the right space.

MEDITATIVE GUIDANCE AND THERAPIST'S STATE

In this practice, I will guide you into a **meditative** state.

- Position yourself just in the right position.
- As a receiver, try to define your feeling of the position.

SENSATION OF TOUCH

The touch is very light, with a lot of warmth. It is global; one has the impression that it extends to the feet. At the same time, there is this sensation that I am grasping the **"micro-crystalline field"**: all these cells, this small cellular network with their receptors. It is an enormous field.

THE GLYCOCALYX FIELD

The largest concentration of information in a **glycocalyx** field is found at the level of the **second duodenum** – the most informed part of the intestine. This field is present everywhere on the body, like a global field that encompasses everything. This allows for deep contact.

32. GUIDED PRACTICE: DIAPHRAGM IN BIODYNAMICS

DURATION : 07:05 STUDY SHEET

COURSE SUMMARY

This video offers a guided practice focused on the perception of the diaphragm in biodynamics. Participants are invited to connect with their environment, their breath, and the ambient stillness, thereby creating an experience of deep relaxation and presence. By focusing on the natural movement of breathing and visualising stillness as a point of support, therapists learn to foster a healing process through awareness and tranquillity. Key concepts include the breath of life, bodily harmony, and the importance of feeling grounded and connected to the Earth, conducive to inner balance and overall well-being.

GUIDED PRACTICE: DIAPHRAGM IN BIODYNAMICS

Focus and slightly adjust your position if necessary.

Make contact with the **environment** around you. Immerse yourself in this perception, as if your fingers are extending to encompass your entire body.

Observe your **breathing**. Become aware of the **breath of life** that animates you.

Keep your eyes open and feel the **stillness** present in the room. Perceive this stillness.

The breath of life, breathing, and stillness are present. When these three elements manifest, they take care of you.

The breath of life is this **vital force**, the life you hold in your hands. Breathing is what **moves**, it inhales and exhales.

There is also the space around you, this stillness. Allow yourself to be encompassed by a **therapeutic process** and let yourself heal.

Breathing is there. You can feel the **diaphragm** moving. Do not try to influence it.

Breathing has its own fulcrum, like a door opening and closing, again and again.

Try to perceive this **tranquillity**. Breathing has its fulcrum, a hinge that opens and becomes a still point.

The door opens, the door closes. You are gradually moving towards the sensation of this fulcrum.

Remain aware of the breath of life. It is the living body you hold in your hand, this power of life it contains.

Feel the space around you, behind you, above you, and your contact with the **Earth**.

The different zones A, B, C, D open around you. Breathing is like a **pendulum**.

With a fixed point above, there is **dynamic stillness**. Immobility. This tranquillity, this immobility, this point, merge.

Imagine you are in an **ocean of tranquillity**. You will feel warmth now.

It's as if you have the ability to extend your perception to the horizon. You move into zone D.

You are sitting in this ocean of tranquillity. There is the breath of life and stillness.

Then, there is the **midline**. From the midline to the horizon, a wave emerges.

Like a wave coming from the horizon towards the midline, bringing you back into perfect time.

Let this wave come. For therapists, remain well-grounded in your feet.

The body's warmth increases, creating **homogeneity**.

When all these elements harmonise, you can feel breathing like the wind moving through the air.

You experience a sensation of **wholeness**, homogeneity, calm, and well-being. You can withdraw and share.

33. DAY 3 MEDITATION

DURATION : 13:42 STUDY SHEET

COURSE SUMMARY

In this meditation session, the focus is on bodily preparation and vital connection through breathing exercises. You will learn to master alternate nostril breathing to open the sinuses and practice deep breathing techniques that promote energy release. The Mula Bandha exercise will help you centre your energy in the central channel and establish a deep connection with the earth, allowing you to access your inner 'mountain'. By exploring the importance of the belly as a vital centre, the meditation encourages an awareness of our interdependence with the universe, thus fostering a state of inner harmony.

33. DAY 3 MEDITATION: BODY PREPARATION AND VITAL CONNECTION

ENTERING AQUATIC CONSCIOUSNESS: ALTERNATE NOSTRIL BREATHING

We are going to prepare the body to enter an **aquatic consciousness**. Once the mind quietens, we will connect to a movement that your body knows very well.

Settle in comfortably.

BREATHING EXERCISE

1. Place your **left hand** under your armpit. Your right hand comes in front of you.

2. With your **index finger**, place it in the middle, at the root of your nose.

3. Block your **right nostril** with your thumb.

4. Gently inhale through your **left nostril**. Be aware that you are inhaling more than just a little oxygen; it is an **energy**, a **light**, a **pranic vibration**. Inhale slowly to open the connection towards the sinuses.

5. Block your left nostril and exhale through your right nostril.

6. Inhale again through your right nostril, block, and exhale through your left nostril.

7. Inhale again and try to feel a slight coolness spreading towards your sinuses.

Stop the exercise and let both your hands rest.

DEEP BREATHING AND RELEASE

Open your chest wide by placing your hands under your armpits. This position promotes the opening of the **lungs**, especially during the autumn period.

- Inhale deeply, letting the air descend into the body.
- Exhale through the mouth, blowing out all that is "useless me", the waste of the body and thought.

Inhale again through the nose.

OPENING THE CENTRAL CHANNEL

We will now open the **central channel**, the channel of your life.

- Become aware of the level of the **perineum**.
- Inhale, then contract the anus and perineum (**Mula Bandha**).
- Bring up the air and the diaphragm, raising this energy to the top of the head.
- Tuck in your chin, close your glottis, and try to concentrate this energy in your central channel. Push the **prana** upwards.

Again: inhale, close the perineum and anus, bring the energy up by pushing upwards, become aware of the top of your head, tuck in your chin, close the glottis.

One last time, inhale. Then, exhale all the air, hold, raise the diaphragm. Release.

This activates and slightly warms the body.

CONNECTION TO EARTH AND CENTRE

Sit up straight, in your body. Establish good contact with the **earth**.

Imagine yourself as a **mountain**, a triangle with four faces. You are sitting in your mountain. There is a solid base, a summit, and then the emotional weather of your life.

Despite the storms, the mountain remains stable, immutable. Try to access that part of yourself that is never touched, that **deep nature**.

THE VITAL CENTRE AND INTERDEPENDENCE

Each of us has an important **centre**. We can live without knowing it exists, but we cannot live without it existing.

This knowledge must be acquired, experienced. It is about becoming aware that you are connected to the **universe**, that you are part of this unity, of this space. We are all **interdependent**.

Understanding interdependence is understanding the meaning of **responsibility**. It is becoming aware of your roots, that we are not alone, but that we are part of a whole. Once you understand this, you can find **harmony** within yourself, a harmony that transcends borders.

THE ABDOMEN AS THE FIRST CENTRE

The first truly important centre to discover is to return to the **abdomen**, like a child.

- Observe a baby: it is at its own centre, it breathes and lives through its abdomen.
- Very quickly, the adult moves away from this vital centre.

Become aware of your abdomen. Every time you inhale, the universe exhales and inhales with you. Try to return to this cycle.

FROM ABDOMEN TO HEART, THEN TO HEAD

The abdomen is the gateway to then go into the **heart**, and finally into the **head**.

- Become aware of **love**, of the capacity to be loved.
- Realise that you are happy when you are happy.

Always stay connected to this vital centre. Simply take a few conscious breaths with your abdomen.

Become aware of the navel, not as a selfish centre, but as a connection with **universality**. Keep this connection today, stay centred here. It is a basic connection, a vital plane.

BRAIN AND VITALITY: ESSENTIAL SYNCHRONISATION

The **brain**, in its development, is dependent on this vitality. It needs this energy, it needs the heart to develop.

What happens here (at the centre) happens there (in the brain).

Important fluid currents – **mesenteric**, **splenic**, **vitelline** or **umbilical** – integrate into the heart, which redistributes the blood via the carotid and vertebral system (four major pathways) to return to the basilar trunk and distribute everything into the head.

With each pulsation, we find the pulsation of the brain. The heart and brain are in complete **functional unity**. You cannot separate them, even in your practice.

INTERDEPENDENCE IN OSTEOPATHY

If you do cranial work, you will see that the deep development of these structures depends on this **interdependence**.

The notion of interdependence is crucial. When I do cranial work, I don't just do cranial work. When I do visceral work, I don't just do visceral work. I do **osteopathy**.

- For a baby, I very rarely do cranial work. I put them back in their abdomen, I rebalance them in their abdomen. All the power is there.
- Sometimes, it is necessary to guide, but it is essential to understand how the cranium develops.

These two days, we will study the **embryonic development** of the cranium.

34. DAY 2 QUESTIONS

DURATION : 08:08 STUDY SHEET

COURSE SUMMARY

This video explores the importance of cerebral movement in cranial osteopathy, highlighting how the body, as an intelligent entity, develops autonomously. Key concepts include the link between the viscerocranium, heart, diaphragm, and immune system, showing that dysfunctions at one level can affect other systems. The interconnection between the diaphragm and organic function is emphasised, while also stressing the importance of working on diaphragm dynamics to improve patients' quality of life. Practical exercises and theoretical notions support learning on how to balance interconnected bodily structures.

34. DAY 2 QUESTIONS: BIODYNAMIC EMBRYOLOGY AND THE HUMAN BODY

BRAIN MOVEMENT: FOUNDATION OF CRANIAL OSTEOPATHY

The **movement of brain development** is fundamental and provides profound meaning in **cranial osteopathy**. The human body is intrinsically intelligent and knows the pathways of its own development.

Our role is not to re-educate the brain, but to learn to listen to it. It will instruct and guide us.

We will learn to "do the **brain's Tai Chi**", observing that the **base of the skull** is intimately connected to the brain. The **ventricular system** and the **deep flow of the brain** depend on its dynamics.

Similarly, the entire **membranous system** (tension membranes, falx cerebri, tentorium cerebelli, falx cerebelli, dural membranes, parietal, visceral) is dependent on cerebral dynamics. This compartmentalisation of the brain, from the dense structure of the cranial base to the vault, is a reflection of these dynamics.

THE VISCEROCRANIUM AND ORGANIC CONNECTIONS

We will study the **viscerocranium**, starting with the **septum transversum**. The latter is a tissue integration including the heart, the diaphragm, and extending with the **falciform ligament**.

This structure reveals an early functional integration between the heart and the liver. It continues via the **foramen of Winslow** to form the **greater omentum**.

IMPLICIT RELATIONSHIPS

This means there is a fundamental relationship between:

- **The heart**
- **The diaphragm**
- **The brain**
- **The immune system**

Destabilisation at the level of the heart or the base of the skull can therefore impact the immune system. The superior immune system, **Waldeyer-Heiner's ring** (tonsils), is an example of this.

IMPACT ON THE ENT SPHERE

Dysfunction at this level can have significant repercussions. For example, many children suffering from **ENT** problems present, at a molecular level, the same **immunoglobulin** (type 1) as that found at the caecal level.

Destabilisation of the **peritoneum** can thus affect the entire ENT sphere. In practice, rebalancing the ENT sphere often requires rebalancing the peritoneum, particularly at the caecal level, which contains lymphatic tissues identical to those of the tonsils. It is interesting to note that in the past, tonsillectomies and appendectomies were often performed synchronously.

THE OESOPHAGUS AND ITS INTERCONNECTIONS

The oesophageal relationship is also crucial. The **oesophageal mesenchyme** can disrupt the function of the diaphragm.

The stomach and oesophagus are suspended up to the base of the skull via the **carotid foramina**, the **pterygoid plexuses**, and the **pterygoids**. Any destabilisation of the base of the skull can therefore alter **cardio-tuberosity** function and, consequently, diaphragmatic movement, which is essential to our physiology.

THE MESOBLASTIC WALL AND THE DIAPHRAGM

The **lateral mesoblastic wall** (lateral wall) is also very important. The **costal grid** is entirely dependent on the proper functioning of the diaphragm.

Consequently, the entire dorsal area is connected to these dynamics. Furthermore, we have noted a junction with the **kidney**, the **psoas**, and the **duodenum**, highlighting the overall integrity of the system.

THE IMPORTANCE OF THE DIAPHRAGM

We must learn to work the diaphragm with more subtlety, whether remotely or not. By expanding this diaphragm, the patient's quality of life is significantly improved.

When exercising the diaphragm, an essential reference point is its **fulcrum**. It is possible to suspend this fulcrum in the dynamic stillness of space.

It's like a door that opens and closes, with a hinge, a **"Hitchpoint"** or fulcrum. This starting point may seem mobile, but on deeper investigation, it becomes a fixed point, your reference, but always suspended in the dynamic stillness of the system.

THE HEART, THE PERITONEUM, AND EMBRYONIC MOVEMENTS

A question arises concerning the **meso-oesophagus** and the **vena cava**. The latter is very close to the heart, almost on the same axis, and not far from the liver.

The first great movement of the **notochord** and the second, circular, movement of the **amniotic cavity** involve a visceral centre that we identify with the heart and the peritoneum.

THE IMAGE OF COILING

The image is one of **coiling** around the **cardio-pleuro-peritoneal** system, like a joint. The **neural tube** develops strongly forwards, drawing the heart, but it also grows backwards, bringing the allantois through an anterior restriction field.

This means there is a closing movement of the body by the **embryonic pedicle**. It is at this moment that the allantois and the heart integrate, becoming a fulcrum.

THE ORIGIN OF THE SEPTUM TRANSVERSUM

At this stage, two elements are crucial: the heart pushing on the southern part and the upper part of the **vitelline vesicle**. This push orientates and organises the **septum transversum**.

Its tissue origin was already present (the cells were there), but this encounter between the brain and the heart, and the heart and the vitelline vesicle, provides the impetus for the creation of the septum transversum. The heart thus acts as a joint between the brain and the diaphragm.

35. THE NEURAL PLATE

DURATION : 21:32 STUDY SHEET

COURSE SUMMARY

This video delves into the fascinating dynamics of the neural plate and the notochordal process, essential for embryonic formation. By studying how the notochord influences the curvature of the neural plate and the formation of the neural groove, students will discover the fluidic and vascular interactions that shape the development of the neural tube. The importance of elements such as mesenchyme and molecular compounds like Sonic Hedge-Hog (SHH) and Bone Morphogenetic Proteins (BMP) will be discussed, highlighting their role in dorsalisation and cellular induction. In practice, osteopaths will learn to use this knowledge to rebalance the nervous system and release tensions, while also measuring the impact of the environment on embryonic development.

THE NEURAL PLATE

INTRODUCTION TO THE NOTOCHORDAL PROCESS

When examining the **sacrum**, one observes the **notochordal process**, a descending wave that acts as a fulcrum. This notochordal process directly induces the impulse on the **neural plate**, orienting and organising the force applied to it. In just three days, the growth of the embryo manifests, with a descent of **Hensen's node** relative to overall growth.

FORMATION OF THE NEURAL GROOVE AND VASCULAR SYSTEM

The beginning of the **neural groove** appears, accompanied by the **pharyngeal** and **cloacal** membranes. The notochordal process plays a crucial role, allowing the neural plate to curve and form a groove. This process occurs synchronously with a lateral and fluid reorganisation.

At the time of neural groove formation, a fluid organisation occurs, marked by the appearance of **fine aortic capillaries**. For this lateral fluid process to be effective, several components must be present:

- A trajectory
- Particles

- Vacuolations
- Metabolic concentrations

These elements, present in the **mesenchyme**, create a lateral vascular trajectory. Thus, the tube gradually closes while establishing a vascular trajectory in the **mesoderm**.

INTEGRATION OF AMNIOTIC FLUID AND NEURAL TUBE CLOSURE

Simultaneously, the **amniotic cavity** above contains fluid. As the neural plate begins to form a tube and a groove, the amniotic fluid is integrated into the neural tube. This process is accompanied by the integration of the **external coelom** into the **internal coelom**.

When the neural tube closes completely, with the closure of the **anterior and posterior neuropores**, the integration of the coelom is total. This gives rise to a visceral box, an abdominal cavity, and a cerebral cavity, all formed simultaneously. This moment, on the **28th day**, is marked by remarkable synchronicity.

SYNCHRONICITY OF EVENTS AND THE ROLE OF THE NOTOCHORD

This moment of closure is crucial. With the integration of the coelom and the aortas, tissue forms around the notochord, in metabolic retention fields. This closure is constant, and for osteopaths, it is interesting to observe these phenomena occurring at a distance but at the same time.

The precursor of this groove is the notochord. Its closure leads to basic molecular levels that slow down growth. The **Sonic Hedge-Hog (SHH)** phenomenon intervenes, influencing the dorsalisation of the system, accompanied by **bone morphogenetic proteins (BMP)**.

MOLECULAR PHENOMENA AND GENETIC INFLUENCES

Neural Crest Cells also play a role, manifesting morphological, genetic, and molecular dynamics. The observed movements, described by Deschmidt and others, reveal phenomena of inhibition or induction on molecular bases of a genetic type.

The notochord influences the cells associated with it, causing an induction phenomenon. This information will be crucial for neurulation, influencing the **ventral midline** and causing the **dorsal midline**. The growth of the dorsal midline re-influences the anterior midline, thus creating a ventral, dorsal, and anterior **midline**.

MOVEMENT, ENVIRONMENT, AND THERAPY

Morphogenetic poles, such as types 4 and 7, participate in dorsalisation. These molecular phenomena are supported by scientific bases, explaining the organised programme at this level. Movement reorients development, highlighting that the environment, rather than genes, is the driving force of this process.

From a practical perspective, it is essential to understand the environment in which the patient evolves. The notochordal axis serves as a reference for releasing the fluid side of the brain, hydrating, and eliminating energetically dried-up areas.

AMNIOTIC FLUID AND ITS SIGNIFICANCE

We will focus on the first moment of neural tube closure. This involves a rebalancing of the nervous system, both sympathetic and parasympathetic. It is crucial to remember three elements: the **ectodermal** tissue and its epithelial cells, the movement induced by the notochord, and the lateral growth that forms the groove.

We begin to distinguish the cells that will form the neural tube, encompassing the spinal canal and the brain, while laterally, other cells will form the epidermal epithelial tissue.

NEURAL CREST CELLS

Another important structure appears: the anlage of the **neural crest** tissue. Neural crest cells, in red, blue, and black, not only contribute to neural tube closure but also form the peripheral nervous system. They interact with the mesoderm to create ecto-mesodermal tissues, such as the face and skull bones.

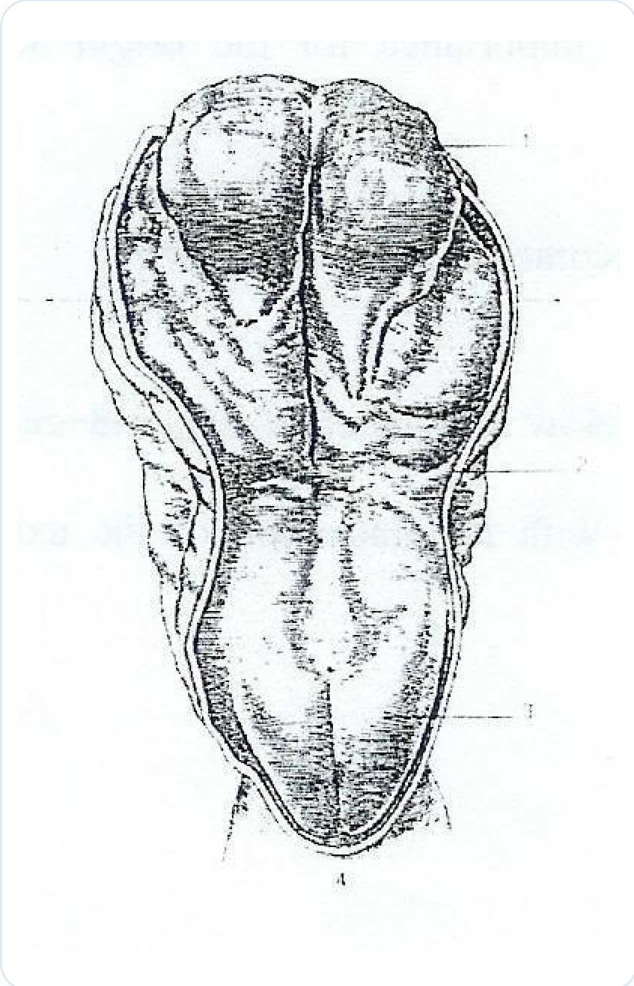


FIG. — Embryonic neural crest

The migrating neural crest draws in the rhombencephalic neural crests to form the facial system. The cephalic brain divides into **prosencephalon**, **mesencephalon**, and **rhombencephalon**, each creating a neural crest.

ROLE OF INFORMATION AND MOVEMENT

This tissue is informed and transmits information, connecting the belly to the periphery and vice versa. Sensors, such as those for atmospheric pressure and light, provide essential information. The ectoderm, upon receiving information from the notochord, does not grow randomly.

The tissue cells differentiate into three types:

- Those forming the neural tube.
- Those forming the neural crests.
- Those forming the epidermal epiblast.

These tissues are fundamental for the development of the peripheral nervous system, mesenteric plexuses, and other bodily structures.

LINKS BETWEEN AMNIOTIC FLUID AND DEVELOPMENT

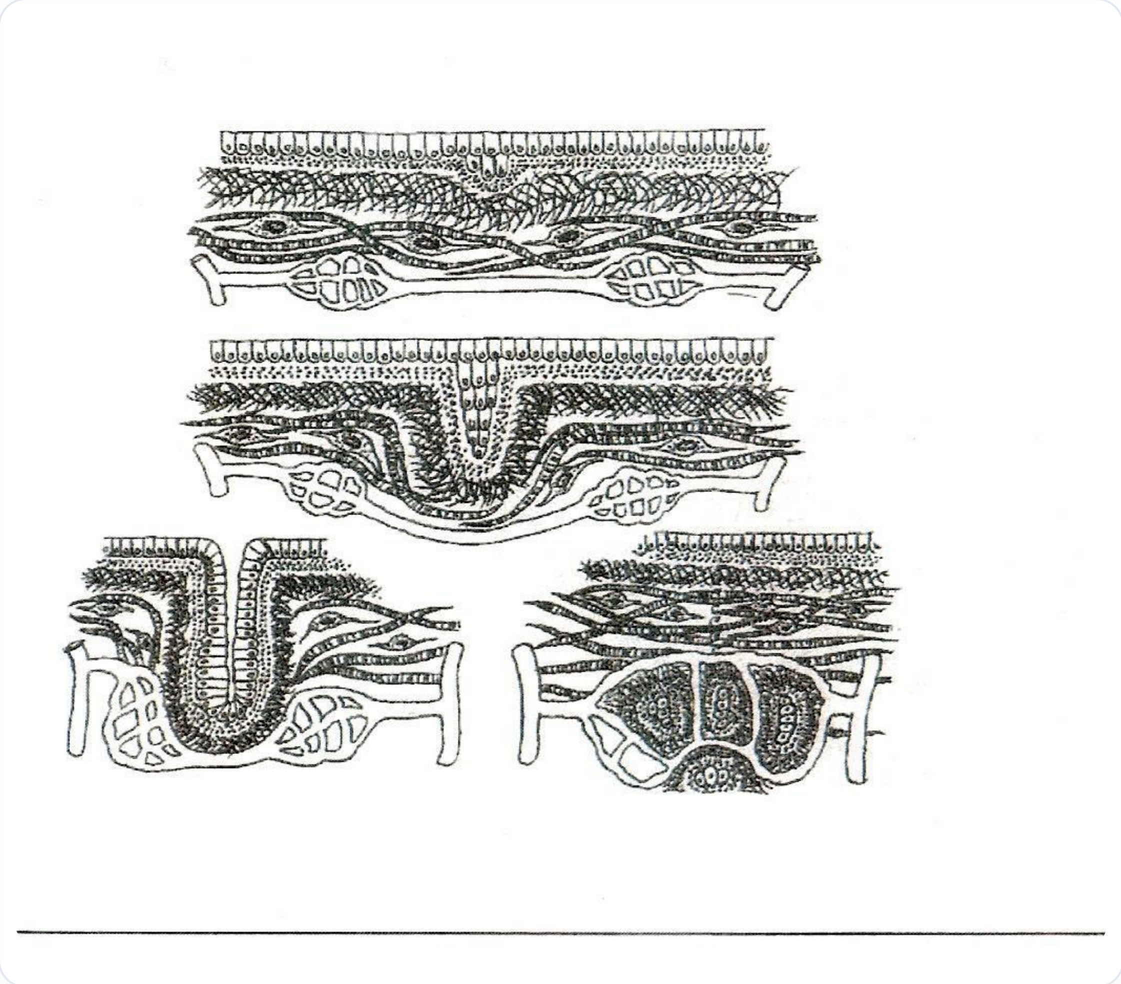


FIG. — Intestinal villi development

Amniotic fluid, rich in **building proteins**, plays a crucial role. This fluid, present inside and outside, is linked to the quality of the skin and cerebrospinal fluid. The latter is essential for body movement and cleansing.

Embryonic development involves several cavities, including the **blastocoel** and the **amniotic cavity**. The child ingests amniotic fluid, integrating information about its development. After the 28th day, once the neural tube is closed, the child begins to eliminate its first pro-urine, creating a link with the aortic system.

NEURAL TUBE CLOSURE: ANATOMY AND IMPLICATIONS

Sonic Hedgehog (SHH) is crucial at the genetic level, as are **Cell Adhesion Molecules (CAM)**, which play a role in cell adhesion. Neural tube closure occurs at a time when the cephalic part is more developed than the caudal part.

This closure is decisive for the development of the embryo, connecting the prosencephalon, mesencephalon, and rhombencephalon. The notochord, below, influences the development of the nervous system.

THERAPEUTIC TOOL: REBALANCING THE SYSTEM

We will integrate all metabolic fields for the ectomesenchyme, taking into account morphological **building blocks** and growth factors. By visualising the closing movement of the neural tube, we can use this first encounter between the left and right sides as a therapeutic tool to rebalance the system.

Once the anterior neuropore is closed, it becomes the **lamina terminalis**, marking a significant impulse in bodily development.

36. THE NEURAL PLATE IN PRACTICE

DURATION : 18:22

STUDY SHEET

COURSE SUMMARY

This video addresses the fundamental concept of bodily rebalancing in biodynamic osteopathy, with an emphasis on the body's homogeneity. Practitioners learn to position themselves correctly to allow for a subtle perception of imbalances in the body, while remaining centred. The video explores how practitioners can identify different densities and sensations in the physical, etheric, astral, and mental systems, while facilitating the patient's self-organisation through a gentle and respectful approach. Practical rebalancing and observation techniques are detailed, offering students valuable tools to improve their practice and sensitivity as therapists.

36. THE PRACTICAL NEURAL PLATE: REBALANCING AND PERCEPTION

I. REBALANCING THROUGH HOMOGENEITY

We will seek the **important plane of closure** and carry out **rebalancing** work. The body is approached as a unit, divided into four parts.

The finger position is above, to the left and right, at the occipital level. The hands are positioned above, below, to the left and right, sub-occipitally.

A well-functioning and integrated body exhibits **homogeneity**. This means there is no noticeable difference between top and bottom, left and right. Restoring this homogeneity is a fundamental technique in **biodynamics**.

This is one of the first things to learn: to restore and perform a **rebalancing**. This work is particularly important for the **nervous system**, as it organises a large part of our body, including the vegetative, emotional, and limbic systems.

What we are doing here is somewhat the "starter" for this system.

II. POSITIONING AND PERCEPTION

When we position ourselves, we adopt a **precise** and very gentle **posture**, working mainly with two fingers. We settle gently, without pressure. Immediately, the **micro-crystalline field** appears.

When I speak of the micro-crystalline field, I am referring to the perception of the body as a unit, a complete individual. While maintaining this global vision, I position myself at this level and observe.

A. THE IMPORTANCE OF PRACTITIONER CENTRING

I must first be very **centred** myself. If I am off-centre, my perception will be altered. That is why a period of centring is essential.

Meditation is important for assessing my own state. There are days when one is more in form and others when one is less so. One must recognise one's state so as not to influence the person being treated. You must not induce, but only receive.

B. OBSERVING IMBALANCES

As you position yourself, you may feel **densities** or sensations. For example:

- One side more swollen.
- Excessive upward force, typical of people who are very "in their heads".
- An empty space.
- An overflow.
- Dissociations or sensations of spiralling, twisting.
- Areas of excessive heat.

We will begin by simply placing our hands and feeling the homogeneity or lack thereof in the system. Then, we can delve deeper, by questioning the different levels of the body.

C. SEEKING LEVELS OF AWARENESS

We will seek the different **bodies** or types of information combinations:

- The physical body
- The etheric body
- The astral (emotional) body
- The mental body

I find myself with different combinations, potentially 64 according to the **I Ching**. I look for the spaces that appear. Does my body react physically? If not, is it an etheric problem? Astral (purely emotional)? Mental?

The body can give information through a word, a gesture, a look, or even release patterns or past memories. This is the time to listen.

III. PRACTICAL APPLICATION OF REBALANCING

For example, I feel a greater **density** in the lower left, a slight descent of the left system. I let my sensation appear without delving into it.

I step back, I observe, and I focus on health. I just provide points of support. I am on a structure, a kind of remnant, a scar. I have a good **fulcrum** with my body, a good point of support.

I am in the globality. I do not focus on one section, but on the whole. I adjust myself so that my points of support, though gentle, are the most effective, and that I myself am in balance.

A. TAKING HOLD AND RESPECT

I receive the imbalance and allow the body to organise itself, observing its respiration. I hold the head like a bowl of water filled to the brim. This leads me to a deep respect for what I touch.

You are touching a powerful system, and you are just giving it support, very gently. Already, the state has changed. What I am doing is giving space, well-being.

I remain focused on health. The body rebalances gently. You will feel that it is both pleasant and powerful at a deeper level. This is the first layer of work here, I observe.

B. THE EVOLUTION OF PERCEPTION

There is always this descending respiration, its exhalation. I am halfway there, I wait. Then, I enter another rhythm. I open myself to the space around the person.

I perceive the person in the room, as if I saw myself with them in the universe. There is a change of rhythm at the heart level. It still seeks the image, then it stabilises. Another level appears.

I see it, but I remain focused on health until a state of homogeneity is reached, until this base is activated.

C. THE PATIENT'S SENSATIONS

At first, I had the impression of waves passing through one side, then the other of the body. After perceiving the globality, it was more like jerky movements, going right, left.

In the end, I felt that my breathing allowed me to relax. Often, at the end of a treatment, after some work, this rebalancing is performed. It is a key point for balancing the nervous system, as that is where it forms.

I position myself at this level, and I must have the same density as below. I place myself in this globality and observe what appears. For example, with her, it was a bit like that, a bit blocked. I stay, I do not intervene.

IV. TRUST IN HEALTH AND NON-INTERVENTION

How do you distinguish if you are on health or not? If you do not get drawn into the lesion and you remain present. The **fulcrum** of health is a fulcrum of presence.

It is being in the globality, in trust towards the whole, and knowing that health is what offers the possibility of **self-healing**. I invoke this principle, I let it act because it is more powerful than me.

At first, I placed myself in the globality, then I looked for another plane, I observed her respiration, and I went underneath. We did what we started yesterday. I placed myself in this perception, and another perception appeared. I again allowed the system to reorganise itself.

A. THE NATURAL CLARITY OF THE SYSTEM (ANALOGY OF CLEAR WATER)

Health is immutable. It's as if you put mud in clear water. You no longer see the clarity. But if you let the water settle, the clarity returns. The nature of the water has not changed.

You must rise into this nature. There is a part of you – it's a Buddhist principle – that possesses this clarity, this health, this power. I place myself at this level. This approach has a reorganising power.

The only thing I do is provide points of support so that the system seeks and reorganises itself. I trust these principles. My question is: do you trust this?

B. INITIATION TO LISTENING

This is an initiation to listening to this movement, this power. An initiation to receiving information and participating in it.

For now, we have only seen the body and the beginnings of the nervous system, with the first closure. That already seems like a lot. Finally, we arrive at birth and we redo all the birth work in osteopathy with this embryological vision.

It's no stranger than anything else. Fortunately, we don't consciously remember it! However, your unconscious understood very well and was uncomfortable. It reacted immediately: I breathed, I'm hungry, I'm cold. It's quite a violent experience.

V. THE UNCONSCIOUS AND RESISTANCE TO CHANGE

This makes a good introduction for the "teaser". A part of you has observed that all change is dangerous. Every time you are confronted with a change, even a beneficial one, a part of you reacts.

"We already know this benefit, we already know the parameters. I know your idea, go away. I'm not changing." And you are immediately brought back into your security patterns, into your "comfort zone", even if it is uncomfortable.

Sometimes, change is a good thing. There is a way to work with the unconscious, because we are confronted with it. Tissues do not lie. When we are on the tissues, we are confronted with our own **inner lies**.

It is important to discover these lies that we all carry and to dare to confront them. But that takes time.

A. THE UNSAID AND THE CONSTRUCTION OF SELF

Sometimes, things should not be said, it's not the right time. It's omission. They are not said at the moment, but perhaps later, or perhaps never. These **unsaid things** that we carry are the foundations upon which we build ourselves.

It's a subtle lie. We all have them. It's good to seek this **authenticity**, even if it's not always pleasant.

B. OBSTACLES TO MEDITATION AND PRESENCE

A part of you does not want to change. This is the first obstacle in meditation: laziness, or something else. There are different types of meditation.

I first did "floating" meditations that were very nice. Then I had the chance to meet a master who taught me meditation by starting with the basics: working the mind, facing its wandering.

Learning to return to one's object of attention, to train oneself in that. It's sometimes boring. Then, one begins to perceive a quality of **presence** that appears. Because when you know you are no longer there, you can return to your presence.

I offer you a meditation that brings a quality of presence, of attention, something that is good for the body, mind, and soul. It is interesting to see that studies on meditation show the activation of a part of our brain: the **prefrontal cortex**.

The prefrontal cortex has the ability to open us up, because we often operate in automatic patterns. It allows us to leave that and access another plane, another moment of the brain. It has been discovered that a specific area there activates empathy, but a just empathy, a compassion.

37. THE PRACTICAL BRAIN

DURATION : 13:08 STUDY SHEET

COURSE SUMMARY

This video, 'The Practical Brain', presents a detailed biodynamic approach centred on the brain and its expansion. The instructor teaches how to identify levels of expansion in the patient, exploring effective revitalisation strategies, such as working on the heart and digestive axes. This is followed by an exploration of cephalic and pontine flexion movements, with specific precautions to be taken during touch. By focusing on the sensation of expansion, practitioners will learn to accompany the movement of encephalisation, understanding the rhythm and synchronicity linked to biodynamic embryology. This session is both theoretical and practical, promising students an enriched understanding and developed skills to improve their practice.

37. THE PRACTICAL BRAIN: A BIODYNAMIC APPROACH

I. INITIAL POSITIONING AND SEEKING EXPANSION

I position myself above the patient, specifically at the level of the **insula**. My hands are placed here, my thumbs slightly crossing, and I ensure I have good support for my elbows and wrists on the table.

The **touch** is extremely gentle. I aim to feel the **expansion** of the field.

A. IDENTIFYING LACK OF EXPANSION

The very first thing to assess is the level of **expansion**. If the expansion is insufficient and I am not sufficiently "pushed back", it is a sign that something is amiss.

Often, one expects an immediate and strong expansion, but this is not always the case.

B. REVITALISATION STRATEGIES IN CASE OF INSUFFICIENT EXPANSION

If expansion is insufficient, several avenues can be explored to **revitalise** the system:

- **The heart**: Easing the heart is often a very effective approach.
- **Digestive axes**: Working on the digestive axes (mesenteric, mesaric) or the coeliac axis can bring the necessary expansion.

- **The sacrum and pedicles**: Moving up to the sacrum and pedicular levels can also help.

Generally, working with the heart or digestive axes yields quicker and more effective results.

II. THE FLEXION MOVEMENT AND ENCEPHALISATION

Once expansion is present, I feel an expansion followed by a **flexion** movement forwards. This is cephalic flexion, which brings the brain towards the heart.

Behind, I must perceive a **cervical flexion** movement. If I feel a slight deficit at this level, it indicates the need for a cervical or structural correction.

A. PONTINE FLEXION AND TELENCEPHALISATION

I then focus on my little fingers and seek **pontine flexion**. The movement descends.

Then, a new expansion appears. At this stage, I can open my hand. This is the beginning of the **telencephalisation** process, which is recovered from the occipital to the temporal level.

B. COMMON ERRORS AND PRECAUTIONS

Most errors occur when directly touching the brain. It is imperative not to touch the brain if one is not "under that frequency".

- I position myself above, without touching with my thumbs, maintaining a good distance.
- I create a very gentle **fulcrum** with my thumbs and a significant point of support.

C. THE SENSATION OF EXPANSION

Good expansion is felt as a plane of warmth and a gentle pushing sensation.

III. SEQUENTIAL MOVEMENT AND ACCOMPANIMENT

I perform a very gentle movement. This is not yet my "insular H-bridge". I make the flexion movement forwards. The movement descends well, but can slightly twist (for example, to the left). I very gently accompany this twist.

At this stage, a small correction may be necessary, potentially at the cardiac level. There may be a moment when the movement is not perfect, but that is not a problem.

A. DESCENT OF FLEXION AND PONTINE PLANE

I then move towards the descent of flexion, which the patient should feel distinctly. It descends without issue.

Once this is established, I move to the **pontine plane**. Here too, a small intervention may be required.

B. THE TELENCEPHALIC PLANE AND ASCENDING EXPANSION

After the pontine plane, I shift my attention to the **telencephalic plane**, the telencephalic anlage. The movement should come towards me. I observe an expansion.

I position myself on the **"Hitchman"**: the movement turns towards me, expands, and ascends.

C. THE GREAT MOVEMENT OF ENCEPHALISATION

The movement restarts, descends again, and expresses itself. I very gently descend into the great flexion, the great movement of **encephalisation**. I recover this movement at the occipital level.

It is crucial to accompany this movement, not to impose it. We wait for it to manifest, then we follow it. The movement is already present, like a "drawn track", with a particular rhythm. We only reproduce the existing **pattern**.

IV. UNDERSTANDING RHYTHM AND SYNCHRONICITY

This pattern is not reproducible in the sense of a periodically repeating cycle. It is a system that is constantly there, a **writing in the body**. This is the very essence of **biodynamic embryology**: the anlage of structures (ears, ocular system, optic placode, etc.) is established according to this pattern.

We will add elements to this understanding, but after grasping this guideline.

A. LINKING EMERGING STRUCTURES TO MOVEMENT

Understanding this movement allows us to see how the **ventricles** are established, how the **dura mater** follows, how the **cranium** forms. If something blocks (a level I do not yet know), I can rely on my knowledge of timing and synchronous phenomena.

For example, I can look for an **artery** or a specified electrical level. The patient will be able to feel what is happening.

B. THE IMPORTANCE OF CARDIAC DYNAMICS

It is essential to have a good understanding of the heart's **dynamics**. Sometimes, expansion is greater on one side (for example, on the left).

C. AVOIDING EXCESSIVE INTELLECTUALISATION

It is important not to over-intellectualise the sensation. Too much mental activity can "put too much electricity into the limbic system" and hinder subtle perception.

V. PRACTICAL CASE: ANALYSIS OF A SENSATION

During an assessment:

- Expansion is gentle, not complete. Flexion is good but lacks dynamisation.

- The rhythm is variable. The movement shifts more to a track on the left, with a slowing down.
- It is sometimes difficult to reach the **frontal lobe** in depth.

A. PROGRESSION OF MOVEMENT

Then, we move to the pontine system, with an ascent of the patient's tube. I become a point of support, and it pushes. Lateral expulsion, particularly on the right, starts stronger.

The patient "recovers the package below". After a slightly slow initial start, the movement catches up well.

B. THE PATIENT'S TESTIMONY

The patient describes a difficult start, as if they had to "get going", release a fixation on the left that prevented expansion. Initial flexion was difficult.

When a connection was established, the system "unblocked". This is comparable to learning to drive a **Formula 1** car: at first, one is clumsy, then with learning, it works.

C. KEYS TO ACCOMPANIMENT

- **Environment and space**: Allow breathing, shift the movement.
- **Progressiveness**: Enter progressively and connect to this "writing" of the body.
- **Centring of listening**: Centre one's listening at the brain level, and if it "doesn't work", observe elsewhere.

It is a question of **synchronicity**. One must "create a synchronism" to find the right word and work with this synchronicity.

38. THE BRAIN: THE VENTRICLES

DURATION : 14:52

STUDY SHEET

COURSE SUMMARY

In this video, we delve deeply into the brain's ventricular system, integrating key embryological concepts such as telencephalisation, ascensus, and hemispheric differentiation. You will learn how the growth of the lateral ventricles and other brain structures influences not only neurological development but also the dynamics of osteopathic therapy. The formation of the choroid plexuses, as well as the circulation of cerebrospinal fluid, will also be addressed, offering valuable insights into the interactions between the brain and the rest of the body, such as connections with the peritoneum. This understanding will enrich your clinical approach and stimulate your practice.

THE BRAIN: THE VENTRICLES

INTRODUCTION TO THE VENTRICULAR SYSTEM

We will integrate the **ventricular system** into the developmental stages already studied. Segmentation and the different structures are essential. A lateral view of the image offers a better perspective.

It is crucial to integrate this developmental movement. When examining a patient, keep this global perspective in mind. Even when working on the feet, unsuspected connections, particularly with the peritoneum, can be discovered.

PROCESS OF ASCENSUS AND HEMISPHERISATION

We distinguish the **midbrain**, the **highbrain**, and the spinal cord. The process of ascensus is clearly visible, accompanied by hemispherisation.

Do not forget the overall growth and repositioning of the brain, which resembles a cabbage. Everything happens within this process.

ROLE OF NEURO-PORES AND PROSENCEPHALON

The anterior **neuro-pores** are planes of dynamisation. The complete closure of the inner and outer neuro-pores triggers a dynamic of brain development.

Initially, we essentially have the **prosencephalon**, with the anterior neuro-pore, followed by the mesencephalon and the rhombencephalon. This is a structure containing fluid inside this tube and these vesicles.

TELENCEPHALISATION AND VENTRICLE FORMATION

The process of **telencephalisation** gives rise to the **lateral ventricles**. This is how the third, fourth, and lateral ventricles are formed.

- **Initial vesicles:** Initially, three small vesicles containing fluid.
- **Brain growth:** The brain begins to grow, and we will explore the inside of the ventricles.

Imagine you are at the level of the prosencephalon, which is flexed, so the entire system leans forward.

CLOSURE OF NEURO-PORES AND LATERAL EXPANSION

The closure of the anterior neuro-pore and the posterior neuro-pore occurs between the 26th and 28th day, approximately one lunar cycle.

At that time, growth occurs with a **lateral expansion** of the prosencephalon. From a simple ball, it becomes two balls.

FLUID MOVEMENT AND HORN FORMATION

The **cerebrospinal fluid** follows this movement. It moves forward, then the system moves backward, thus forming the horns of the lateral ventricles.

This movement follows the development of the prosencephalon.

INTERVENTRICULAR FORAMEN AND TELENCEPHALISATION

The **interventricular foramen** is a small expansion that occurs simultaneously with the process of telencephalisation, on both sides. It follows the complete movement, a fulcrum, then returns backward.

This is solely the developmental movement of the telencephalon. Sometimes, one hemisphere develops faster than the other.

PRACTICE AND BRAIN EXPLORATION

During a practice, we will revisit this dynamic to explore the depth of the brain, as if swimming with whales, without touching them.

ORIGIN OF THE PROSENCEPHALON AND EYES

One could say that initially, the prosencephalon is the **third ventricle**. The eye is an expansion from it. (We will discuss this later.)

The eye is made of nervous tissue, not fluid. Its origin is an expansion of the third ventricle.

The interthalamic adhesion connects the two thalami, with fluid around them.

CEREBROSPINAL FLUID COMPARTMENTS

There is not only the intra-encephalic compartment but also the extra-encephalic compartment, whose metabolic role is increasingly being discovered.

CHOROID PLEXUS AND FLUID FLOW

The **choroid plexuses** are areas of fluid production, present in each ventricle: the lateral ventricles, the third, and the fourth ventricle. (The third ventricle is the first to form in this chronology.)

The lateral ventricles are an expansion that follows the development of the telencephalon.

- **Foramen of Monro:** Located between the third ventricle and the lateral ventricles.

POSTERIOR FLUID CIRCULATION

After the third ventricle, the **aqueduct of Sylvius** leads to the fourth ventricle.

- **Foramen of Magendie:** A significant expansion towards the brain, a main opening to the extracephalic compartment.

PACCHIONIAN GRANULATIONS

The **Pacchionian granulations** are resorption sites located along the sinus.

KEY DEVELOPMENTAL MOVEMENTS

Key movements include **cephalic flexion**, **cervical curvature**, **pontine curvature**, and especially this great movement of hemispherisation and telencephalisation that reshapes the embryo.

- **Initial structures:** Prosencephalon, mesencephalon, rhombencephalon.

- The green part is particularly important for understanding this developmental dynamic.

It will be interesting to review the days of development later to correlate this with digestive or other phenomena.

CONNECTIONS AND FULCRUMS

This recalls the different fulcrums and the general affection due to the great cephalisation.

It is also interesting to remember the connections with the tip of the **notochord**, the **supraocciput**, the **basiocciput**, and the **presphenoid**.

THE ASCENSUS CEREBRALI AND THE FORMATION OF THE FACE

The **ascensus cerebrali** is the process of cerebralisation and the formation of the face.

- **Face development:** Initially, we observe the nasal placode, the mouth, and the heart in the small embryo.
- **Fulcrum:** The growth of the telencephalon occurs with a crucial fulcrum at the level of the future **crista galli** or the **nasion**.

Everything returns towards the median axis of the embryo during growth.

TELENCEPHALISATION AND ANTERIOR MEDIAN LINE

The process of telencephalisation has a direct action on the formation of the **anterior median line**.

We observe enormous growth and all the attraction that operates.

CEREBRALISATION AND PALATE STRUCTURES

The process of **cerebralisation** (ascensus cerebrali) will influence all areas of the palate, which we will explain next time. There is a very important fulcrum here.

Telencephalic development leads to the approximation of the palatine, the flexion of the choanae, and the posterior resultant. This is a simultaneous change between brain development backward, corticalisation, and ventricularisation.

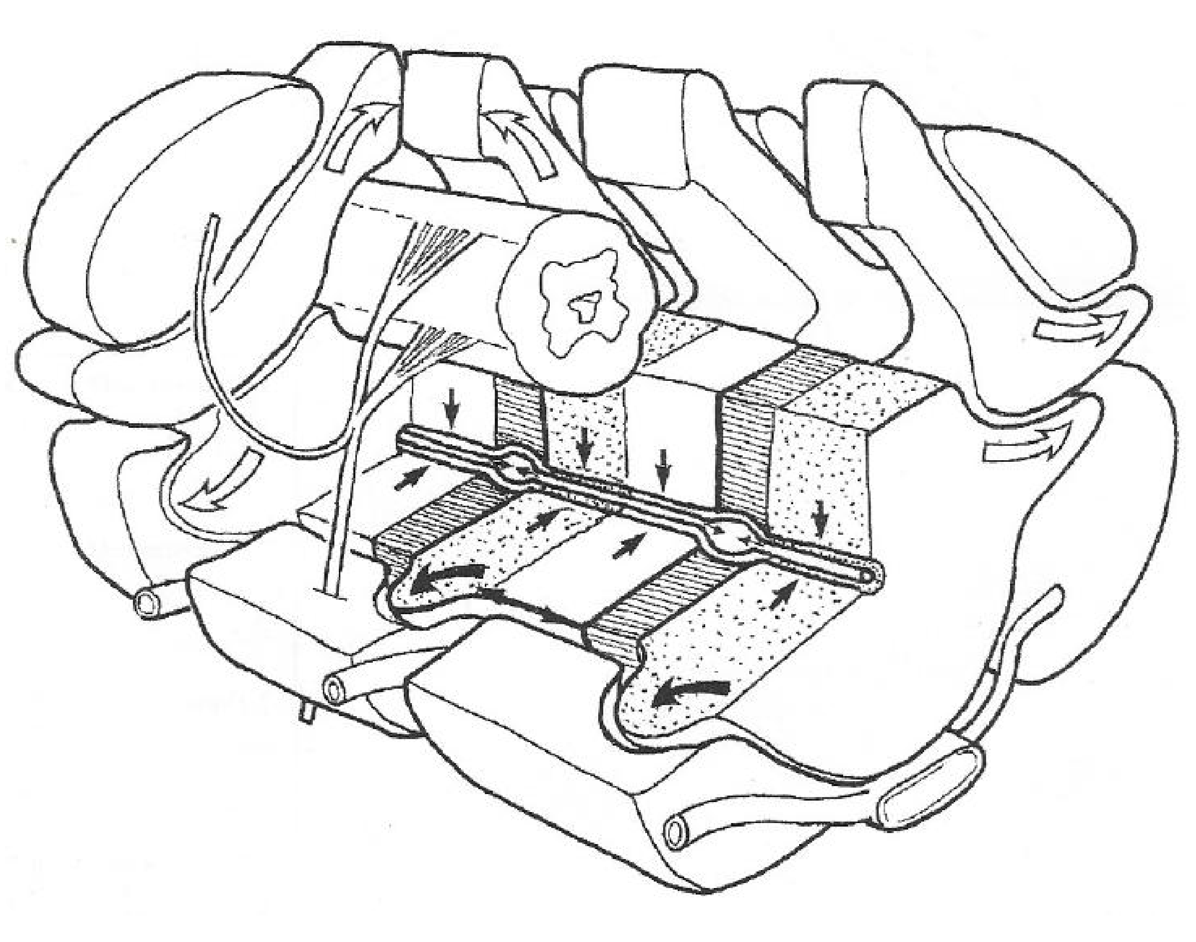


FIG. — Spinal CSF circulation

39. SUMMARY OF MOVEMENT

DURATION : 05:02

STUDY SHEET

COURSE SUMMARY

This video delves into the heart of embryonic movement, detailing the crucial role of the electrical and electromagnetic axis in establishing a differential growth rate. You will learn how the notochordal process initiates cerebralisation, influencing the development of essential structures such as the digestive tract and the cardiac system. The concepts of palpation and the interconnection between different embryonic layers are explored to better understand the dynamics between the heart, brain, and other organs. Enjoy this enriching synthesis that highlights the complex interactions that shape the embryo.

SUMMARY OF MOVEMENT

ESTABLISHMENT OF THE AXIS AND DIFFERENTIAL GROWTH

The initial process involves the establishment of an **electrical** and **electromagnetic axis**. This axis progressively organises a **differential growth rate**.

This growth rate acts as an **algorithm** integrated into the system, causing at a given moment the precise **"hola"** (wave) of the **notochord**.

NOTOCHORDAL AND MESODERMAL PROCESS

The **notochordal** process forms and simultaneously allows for the establishment of the **lateral mesodermal** process. At the same instant, **symmetry** begins to establish itself.

A **fulcrum** emerges at the base of the skull, precisely at the level of the future **sphenobasilar synchondrosis**. This fulcrum is supported by the developmental potential of the **sacrum**.

CEREBRALISATION AND CONCOMITANT DEVELOPMENTS

This development marks the beginning of **cerebralisation**. Cerebralisation induces **cardialisation**, which in turn leads to **diaphragmatisation**, then **hepatisation**.

Posterior growth causes **pneumatisation**, then posterior **adrenalisation**, and so on.

CEREBRAL EVOLUTION AND KEY POINTS

During the **cephalic growth** phase, the **foramen of Monro** opens laterally, clearing the **ventricular** system. This process accompanies cerebralisation and **corticalisation**.

A posterior **neurulation** reappears, relying on a key point called the **Insula**.

The fundamental point of stillness remains the tip of the notochord. However, in the brain, the **Insula** constitutes the fulcrum and the crucial tipping point for development.

ASCENDING AND DESCENDING MOVEMENTS

A process of **ascension** and a process of **descent** organise the entire development. These two movements are interdependent.

When cerebralisation occurs, synchronous events take place:

- The development of the **digestive tube**
- The development of the **cardiac tube**
- The development of the **pulmonary tube**

This ascent of the tube and this descent are crucial. The most important area that allows these phenomena is that of the **heart** and its relationship with the mother.

NOTOCHORD, NEURAL PLATE AND PRECARDIAC ZONE

The notochord induces the formation of the **neural plate** at the anterior of the embryo. Two **fluidic trajectories** are created.

These trajectories coil upwards to form the **precardiac zone**.

DEPTHS OF PALPATION AND HITCH POINTS

There are different **depths of palpation** to consider:

1. **Palpation of the Hitchpoint (fulcrum)**: Linked to brain development.

2. **Palpation of the brain**: In relation to its position and developmental dynamics.

3. **Pure development**: Concerning relationships with the heart, liver, and digestive system.

In terms of depth, several layers can be identified:

- **Superficial zone**: Contact with hair, then skin and bone.
- **First layer**: The **visceral dural layer**, just behind the bone.
- **Fluid layer** protecting the brain.
- **Cortical layer**: To feel its movement.

- **The nuclei**: Descending deeper.
- **The ventricular plane**: Mainly lateral.

40. INTRODUCTION TO BIODYNAMICS

DURATION : 00:42 STUDY SHEET

COURSE SUMMARY

This video introduces the concept of biodynamics, which goes beyond a simple definition by associating it with ontogenetic developmental movement. It explores the dynamics of living organisms, revealing the influence of a creative genius in the organisation of life. The teaching also offers a new perspective on the craniosacral system, considering it not only from an anatomical point of view, but also embryological and dynamic. Students and therapists will thus learn to integrate these dimensions into their medical practice.

INTRODUCTION TO BIODYNAMICS

The term **biodynamic** can be quickly and simplistically interpreted. Within the context of this course, I want you to associate it with **ontogenetic developmental movement**.

Biodynamics therefore represents the **dynamics of the living**. It illustrates how this living is organised by a **creative genius**, a kind of emergence of originality that is, ultimately, quite marvellous.

Furthermore, you now have an enriched perspective. The **cranio-sacral system**, which was initially perceived from a strictly anatomical angle, now reveals itself from an **embryological** and **dynamic** aspect.

41. CHARACTERISTICS OF THE DIFFERENT TISSUES

DURATION : 03:25 STUDY SHEET

COURSE SUMMARY

In this video, we delve into the characteristics of different bodily tissues in relation to tensegrity and homeostasis. Mr. Benjamin presents key concepts such as 'boundary tissue' and 'interior tissue', while analysing their respective roles within organ systems. You will discover how gastric disorders can be linked to imbalances in the lesser pelvis and the importance of rebalancing the pelvic fascia for digestive health.

Furthermore, the video highlights the central role of vascular tissue, acting as both a driver and a brake for tissue growth, underscoring its importance in circulation and cell nourishment. A deep understanding of these mechanisms opens up new perspectives for healthcare practitioners and osteopaths, offering concrete tools to improve their patients' health.

CHARACTERISTICS OF DIFFERENT TISSUES

This course aims to understand the phenomenon of **tensegrity** at a developmental level and the balance of the body's **homeostatic** processes, particularly the organisation of **metabolic fields**.

Mr. Benjamin, a doctor, conducted extensive research on **ontogenesis**. By observing movements, he deduced a concept of the **metabolic field**. He introduced the idea of **"boundary tissue"** and **"interior tissue"**.

THE TWO MAIN TYPES OF TISSUES

Under the microscope, he distinguished two main types of tissues:

- **Boundary tissue:** Located at the periphery, it forms the boundaries between vesicles and other structures.
- **Interior (nutritive) tissue:** This tissue brings nourishment.

The **ectoderm** and **endoderm**, for example, are **lipid-forming** or **connective** tissues. This is a classic histological and embryological interpretation, as well as an approach to Blüchmann's **biodynamic embryology**.

AN EXAMPLE OF A MOLECULAR LINK

Many women suffering from chronic gastric problems, such as **reflux** or **oesophagitis**, often find the origin of these disorders in the **pelvic basin**. This manifests at a molecular level through a differentiation in the concentration of **hormones**, **neuro-hormones**, and **gastrin**.

It is often necessary to rebalance the **pelvic fascia** to improve stomach physiology. This is an example of a molecular link.

THE ROLE OF THE VASCULAR SYSTEM

We will address the **vascular system**, the **endothelial system**, which constitutes a major link in the body. A small scratch here causes bleeding. A small scratch there also causes bleeding.

It is a very deep conducting tissue. One must know how to use it in its **electrical force**, with the interdependence of organisational planes, on symbolic forces. One must almost symbolise, that is, gather and unify the system.

VASCULAR TISSUE: MOTOR AND BRAKE

What is mainly retained is that vascular tissue is a tissue where there is no **congestion**. It is always dependent on this system. Whereas in other tissues, there can be phases of congestion.

However, the vascular system is both a **motor** and a **resistance** for the growth of the tissues it nourishes. An arriving vessel is a motor because it provides the force to the tissue it nourishes. It transports nourishment, it gives.

But at the same time, it does not grow at the same rate as the tissues it supplies. So, it is both the motor for growth and the brake for growth. This may seem paradoxical, but it is not.

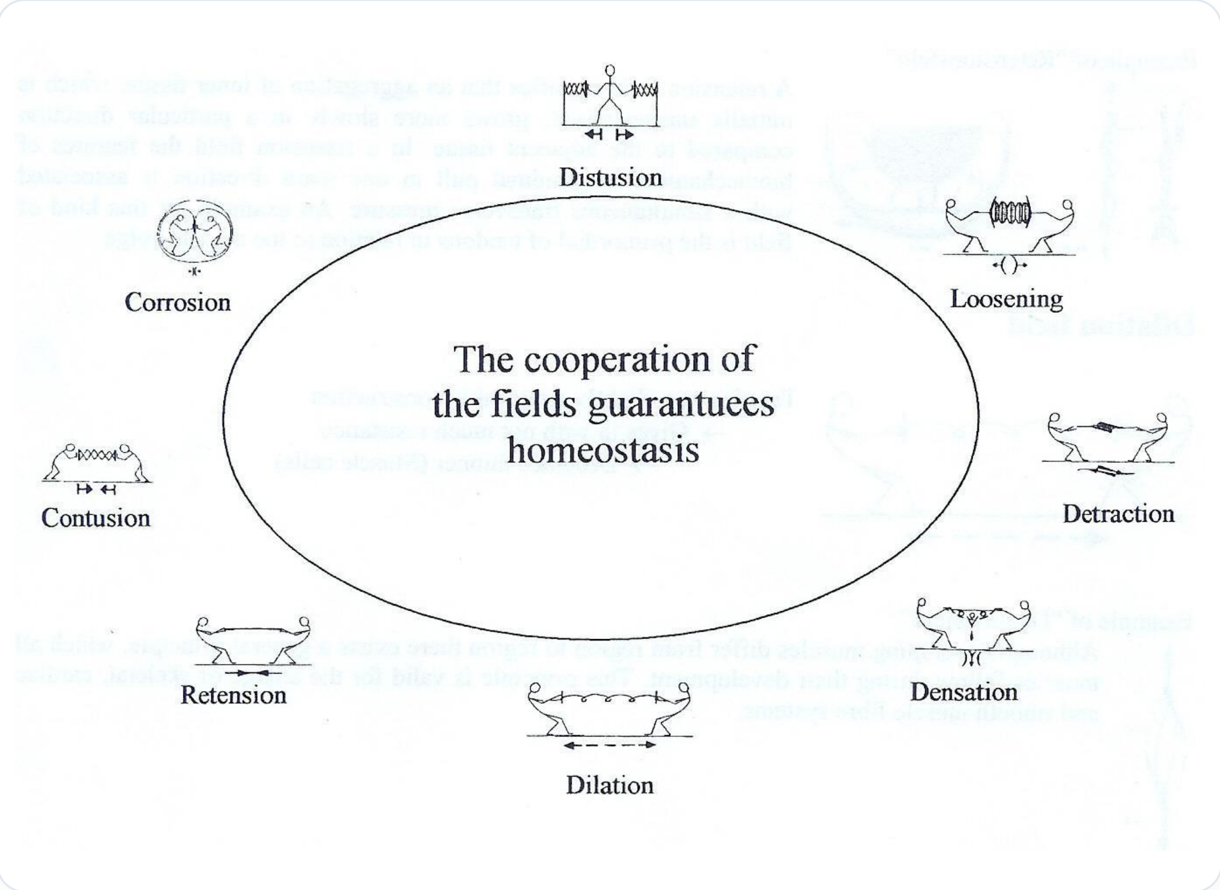


FIG. — Homeostasis by fields

43. CLASSIFICATION OF THE DIFFERENT METABOLIC FIELDS

DURATION : 18:39 STUDY SHEET

COURSE SUMMARY

This video explores the 'Classification of Different Metabolic Fields', focusing on two key concepts: the Corrosion Field and the Aspiration Field. The Corrosion Field is illustrated by the process of fusion of the primitive aortas, where the disappearance of internal tissues creates new structures, such as the branchial arches, and illustrates pathological phenomena such as bedsores. In parallel, the Aspiration Field describes how glands are formed through the aspiration and manipulation of epithelial tissues, taking examples such as the formation of the lungs and the diaphragm. This video provides an in-depth understanding of the metabolic transformations at work during embryonic development.

CLASSIFICATION OF DIFFERENT METABOLIC FIELDS

I. THE CORROSION FIELD: CREATION OF NEW STRUCTURES

A. FUSION OF PRIMITIVE AORTAS

Initially, we have **two primitive aortas** in the middle, with internal nutritive tissues for the developing embryo.

The influential **acneural** referent, which becomes a groove, creates a trajectory for the embryo. This trajectory initiates the formation of two small primitive aortic conduits.

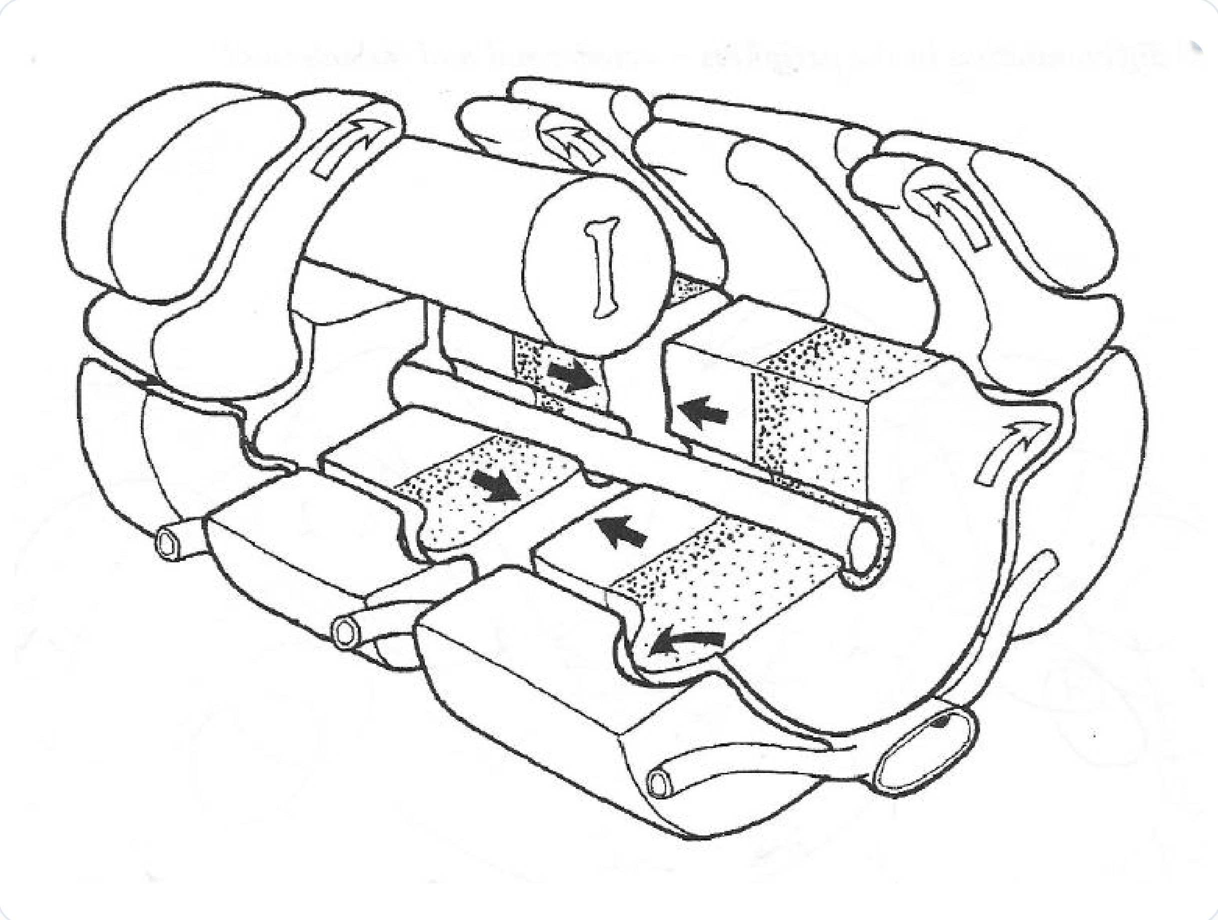


FIG. — Somites and neural tube

The embryo performs a movement of **flexion**, **internal rotation**, and **adduction**, a developmental movement where it moves towards the centre. This approximation is due to the development of the aortas and a process of **telencephalisation** which pulls backwards, bringing everything forwards.

During this phase, the two aortic tubes, initially distinct, converge towards the midline and eventually **fuse**.

B. THE FUSION PROCESS AND CORROSION

During this fusion, the internal tissues gradually diminish until they disappear.

The absence of these internal tissues creates a **corrosion field**. This is a common phenomenon in the body for the formation of various structures.

C. EXAMPLE OF BRANCHIAL ARCHES

A concrete example is the formation of the **branchial arches**. During embryonic flexion, compression appears externally, on the surface ectoderm, forming the branchial arches.

Pressure exerted by the branchial arch against the heart and mandible opens a space: the **mouth**. This is an example of a corrosion field.

D. Corrosion in a Pathological Context: The Bedsore

Another example of a corrosion field, although pathological, is the **bedsore**. Prolonged hospitalisation in bed can cause constant pressure, poor blood supply, and a loss of trophic aspect.

This first leads to overheating, then to a corrosion field. This is the appearance of a new structure due to the destruction of internal tissue.

This concept of a corrosion field is fundamental in the development of the **vascular system**, where structures are constantly changing.

E. Reorganisation of the Vascular System

Initially, we have the **cardinal veins** (anterior and posterior), which then evolve into **subcardinal** veins, then **supracardinal** veins.

This complex reorganisation of the body depends on growth. Compression and decompression fields allow for the appearance of new structures, a process referred to as a corrosion field.

This is not destruction, but a **change in form**.

II. The Aspiration Field (Loosening Kill): Formation of Glands

A. General Functioning of the Aspiration Field

We will now address the **"loosening kill"**, or aspiration and suction field, a fundamental mechanism for the formation of all **glands** in the body.

This type of field always appears along the surfaces of **epithelial** tissues, whether they are of **ectodermal** or **endodermal** origin.

B. Example of Endodermal Epithelial Tissue

Let's take the example of **endodermal** epithelial tissue superimposed on connective tissue rich in vessels, lamps, matrix, and fibres.

Initially, a **congestive** phase is observed along the surfaces, like an accumulation of water. This is similar to walking on a beach where the ground appears congested but can actually give way under weight, like **quicksand**.

C. Formation of the Lung and Diaphragm

Consider the **lung** and the establishment of the **diaphragm**. Behind the digestive tract, on the tissue, there is a weakening.

With the growth and lengthening of the embryo, a space opens, creating an aspiration field. The epithelial tissue is as if sucked into this space, like a suction cup.

The epithelial tissue descends into the connective tissue. It is aspirated and collapses into this connective tissue, forming an aspiration field around it.

D. DEVELOPMENT OF GLANDS: EXOCRINE AND ENDOCRINE

Following this aspiration, two types of cellular movements can occur:

1. **Exocrine glands:** The cells descend and form a gland that secretes a substance into a lumen created by a tube. For example, the **gallbladder** is an exocrine gland.

2. **Endocrine glands:** If the process goes further into the metabolic field, the gland becomes "swollen", disappears, and leaves an isolated gland in the middle. This secretes its substance directly into the **vascular** network.

The **pancreas** is an excellent example of this duality, with an exocrine pancreas and an endocrine pancreas. They originate from the same aspiration field, but some elements have lost their connection to become **cellular islets** producing hormones (insulin) or enzymes (amylase, lipase).

The lungs themselves can be considered a **respiratory gland**, following the same developmental algorithm but with different spaces and chronology. The **pleural vacuum** maintains their shape.

The difference between these glands is not in the initial metabolic field, but in the growth and loss of connection with the epithelial surface.

E. ENVIRONMENT AND GENETIC RESPONSE

Life begins as a vast aspiration field, where the environment interacts with the **genetic** response. The environment releases molecules (**morphogens**) that influence the tissue.

Corrosion fields create new structures, like the bedsore which is a destruction of internal tissue with the appearance of a new structure.

These embryonic processes, determined by the **genome**, space, and time, actively participate in the creation of the different metabolic fields.

III. DILATION, RETENTION, AND DETRACTION FIELDS

A. THE PRINCIPLE OF HOMEOSTASIS

Each metabolic field contributes to this creation, and disturbing one field can disrupt the entire system. This is the principle of **homeostatic guarantee**, where everyone must achieve their balance.

We are in a structure under construction. A **mesoblastic** tissue appears at a given moment, at a certain periphery.

B. FROM DILATION TO DETRACTION

Growth pulls on the tissue, which lengthens while retaining its elasticity by filling with **elastin**. This is a **dilation field**.

If the stretching continues and deepens, the tissue becomes a **retention field**. Initially perceived as a muscle, excessive tension transforms it into a **ligament** or **fascia**.

If the traction prolongs and intensifies further, we enter a **detraction field**. The fluid structure is lost, and the tissue eventually becomes **bone**.

Thus, a continuous dilation field gives a retention field, which in turn, by prolonging, can lead to a detraction field and form bone. This explains the appearance of bone in different ways (membranous or cartilaginous).

IV. CONTUSION AND DISTUSION FIELDS

A. UNDERSTANDING THE FIELDS

Tissue development is influenced by **contusion** and **distusion** fields.

These fields result either from the compression of cells, leading to the formation of **cartilaginous** tissues, or from a metabolic growth mechanism.

B. EFFECTS ON TISSUE THICKNESS

These fields lead to differences in tissue thickness. Each tissue reaches a balance according to its destiny, its bodily situation, its genome and its environment, as well as its growth force.

The role is to maintain this homeostatic principle through the primitive embryonic fields.

C. THE SACRUM AND THE VERTEBRAL AXIS

It is crucial to remember that the **sacrum** is at the base of cortical development.

If the sacrum is mobile, the **vertebral axis** is mobile. If the vertebral axis is mobile, the **medullary** aspect is mobile. If the medullary level is mobile, the **cranium** is mobile.

The sacrum is a structure to be systematically checked.

D. KEY EMBRYONIC SYNCHRONICITY

Synchronisation is essential. The neural tube, for example, articulates with the **cardiopulmonary** cavity.

The **urogenital** axis, by its mesodermal force, is the articulation between the endo and the ecto. The mesoderm is the articulation between the endoderm and the ectoderm.

It is necessary to integrate the **vascular system** to fully understand this scheme, highlighting the importance of the brain, liver, heart, lungs, mesenteric plexus, and renal system.

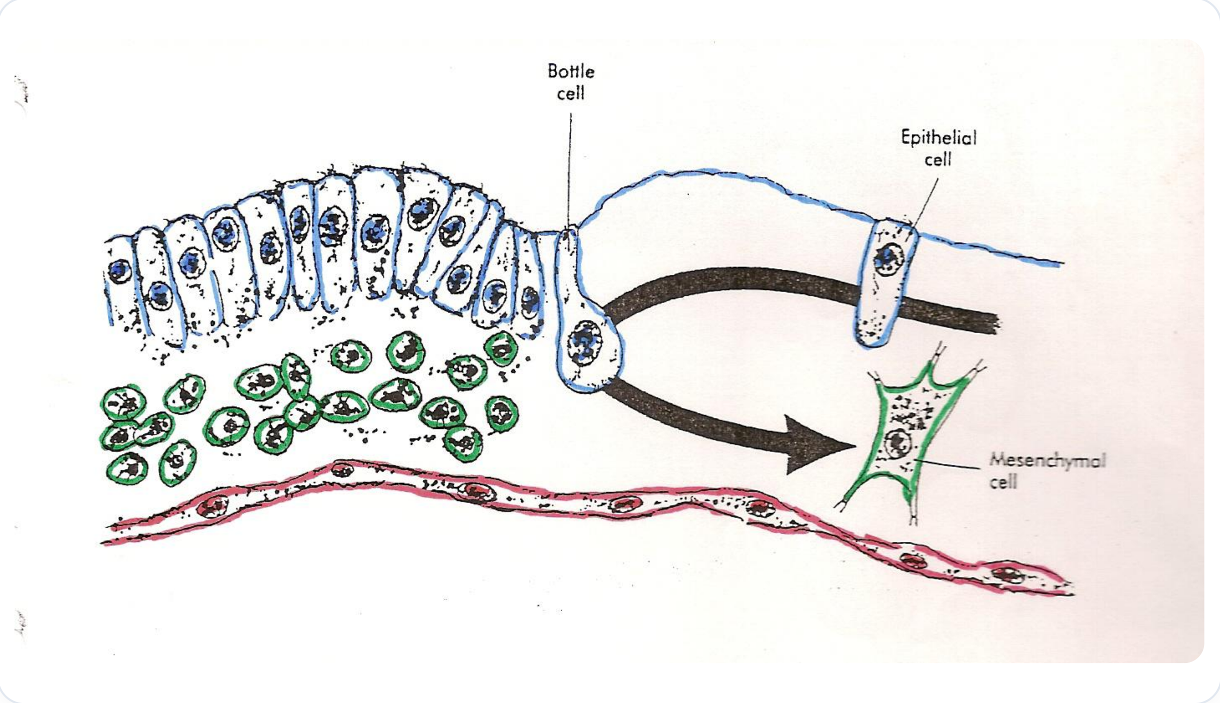


FIG. — EMT (Transition Épithélio-Mesenchymateuse)

44. ACID-BASE BALANCE AND THE HYPOTHALAMIC-PITUITARY SYSTEM

DURATION : 04:50 STUDY SHEET

COURSE SUMMARY

This video delves into the interaction between acid-base balance and the hypothalamic-pituitary system, highlighting the crucial role of the hypothalamus in regulating body tone. Through concepts such as ergotropy and trophotropy, you will learn how acid-base variations influence not only your posture but also your general well-being. Synaptic transmission and its integration at higher neurological levels are explored, offering an understanding of bodily responses and sphincters. By observing these interactions, you will be able to better adapt your practices in osteopathy or manual therapy, allowing for an integrative approach to the body and its chemistry.

ACID-BASE BALANCE AND THE HYPOTHALAMIC-PITUITARY SYSTEM

We will delve deeper into understanding the body in relation to its **acid-base balance**, its **nervous system**, and the chemistry of its tone.

At a certain point in its development, the body integrates a **thalamic** and **hypothalamic** level into its centre.

ROLE OF THE HYPOTHALAMUS IN REGULATION

The **hypothalamus** plays a crucial role, at a deep neurological level, in performing **regulation**. This regulation redistributes a preferential tone, of either an **ergotrophic** or **trophotrophic** type.

Cellular information, which arrives via an **oxidation-reduction** system or **cyclic AMP** and **cyclic GMP**, is integrated at the level of the hypothalamus. The latter then provides feedback.

To simplify, your acid-base terrain will be integrated, via the bloodstream, at the **hypothalamic-pituitary** level. This will transmit information related to your **limbic system** and your **Papez circuit**, re-integrating this into a tonality.

Thus, it will give you a side:

- **Ergotrophic**: oriented towards work, **orthosympathetic**.

- **Trophotrophic**: oriented towards recovery.

This manifests as a posture. Through sympathetic or parasympathetic tone, the hypothalamus must be informed of the acid-base balance, expressing a **trophotrophic** or **ergotrophic** tone.

INTEGRATION OF ACID-BASE BALANCE

At the level of the hypothalamus, a predominant **acidosis** or **alkalosis** between tissues manifests. These are local adaptations integrated into intermediate, **diaphragmatic** levels, connected to continuously stimulated movements.

This means that we have **synaptic transmission** that carries **cyclic AMP** or **gonadocyclic** type information. These two types of transmission are expressed at the cellular level by a dynamic system, i.e., an oxidation-reduction system, or in other words, acid-base.

The whole is recovered at the neurological level, integrated into the **corticospinal pathway** and the **limbic pathway**. All memories, including your diet, are integrated at this level and provide a response.

BODY RESPONSE AND POSTURE

This response manifests in your system via the **paraventricular** or **supraoptic nuclei**, leading to a bodily response.

For example, you can go into a **left-right** or **right-left spiral**, and this will influence your **sphincters**.

At the level of your **pancreatic facial pentagon**, you have different areas with specific sphincters:

- The **pyloric sphincter**.
- The **duodenal sphincter**.
- The **ileocecal sphincter**.
- The **recto-anal sphincter**.

These sphincters integrate according to your movement. If you shift your centre of gravity to the right or to the left, you will have a different action on these sphincters.

If you are blocked, this affects your sphincters, which can be either open or closed. You can be in **diarrhoea** or **constipation**, or in a **precancerous** or **pro-inflammatory** state. All of this, whether you are in **para** or **ortho**, has repercussions in specific chemistry.

The body tries to balance the whole and to equilibrate itself, putting you in a state of constipation or encouraging you to open your centre and have diarrhoea.

All of this is expressed by this chemistry integrated at the hypothalamic level.

We can observe in your **chemo-postural** plan phases of **external rotation** and phases of **internal rotation**.

Observation of posture and balance allows us to read the body's chemistry. This is why the body continuously tries, through the respiratory act, to adapt to the environment programmed by its hypothalamic-pituitary system, located at the base of the skull.

45. DEMO: VENTRICULAR SYSTEM

DURATION : 03:45

STUDY SHEET

COURSE SUMMARY

This video offers a practical demonstration of the biodynamic approach applied to the ventricular system, focusing on tactile perception and progression through different tissue layers. The instructor guides therapists through a process of body exploration, starting with a delicate contact with the surface of the head and descending to the bony structures, while cultivating an increased sensitivity to changes in texture and temperature. Key concepts discussed include oscillatory interaction in the cerebral hemispheres, as well as the creation of a therapeutic space that promotes better breathing and body engagement. This practice proves enriching for understanding the body's internal and evolutionary dynamics.

DEMONSTRATION: VENTRICULAR SYSTEM

CONTACT AND INITIAL PERCEPTION

I begin by positioning myself from the **periphery** to the **depth**, my thumbs spontaneously orienting along the axis of the **ventricles**. My first point of contact is with the **texture of the hair**.

Without excessive movement, I wait to feel the **substance** of the head and the **warmth** that begins to emanate from my hands.

This is a particular moment where one perceives a **difference**, a clear emergence. Suddenly, one feels the **bone**. My hands, rich in **receptors**, become active.

PROGRESSION THROUGH TISSUE LAYERS

I will use the same **progressive** approach. My hands emit warmth, and I descend delicately, layer by layer: the hair, the skin, the **fascia**, then the bone.

I am still on the bone. But now, the perception changes; it begins to "pass through" to the left. I gently enter into this sensation, this perception, this **frequency**.

The bone no longer seems a **rigid** structure. It becomes like **fluid**, allowing for deeper penetration. It is important to remember that we are passing through **mesoderm** here, a mesodermal tissue that extends to the **brain**.

REACHING THE DEEP SPHERES

I wait for the body to manage a subtle **oscillatory** behaviour. I begin to feel it. I take a step back to better appreciate the **depth**.

We are at the superficial layer, the **cortex**, the shadow of the **arachnoid** with its arachnoid pillars. At this level, I begin to perceive the **hemispheres**.

We will go a little further, playing with the **ventricles**. We will feel the brain as a whole.

THE THERAPEUTIC INTERACTION

She feels a new sensation. I will induce a slight movement, like a **dolphin**, on the right side which is breathing a little less well.

I simply offer her **space** to allow her to breathe better. Then, I slowly withdraw, layer by layer.

RECIPIENT'S PERCEPTION

Here is a description of the perceived sensation:

- **At the bone level:** An initial impact sensation, then great lightness.
- **Below the bone:** The movement of the hemispheres was very pleasant. A clear distinction was perceptible between the left and right, particularly at the level of the ventricles.
- **Finally:** A more **electrical** frequency, a very rapid but not sharp vibration.

46. GUIDED PRACTICE: VENTRICULAR SYSTEMS

DURATION : 10:20 STUDY SHEET

COURSE SUMMARY

This video offers a guided practice focused on exploring the ventricular systems through subtle touch. The instructors encourage adopting appropriate positioning and attitude, emphasising the importance of feeling the space around the patient while maintaining a light contact. Participants learn to mentally map the different layers of tissue, from the hair down to the bone structure, while becoming aware of variations in density and fluidity. Progressing towards the cerebral depths, practitioners develop a fine perception of the movements of the hemispheres and the different neural structures, which enriches their ability to work with the nervous system and find solutions adapted to the patient's needs.

GUIDED PRACTICE: VENTRICULAR SYSTEMS

PRACTITIONER POSITIONING

Take the time to place your **thumbs** in the orientation of the **ventricles**, with a delicate touch.

Have a light hand and good contact. Let your elbows and your whole body, your whole mind, serve your hands for the patient.

The thumbs should be light. It is important to have a homogeneous hand, without exerting too much pressure at the top or bottom.

Open your hands wide and first try to have a vision of the **space** around you and your patient.

THE NOTION OF SPACE

You always work with this field around the patient. Even if you go "inside", your mind must remain spacious within the system.

Even if you penetrate the tissue, always give it this notion of space. It is a constant need for space.

The **embryonic force** needs space to express itself. It needs growth and space.

CONNECTING TO SENSATION

Be present with your hands. Feel like a thread connecting your hands to your brain.

Feel this connection from your hands to your brain. Let your brain trust its ability to receive.

Trust your theme and try to perceive the first layer well, which is the **hair**.

MAPPING THE LAYERS

Feel the hair under your fingers and create a **mental map**.

Then, little by little, you will feel that the hair is attached to the **skin**. Then place yourself more directly on the skin.

Try not to move, stay in your perception, just in the sensation.

You will already feel the difference between this first layer and the skin. Feel the skin.

PENETRATING THE DEPTHS

Next, you will gradually bring your field in, very gently, to now feel the **bony structure**, the **outer bone**.

The skin itself is attached to the bone. Place yourself on the bone and try to feel the difference on the **outer cortex**.

With a little time and concentration, you will feel that you can penetrate a little further. Another **density** will appear.

SUBCUTANEOUS PROGRESSION

You go inside, you pass through very gently. You can feel what is happening behind you.

You arrive at something more **fluid**. You have moved from hair to skin, then to the outer bone, and now to the inner part.

You still have a layer which is a bit what we call the **layer...** Use your warmth.

TOWARDS THE VENTRICULAR SYSTEM

You gently enter the **cortex**. The brain emits electricity, it is an interface, another frequency.

It's a slightly electric, tingling, vibratory sensation. And there, you slowly descend towards the **ventricular plane**.

You cross the different wired layers, the **grey matter**, the **nuclei**. Concentrate towards your thumbs.

You will feel that the **right hemisphere** and the **left hemisphere** have slightly separate movements. Accompany the ventricle like a dolphin moving forward.

Try to enter with a slight delay. Do not breathe at the same speed. Let yourself be guided by this process.

SYNTHESIS OF PALPATED LAYERS

You perceive structures such as:

- The **central nuclei**
- The **white matter**
- The **cortex**
- The **pia mater** (bimère)
- The **equitaine layer**
- The **arachnoid** (arglide)
- The **inner cortex**
- The **outer cortex**
- The **fascia**
- The presence of **hair**

47. NEURAL CRESTS

DURATION : 11:21 STUDY SHEET

COURSE SUMMARY

This video delves into the crucial role of neural crests in embryological development, focusing on their impact on the formation of the peripheral nervous system, the face, and associated structures. Through key concepts such as cell migration and influence on the skin, it reveals how neural crests interact with the autonomic nervous system and affect organ health. Therapeutic implications are also explored, allowing therapists to reconcile anatomy, physiology, and epigenetics to better understand and treat their patients. The differentiation processes and epigenetic signals that guide neural crest formation are presented, offering avenues for innovative therapeutic approaches.

NEURAL CRESTS: IMPACT AND IMPLICATIONS

Neural crests are essential structures that appear during the closure of the neural tube. They extend along the neural groove and are crucial because they give rise to the entirety of the **peripheral nervous system**, including all nerves and ganglionic chains.

IMPACT OF NEURAL CRESTS ON THE ORGANISM

IN THE HEAD AND FACE

Intense **invasion** and **migration** of neural crests are responsible for the formation of a large part of the face, from bone to skin. At the base of the skull, tissue mixes and organises with the **mesoderm**, forming the **ectomesoblast**.

This migration is visible with the **rhombomeres**, which organise around the optic capsule and the mesencephalic area. Rhombomeres, numbered 1 to 7, invade the superior, median, and inferior pharyngeal arches.

THE SKIN

The **skin** is a tissue strongly influenced by the nervous system, partly derived from neural crests. It reacts to **solar radiation**, **pressure**, and **behaviour**. The face, in particular, reflects this state, allowing much information about internal health to be read.

THE AUTONOMIC AND VISCERAL NERVOUS SYSTEM

The ganglionic chain, the pre-visceral chain, and the intra-visceral ganglia originate from the neural crests. They also influence the **sympathetic** and **parasympathetic nervous systems**, informing organs such as:

- The heart
- The lungs
- The salivary glands (parotid, submandibular)
- The eye
- The digestive tract

Information from the neural crests can be visibly expressed on the face, allowing the internal state of organs such as the stomach or kidneys to be assessed. The quality of the skin, for example, reflects this information via the **dorsal ganglia** and the **enteric plexus**.

THE TRIGEMINAL NERVE AND CRANIAL NERVES

The migration of embryonic tissues, particularly neurological epithelial types, forms the ganglionic chain and participates in the formation of other structures through connections (integrins, laminins) promoting **melatonin**.

The **cranial nerves**, including the **trigeminal nerve**, whose 5 branches originate from this concentration, convey important information. A major branch transmits visceral information to the stomach and the vagus nerve.

Cranial distribution, including the **archeocranium**, **paleocranium**, **neurocranium**, and **neocranium**, is also conveyed by the neural crest. The formation of the trigeminal nerve depends on the **trigeminal ganglion** and a field that forms between the optic and nasal placodes. The trigeminal nerve results from three convergence fields induced by the dural sheaths, following information from the meninges.

OTHER IMPACTS OF NEURAL CRESTS

Neural crests influence the cascade development of various structures, such as the **inner ear**, **cartilages**, and **bones**. The eye, in its entirety, and its migratory face are independent of the neural crest. The latter positions itself at the level of the cornea, creating a neurological and facial continuity of mesodermal type, in relation to the entire body, including the heart.

DIFFERENTIATION AND EPIGENETICS PROCESS

It is essential to understand the processes of **differentiation**, **remodelling**, **proliferation**, **migration**, **delimitation**, **spiritualisation**, or **transduction**. This cascade of phenomena, although complex at the molecular level, has a direct impact on physiology.

Epigenetics plays a fundamental role in these processes, with specific signals that guide the development of the face and the formation of neural crests. These become **mesodermal** and then purely **neurological**.

They invade the **prosencephalon** (forebrain), **mesencephalon** (midbrain), and **rhombencephalon** (hindbrain), impacting the sympathetic and parasympathetic systems. All of this is linked to the neural crest.

ORIGIN AND THERAPEUTIC IMPLICATIONS

It is crucial to distinguish the embryological origins of different structures:

- **Base of the skull:** derives mainly from the **paraxial mesenchyme**.
- **Base of the dental arches:** derives mainly from the **neural crest**.
- **Spinal column:** derives from the ribs of the **paraxial mesenchyme**.
- **Limb skeleton:** derives from the **somatopleure** (limb buds).

These distinctions have major therapeutic implications:

1. **Face and teeth:** must be treated taking into account their origin from the neural crests.

2. **Base of the skull and spinal column:** their treatment must be homologous to their mesenchymal origin.

3. **Limb lesion:** involves the regulation of the peritoneum and the reintegration of the limb into the abdomino-thoracic cavity. This falls under a pulmonary plan and may require working on an apnoea brake at the lung level.

The articulation point between the **parietal peritoneum** and the **visceral peritoneum**, at the level of the pulmonary hilum, is an example of an area where neural crests play a role, acting as a suction field that invades the entire body.

This understanding of neural crest migration, particularly at the skull and shoulder level, offers avenues for therapeutic reflection.

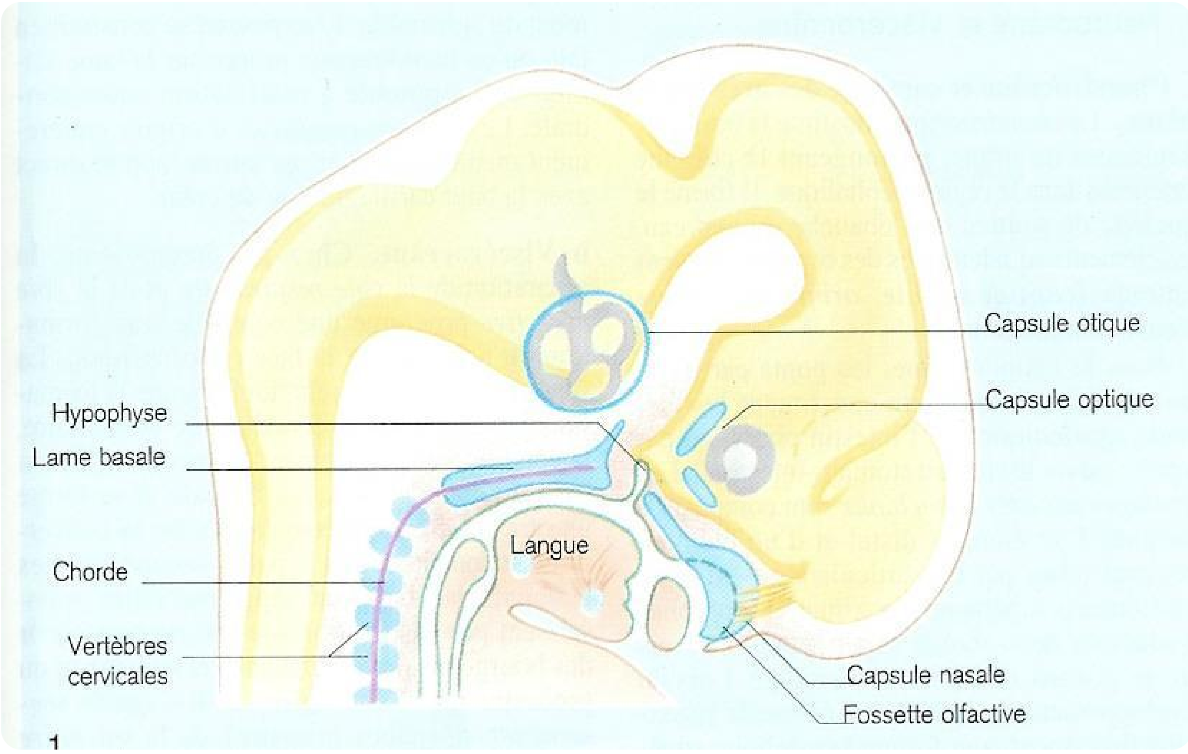


FIG. — Human embryo sagittal

48. NOTES

DURATION : 03:47

STUDY SHEET

COURSE SUMMARY

This video explores the development of the skull through the lens of biodynamic embryology, with a particular focus on the central role of the brain and cranial structures. The concepts of the midline of anticipation and blockages related to 'micro-crystals of time' are discussed, offering therapists keys to help their patients release emotional restrictions and better anchor themselves in the present. With a reflection on the acceptance of death and breathing as an act of creation, this session offers concrete tools to support personal and professional transformation.

DEVELOPMENT OF THE CRANIUM AND THE NOTION OF TIME

THE ENGINE OF CRANIAL DEVELOPMENT: THE BRAIN

We will address the **development of the cranium**, integrating the **meninges**, the **vault**, and the **base of the cranium**. There is a logic and a sequence of information stemming from the movement of **brain** development.

The primary engine of this cranial development is not differential growth rate, but rather the **brain** itself.

Yesterday, we explored this extraordinary development of the brain, this movement of **flexion** and **cerebralisation**. This process then leads to the straightening of the embryo.

THE MIDLINE OF ANTICIPATION

All this force converges towards the **anterior midline**, which can be considered a line of anticipation.

On this line, we find the **gonadisation process**, essential for reproduction. We also find the **thymic process** associated with the sternum, which shapes our **immunity**.

It is a complete schema that allows us to project ourselves into anticipation.

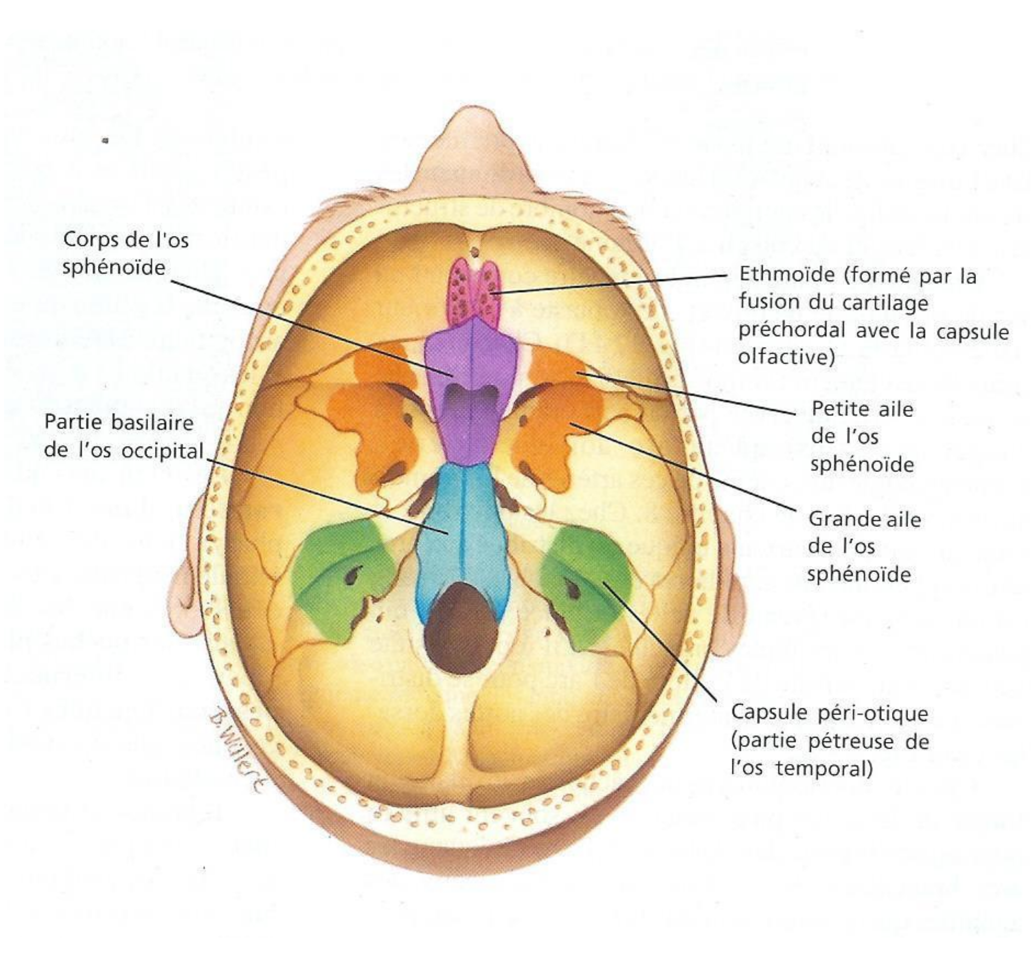


FIG. — Fetal skull base

BLOCKAGES AND "MICRO-CRYSTALS OF TIME"

Sometimes people find themselves blocked. Sometimes, a light pressure in one or two places is enough to open these **"time spaces"**.

We tend to close in on this midline, creating **micro-crystals of time**. This is concentrated time that needs to be released, because anticipation becomes impossible.

We then remain frozen in a past that no longer exists.

LIVING IN THE PAST: AN ANALOGY

Many people live in the past in this way. To illustrate this, I sometimes use this image:

It's as if, in your pocket, you had a small vial from the past, containing a very bad smell. And every day, you open it and breathe in that bad smell.

I tell them: "But let it open, let it go."

ACCEPTING DEATH TO LIVE LIFE TO THE FULLEST

This is where the **acceptance of death** comes in. When we understand death, we become more serene with our own time. Suddenly, the essential is revealed.

For this, it is necessary to study the processes of the **"bardos"** (referring to the intermediate states after death in certain traditions). This means reaching a state where one can easily free oneself from these patterns.

Embryology and the processes of death are intrinsically linked.

RESPIRATION: A CONTINUOUS ACT OF CREATION

How do we pass? I believe that with each **breath**, we traverse the time of creation.

Each breath is an opportunity to catch our breath, a moment of transformation to which we are invited.

Accepting to let go of what arises is crucial.

THE FEAR OF DYING AND TRUST

It's always difficult, because what in me is so afraid of dying? What have I acquired throughout my life that makes me so vulnerable to death?

Often, these are the brakes that hold us back. We must know how to trust.

This is where a specific word, gesture, or look must emerge to give this path of trust to the person.

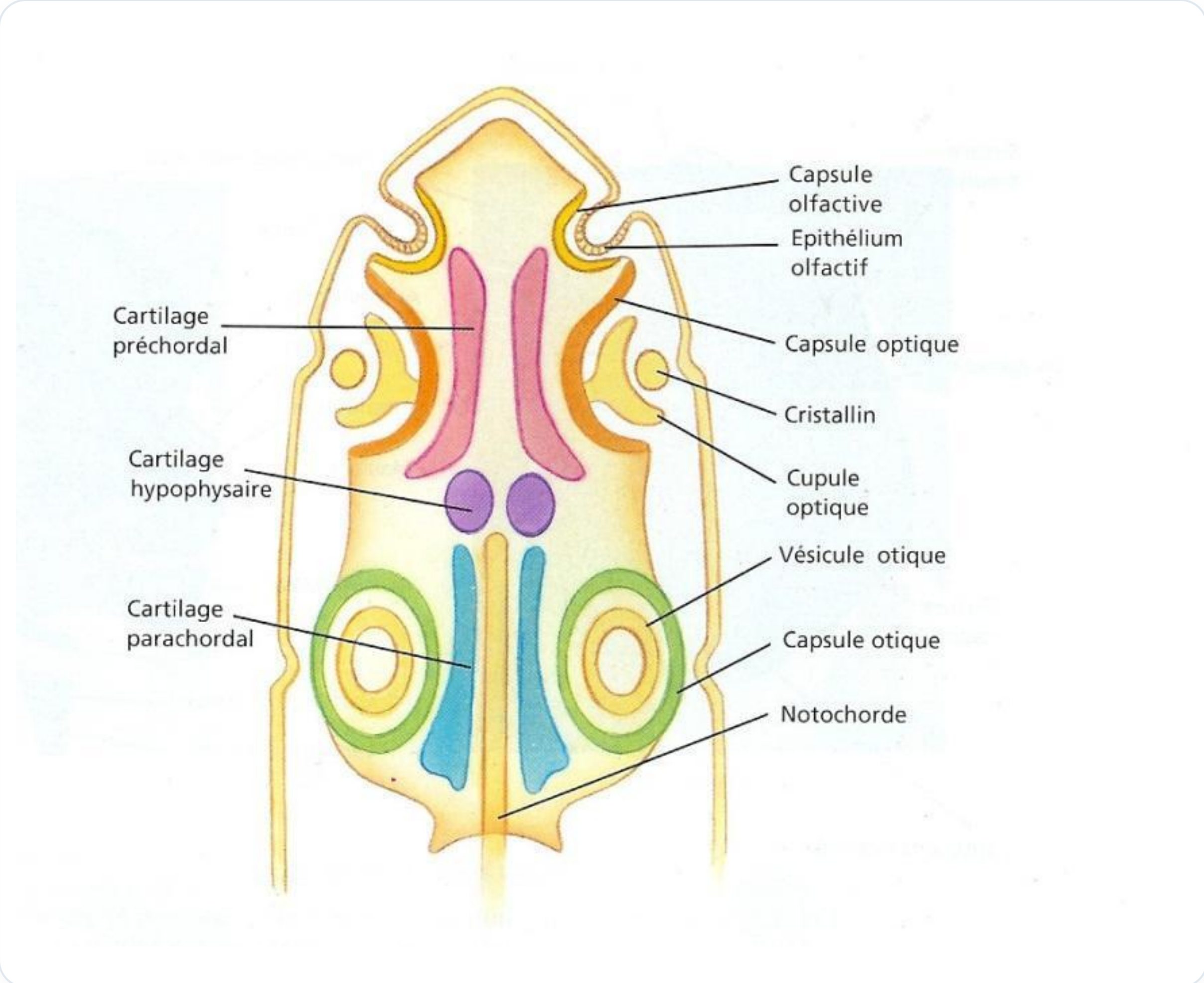


FIG. — Embryonic head development

49. MEDITATION

DURATION : 01:11

STUDY SHEET

COURSE SUMMARY

In this video, meditation is presented as an essential moment of rest and satisfaction after a busy day. The analogy of returning home, taking off one's shoes, and settling in comfortably, illustrates the importance of allowing oneself a moment to pause. The teacher invites participants to cultivate a quality of contentment, awakening an inner flame that recalls the importance of appreciating the present moment. The video thus encourages a personal journey towards meditation, highlighting its benefits for mental and physical well-being.

49. MEDITATION

Meditation can be compared to a busy day at work.

Imagine you've had a long day working. When you get home, you can take off your shoes and settle comfortably into your armchair with a nice beer.

At that moment, you feel a **satisfaction** at having accomplished something significant. You are pleased with the work done and you take a moment to rest, both physically and mentally.

This is an opportunity to develop a small **quality** of contentment within yourself.

Light that small inner flame that reminds you to appreciate every moment and what you have.

Thus, we can begin our journey towards **meditation**.

50. THE CRANIUM

DURATION : 01:00:35 STUDY SHEET

COURSE SUMMARY

This video explores in depth the influence of brain development on the intracranial membranes and the venous system, focusing on the formation of the reciprocal tension membranes. You will learn how the base of the skull is crucial for the craniosacral mechanism, and how the teeth and tongue act as postural sensors that influence balance and respiration. The interaction between the inner ear, posture, and bodily rhythms is also discussed to provide a holistic view of neuro-digestive balance. Finally, the video addresses the metabolic processes related to cartilaginous formation and tissue chains, illustrating their significance in overall bodily dynamics.

THE CRANIUM

THE INFLUENCE OF CEREBRAL DEVELOPMENT ON INTRACRANIAL MEMBRANES AND THE VENOUS SYSTEM

Cerebral development guides and organises the phenomena of the **intracranial membranes**. This process is essential for the formation of the **reciprocal tension membrane** system.

This system is crucial for the pathway of **vessels**, particularly important veins, serving to drain and release **CO2**. Many problems arise from **stases**, with key points such as the **pterions** or certain **transverse occipital venous sinuses**, where eight layers meet. This is a vital crossroads.

THE CRANIO-SACRAL MECHANISM AND THE TEETH

The **base of the cranium** is a fundamental reference point of the **cranio-sacral** mechanism. The **teeth** play an essential role as postural sensors, especially when the child straightens up and begins to have teeth.

They provide support for **correct breathing**. The absence of teeth in the elderly, for example, alters posture and breathing, illustrating their importance.

ROLE OF TEETH AND TONGUE IN POSTURE AND BALANCE

Teeth are sensors that contribute to the child's straightening and improved posture. Not missing a tooth is crucial, as its absence can shift the tongue to the side.

The **tongue** is a key element of the **lingual axis**, which is part of the longitudinal axis and the embryonic chain extending to the thumbs and big toes. The tongue, posture, and teeth work together to maintain **balance**.

THE INNER EAR, POSTURE, AND BODY RHYTHMS

Stability is sought via the **vestibular system**, with an orientation of 23 degrees for the inner ear. This angulation horizontalises and aligns with that of the heart and head tilt.

The body orientates itself in relation to the universe on angular planes to regain stability. We have **cardiac rhythms** (100,000 beats/day), **respiratory rhythms** (25,000 breaths/day), and **digestive rhythms** (1 to 2 litres swallowed/day).

THE BASE OF THE CRANIUM, THE PITUITARY GLAND, AND NEURO-DIGESTIVE BALANCE

The base of the cranium is a major articulation between the brain and the aorta. A crucial **embryonic flexion**, resulting from the bending of the embryo and the encephalon, compresses the mesenchyme under the mesencephalon, forming the **sub-mesencephalic septum**.

The heart, aortas, and amniotic cavity organise this space. A small endodermal pouch, **Rathke's pouch**, forms the anterior pituitary, composed of a neurological part and a digestive part. The **infundibulum**, coming from the brain, joins it to form the neurological pituitary.

The pituitary gland, the endocrine and neurological conductor, rests on the base of the cranium, a point of support. This base of the cranium is a balance between the **neurocranium** and the **viscerocranium**, reflecting the environment of the brain and the digestive system.

METABOLIC PROCESSES AND CARTILAGINOUS FORMATION

Torsion phenomena or imbalances in the movement of the base of the cranium can affect pituitary function. Function is a repetition of a primary embryological movement that organises **motility** and **motoricity**.

The formation of Rathke's pouch is also a field of **metabolic contusion**, where two tissues contract. Initially, cartilaginous tissue develops the hypophyseal, parachordal, and prechordal cartilages, which will form the hypoxephital base, the **sphenoid**, the **ethmoid**, and the **petrous pyramids**.

TISSUE CHAINS AND THEIR SIGNIFICANCE

The **occipito-lingual chain** is crucial for the complete dynamics of the body and constitutes a midline. The lingual complex and the tongue (heart bud) are linked to the **thyroid**, which electrifies our words.

The **frontal sinuses** are involved in self-expression and self-protection. **Mantras** are frequencies that neutralise incessant inner discourse, bringing us back to archetypal sounds.

NEUROLOGICAL CHAINS AND THEIR ARCHETYPAL ROLE

- **The lingual chain:** Interacts with the occipito-lingual chain.
- **The sphenoidal and axial chain:** A descending chain of life anticipation.

DEVELOPMENT OF THE CRANIUM AND BRAIN

The **neurocranium** and **viscerocranium** are separated by the base of the cranium, a vital element for development and articulation between the aorta and the brain. This is a precursor image of the cranium, with **cerebralisation** as the driving force.

The differential growth of the brain leads to the flexion of the embryo, with the **prosencephalon**, **mesencephalon**, and **rhombencephalon** providing the dynamics.

EMBRYONIC FLEXION AND THE BASE OF THE CRANIUM

The forward flexion of the encephalon makes the mesenchyme under the mesencephalon perpendicular to the longitudinal axis. This sub-mesencephalic compression forms the sub-mesencephalic septum, which is the origin of the base of the cranium.

Rathke's pouch, of endodermal origin (intestine), becomes the anterior pituitary, showing that the base of the cranium is aware of the environment of the brain and the digestive system.

THE MEMBRANES OF THE BRAIN: FROM DURA MATER TO BONE

The **neural tube** is surrounded by an **ectomeninge**, the anlage of the dura mater. This tissue develops along the axis of the neural tube, closing at the anterior and posterior neuropores.

With **telencephalisation**, the meninges grow and the midline approaches to form the **crista galli**. The tentorium cerebelli folds at the back, and the falx cerebri forms at the top, following the development of the brain.

DURAL BELTS AND CRANIAL OSSIFICATION

The dura mater, with its double layer (visceral and parietal), is an envelope of the brain. Areas of **densification**, called **dural belts** (anterior, middle, posterior), orient towards the base of the cranium, a place of intense membranous tension.

These mesodermal condensations, under the influence of metabolic growth fields, transform into **bone**, forming the cranial vault.

THE PROCESS OF TELENCEPHALISATION AND THE CRISTA GALLI

Initially, the anterior and posterior brain meet and compress the base of the cranium. The future **interorbital ligament** is present. During telencephalisation, the system is pulled upwards, straightening the **pterygoids** and developing the falx cerebri.

The crista galli, a crucial place for body and mind, is formed by the folding of tissue and the approximation of the pterygoids. It is a point of support for cerebralisation, the face, and the orientation of the eyes, leading to **clairvoyance**.

HEMISPHERIC SPECIALISATION AND ITS IMPLICATIONS

The embryonic flexion force transforms into a straightening force, developing the **cerebral hemispheres**. These attach to the crista galli, tightening up to the Egnon.

Each step, from **Nazion** to **Asterion**, releases movements up to the cranium. Good **dental occlusion** is essential for the freedom of cranial movements, especially in the frontal areas.

BONE FORMATION AND THE ROLE OF THE MESODERM

Bone is built within the membrane, and the brain is the engine of this development. The mesodermal tissue around the neural tube densifies to form the visceral dura mater, then the parietal, and finally the bone.

This mesoderm undergoes metabolic fields of growth and expansion, hardening to form the vault. Bone is the result and the finality of the **cerebralisation** process.

THE TAI CHI OF THE BRAIN AND THE TONGUE

The **ascensus** process of the brain, with its rapid growth, brings structures closer to the midline and spreads them laterally. The primitive digestive tube rises, and the tongue follows its development.

The depth of the tongue is a reflection of the depth of cerebralisation, a homologous symmetry.

THE BASE OF THE CRANIUM: A POINT OF EMBRYOLOGICAL CONVERGENCE

The base of the cranium is a point of convergence of information:

- **Bony:** hypophyseal and parachordal cartilages.
- **Endodermal and ectodermal.**
- **Brain tension membranes.**
- **Lines related to the branchial arches.**

It is a **hypothalamic-pituitary** convergence point, exchanging chemical and tonic information. The vascular roof of the pituitary is filled with vessels.

DURAL SHEATHS AND CRANIAL CONSTRUCTION

The primitive **dural sheaths** organise along the axis between the aorta and the brain, a major confluence. We also find the falx cerebri, the dural sheaths, and the bone that is built from a common tissue.

Cephalisation, the growth of the brain, organises this entire process, including the ventricular system and the tension membranes.

TISSUE CONTINUITY AND RADICULAR PROBLEMS

The external and internal dura mater, the **arachnoid**, and the **pia mater** constitute a continuity with the epineurium, perineurium, and endoneurium. Small mesodermal blood capillaries in the endoneurium are causes of stases.

Pumping and releases decongest radicular problems. **Hoffman's ligaments** and **Forestier's opercula** create valves for fluid balance, protecting nerve roots.

CRANIAL OSSIFICATION AND DURAL WINDOWS

The cranial vault is the result of densification. Metabolic fields participate in the **synchondrosis** of the base of the cranium. Detraction forms the bones in the direction of the displacements of the **trabeculae** and **osteoblasts**.

Ossification nuclei form at the periphery. The development of the brain, framed by the dural sheaths (like windows), dictates the chronology of ossification: frontal, parietal, superior occipital, and superior temporal.

The frontal, parietal, and occipital windows are the future sutures.

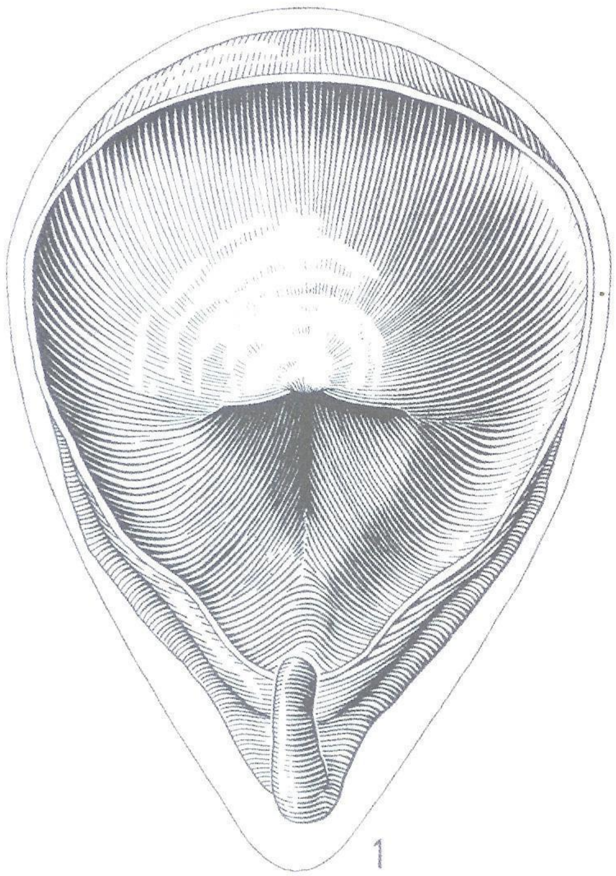


FIG. — Urinary bladder

51. DEMO: THE DURAL SHEATH

DURATION : 06:08 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth demonstration on the examination and rebalancing of the middle dural sheath, aiming to improve the body's freedom of function. Through practical techniques, the instructor meticulously examines cranial structures and detects any anomalies, focusing on fluidity and the electrical field around the coronal suture. The importance of this work is particularly emphasised in children, where mandibular development and sucking dynamics are crucial. In parallel, the video addresses the interconnection between the vascular system, embryonic movements, and the craniosacral axis, while proposing methods to ensure the freedom of various diaphragms, including those in the feet. The session concludes with checks of the

DEMO: THE DURAL SHEATH

EXAMINATION OF THE MIDDLE DURAL SHEATH

I position myself on the **coronal structure** to examine the **middle dural sheath**. I meticulously check each bevel, ensuring no tooth is misplaced or missing.

I observe if the structure presents **areas of increased density** or **rigidity**. Sometimes, this can be the case.

This is a crucial process to ensure the **freedom of function** of the symphysis in relation to the occlusal system. If I detect an anomaly, I intervene directly to manipulate it.

This manipulation can sometimes trigger a **release of an entire electrical field**.

REBALANCING VIA THE DURAL SHEATH

Next, I proceed with **rebalancing** work. I take the dural sheath and focus on it.

I know that this dural sheath is oriented towards the **base of the skull**. I therefore position myself taking into account the entire body, adopting a global vision.

This includes the **Bregma** area and extends to the level of the articulation, covering the entire coronal area. Not the Bregma directly, but gently, at the level of the **anterior fontanelle**.

I wait for a moment for the person to feel a **release**. It will be like a packet of butter melting. This is the transition from a sensation of **density** to a sensation of **fluidity**. I am looking for this **"fluid body"**.

SEEKING FLUIDITY AND ELECTRICAL FIELD

As in our practice this morning, where we moved from hair to liquid, here, I focus on the sheath, but in its entirety. I am looking for this fluidity at the bone level.

I feel that the density is different, less present. At this stage, I begin to gently enter the **electrical field** of the coronal suture.

Once I have this connection, I position myself and look for this link. I **re-harmonise** the base of the skull through the peripheries in another way.

IMPORTANCE IN CHILDREN: MANDIBLE AND ANTERIOR SHEATH

This is very important work in **children**. I work a lot on the **mandible** in babies.

They must go from the womb to the mouth, and they need powerful **sucking** and **lingual** and **maxillary activity**. This is essential for their development.

I can also work on the **anterior sheath** in relation to the **transverse sinus**. I position myself and seek to restore freedom to this tissue.

I work in such a way as to open up the space so that the **drainage system** can be released. You are not always aware of this embryonic movement and its totality.

VASCULAR SYSTEM AND EMBRYONIC MOVEMENTS

This will be the subject of another work where I will present the **vascular system**. I will show how to clear the dried field more precisely with **facial movements**, **heart movements**, and **abdominal movements**.

It is important to remember the **dynamics of brain development**, all this movement, including **ventricular movement**. I place my thumbs to work the different dural sheaths in relation to the base of the skull, and reintegrate them each time.

CRANIOSACRAL AXIS AND DIAPHRAGMS

It is natural to work the **craniosacral axis** and check for the absence of **whiplash**. It is crucial that this craniosacral system has integrated freedom in the different **diaphragms** and levels.

A very important diaphragm is the one at the level of the **feet**. It rebalances the small pelvises. Many problems of **congestion** of the small pelvises can be linked to feet that are not free.

KNEE TECHNIQUE AND REBALANCING

The **knee technique** is also essential. There must be good freedom and good re-alignment of the knees. The knee must be very free at this level.

It's a simple technique: adjust the knee, by compression, to restore its freedom. The important vision, the important way.

I very often finish by checking if the **"rebalancing"** is correct. I work a lot by observing the space.

52. CONCLUSION PHASE 1

DURATION : 02:03 STUDY SHEET

COURSE SUMMARY

This video concludes Phase 1 of the biodynamic embryology teaching, exploring the three fundamental midlines and their relationship with the urogenital zone. The importance of grounding is emphasised, connecting the mineral kingdom to physical reality and integrating the emotional and rhythmic aspects of the plant and animal kingdoms. By establishing yourself in your neurosensory realm, you learn to differentiate between judgement and discernment, thereby fostering a deeper connection with your emotions and your environment. This harmony leads to improved breathing and grounding, beneficial for your professional practice.

CONCLUSION OF PHASE 1

We carry within us these **three fundamental midline formations**.

We have already integrated the first midline, which is the **notochordal midline**. This notochordal line is the starting point for the establishment of the **posterior midline**, which, in turn, gives rise to the **anterior midline**.

THE UROGENITAL ZONE: ANCHORING AND REALITY

Within us is this **urogenital zone**, the place where we can sit and anchor ourselves to the earth.

This is the **matrix**, the **matter**, the **reality**. I can be well-seated in my heels. As was said, one must learn to breathe right down into one's heels, which is synonymous with **good deep breathing**.

THE REALMS OF BEING: A HARMONIC PROGRESSION

THE MINERAL REALM

This represents the **mineral realm**. When you are well-anchored in your mineral realm, when you are well-rooted in your reality, and when your **urogenital reproductive system** is in harmony, then you are well-connected to your **vegetal realm**.

THE VEGETAL REALM

This means you are in balance in your **metabolic realm**. The entire vegetal realm is also intrinsically linked to your **emotions**. Emotions are fluid, they represent the **liquids** of our body.

THE ANIMAL REALM

Once this balance is established, you can feel good in your **rhythmic realm**.

When you are in phase with your rhythms, when you are solidly anchored in them, you are then in harmony with your **animal realm**. Everything rhythmic resides here.

THE HUMAN REALM: NEUROSENSORY AND DISCERNMENT

And when all of this is in place, you can ascend to your **human realm**, the **neurosensory** realm.

This is where one learns to breathe well. When I am well-established in my neurosensory realm, I am no longer in judgment, but in **discernment**. My perception opens up.

When I am in my discernment, I am well within my rhythms. When I am well within my rhythms, I am well within my emotions.

I then fully return to my Earth, in perfect harmony with my emotions.

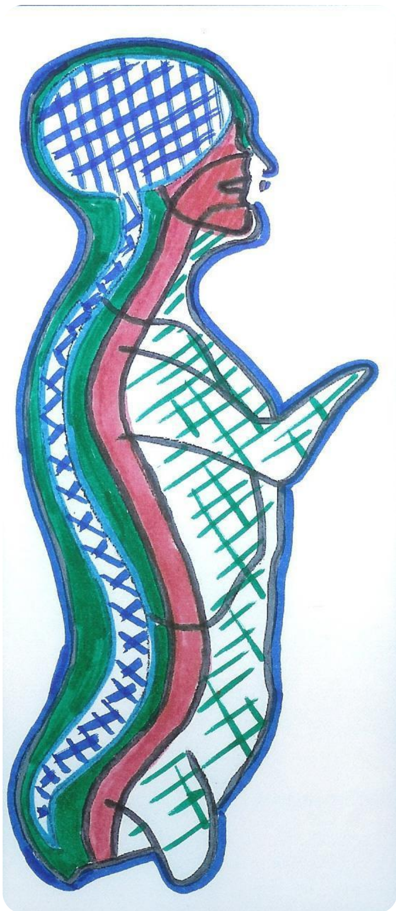


FIG. — Notochord and Neural Tube